

Formality as a Financing Decision

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Abstract

Informality is prevalent and has wide socioeconomic consequences that previous policies have failed to address. This paper asks: Is informality sensitive to financial conditions? The answer is yes: credit supply causally increases the formality share. We use a unique dataset (the universe of commercial bank loans matched with economic censuses) and a shift-share instrument to find that credit supply increases formality through more formal entry and formalization, and fewer firms falling back into informality. The effect is mainly through larger loans and lower collateral requirements. A model rationalizes these findings. The main intuition is that firms that formalize are those whose growth needs are large relative to their wealth. Counterfactuals indicate that financial inclusion is the only lever in the menu that formalizes (four times the tax effect) and simultaneously grows. Our contribution lies in showing that bank loan terms, previously ignored as a formalization lever, causally affect informal firms and formality shares to a greater extent than tax margins.

Keywords: informality, firm dynamics, financial frictions, credit supply, formalization.

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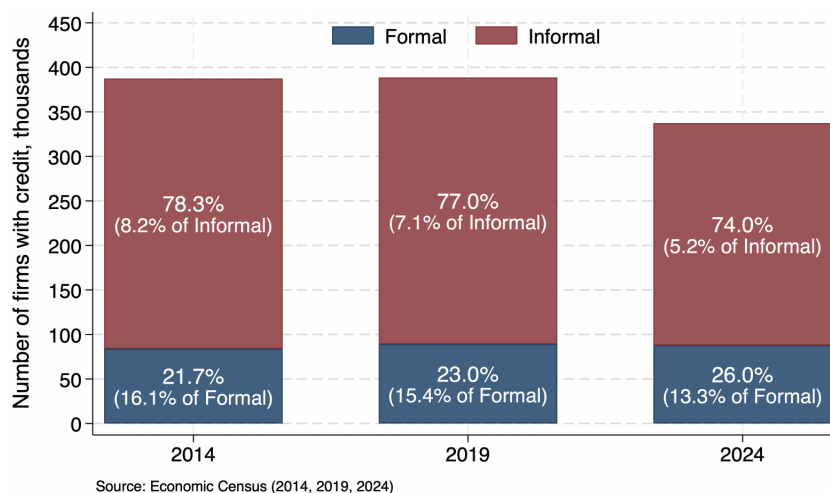
1 Introduction

Informality is a prevalent phenomenon with wide socioeconomic consequences that previous public policies have failed to address, and therefore a relevant topic of study. Informality is present across regions and periods, and in the developing world informality rates can reach well above 50 percent. In Mexico, according to the ILO data, informal employment reached 58 percent in 2019, and the pattern repeats across countries at similar income levels (Appendix Figure 9). Informality erodes tax collection, shrinks the reach of social insurance, concentrates employment in small, unproductive units, while at the same time it provides subsistence income to people who might otherwise face unemployment. Public policy has tried to tackle it mainly through the fiscal margin, with simplified tax regimes¹, and payroll reforms. However, the quasi experimental record of those levers in Mexico shows effects close to zero (Campos-Vazquez et al., 2020).

This paper asks the following research question: Is informality sensitive to financial conditions? The following three pieces of descriptive evidence suggest that the financial conditions and the informal sector affect each other.

First, informal firms are inside the banking system. Figure 1 reveals that among all the firms holding a bank loan in Mexico, 74 percent are informal². However, because the informal sector is so large, the share of informal firms with a bank loan represents only 5 percent, against 13 percent for formal firms. Therefore, informal firms are directly exposed to bank credit conditions.

Figure 1: Borrowing shares by formality



Second, formality status is not permanent and it can change, potentially due to changes

¹The **Simplified Trust Regime** (RESICO, by its acronym in Spanish) is an example of this policy.

²For the purpose of this number, an establishment is informal when it reports zero social security contributions in the census while employing workers and earning income, the census classification used throughout the paper (Section 3).

in the economic environment, like the financial conditions. Table 2 shows that between 11.6 and 21.8 percent of formal firms fall back into informality in five years, while between 3.5 and 6.5 percent of informal firms formalize. In contrast with previous work that treats formality as an absorbing state from which firms cannot return into informality (Meghir et al., 2015; Ulyssea, 2018), we see formality as a two-way border³.

Third, the relationship is visible at the macro level as well: informality and financial depth are negatively correlated across countries (Appendix Figure 9).

The unique richness of the dataset used in this paper allows us to illuminate a blind spot that previous work could not reach. Credit registries and social security records observe only formal outcomes by construction, so studies of credit supply shocks measure outcomes inside formality and stop at its border. In contrast, we shed light beyond this border by matching the universe of commercial bank loans, nearly 1.3 million borrowing firms across 119 banks between 2005 and 2025, to four economic census waves that count formal and informal establishments and follow their status over time, all within the same local labor markets.

Despite the appeal of the mounting descriptive evidence, no causal effect can be claimed. The challenge to claim causality is endogeneity in the form of reverse causality (formality affecting financial conditions and not the inverse) and unobserved common characteristics (only productive firms formalize and hold a bank loan). To overcome this challenge we rely on a Leave-One-Out Shift-Share Instrumental Variable (LOO-SSIV) with exogenous shift. Under this approach, bank level supply shocks (the shift) are estimated from variation within firms that borrow from several banks, in the tradition of Khwaja and Mian (2008) and Amiti and Weinstein (2018), excluding each target market’s own loans. Each market’s instrument then aggregates these Leave-One-Out shocks with the market’s predetermined bank shares, following the quasi experimental shift share framework of Borusyak et al. (2021).⁴ The instrument is strong, with a first stage F statistic above 310, and a one standard deviation shock roughly doubles five year market credit.

The central finding is that credit supply causally increases the formality share through three main mechanisms: more new firms entering as formal, fewer existing formal firms falling back to informality, and more existing informal firms formalizing. So a supply driven ten percent credit expansion adds about six formal firms in the average market and roughly

³The Formal-Informal and Informal-Formal transitions are not an artifact of the definition of informality. Appendix A documents that these transitions exist under alternative definitions of informality: accounting for the intensive margin (Table 14), using the social security number (Table 15), or using the tax number (Table 16).

⁴Inference uses the exposure robust standard errors of Adão et al. (2019) together with wild cluster bootstrap probabilities (Cameron et al., 2008). The design rests on roughly sixteen effective bank shocks, no single bank carries more than 17 percent of the exposure weight, and the results survive rebuilding the instrument without the largest lender, at a cost in precision.

3,300 across the 556 estimation markets (an elasticity of 0.05), while the informal stock stays flat. Formal entry rises, lifting the five year formal startup rate by 0.23 percentage points, about a third of the secular startup decline that [Pugsley and Şahin \(2019\)](#) document for the United States over an equivalent horizon. Exit rises with an informal tilt, so turnover accelerates. Formalization speeds up with an elasticity of 0.052 and movement back into informality slows by 0.039, so a credit supply shifts the traffic toward formality on both sides.

A credit supply deepens, rather than widens, by mainly increasing the loan size and reducing the collateral requirements. Decomposing the response, 64 percent operates through larger loans, 20 percent through more loans per borrower, and 16 percent through new borrowers. Prices barely move while quantities and terms do. The average lending rate rises by only four basis points, on a base of fifteen percent, whereas collateral requirements fall by about 17 percent. That prices do not fall, as a pure supply expansion would suggest, is consistent with the marginal borrowers being riskier.

On the real side, formal employment rises where tax reforms had not affected it, though the magnitude reflects the largest markets. Moreover, value added reallocates away from the informal sector.

The main results are robust to a battery of eleven robustness checks. Formality implications, credit effect decompositions, and informal value added effects are robust. On the contrary, the employment and wage magnitudes are a feature of markets carrying substantial credit weight rather than of the typical market, and the aggregate value added response does not survive controls for lagged trends, so the paper does not lean on either. The robustness exercises including reversion in bank lending, an instrument rebuilt without the largest bank, and a bank level design with independently identified shocks, are collected in [Section 6.3](#).

A parsimonious model rationalizes these effects and disentangles the mechanisms that make a firm formal. The main intuition of the model is that the firms that formalize are those whose growth needs are large relative to their wealth, so credit supply operates through marginal firms at the formality border. In particular, we build a firm dynamics model in which formality is a financing decision. Firms choose each period between formality, informality, and exit. Formality carries taxes and higher operating costs but better financial terms (lower marginal and fixed loan costs and more pledgeable collateral) while informality saves the taxes and has lower operating costs but faces detection costs that grow with size and worse financial terms. The simplified version of the model delivers the paper’s sorting result in closed form. The full model is estimated by matching the census cross section and the Shift-Share coefficients jointly, thirteen parameters against twenty three moments, so the incidence of the empirical shock is an estimated object.

The estimation of the model indicates that the friction is a formality cost wall that is

paid only by the most productive firms. Operating formally costs about 62 thousand dollars a year, against 3.5 thousand informally, and borrowing formally has a fixed cost of nearly 107 thousand dollars, against 1.6 thousand informally. At the same time, the median entrant starts with about 810 dollars. A credit supply reduces this cost wall, allowing marginal firms to be formal (enter and transition). Counterfactuals indicate that granting the formal borrowing terms to informal firms formalizes 3.7 points of the firm distribution while raising output by 5.1 percent and credit by 31 percent. Repressing informal credit also formalizes, by 2.7 points, but destroys entry and shrinks output. Equalizing taxes moves less than one point. Financial inclusion is the only lever in the menu that formalizes and grows at the same time, and its formalization effect is about four times the tax effect.

The paper contributes to three literatures, developed in Section 2. To the credit supply literature it adds the margin that registry based designs cannot see, showing that bank shocks recompose activity across the formality border. To the informality literature it adds a financial lever that moves the margin fiscal levers did not, and a model in which formality is a dynamic status with two-way switching rather than an absorbing state. To the literature on firm dynamics with financial frictions it adds quasi experimental discipline on the financial block and evidence that the binding friction is the fixed cost of financial participation rather than the marginal price of credit.

The paper proceeds as follows. Section 2 reviews the literature. Section 3 describes the data and Section 4 the descriptive facts. Section 5 develops the identification strategy, Section 6 presents the results, and Section 6.3 the robustness battery. Section 7 develops the two period model, Section 8 the quantitative model, Section 9 the estimation, and Section 10 the counterfactuals. Section 11 concludes.

2 Literature review

This paper speaks to three literatures. The first traces credit supply shocks to real outcomes. A within firm identification strategy introduced by Khwaja and Mian (2008) separates bank supply from firm demand using multibank borrowers, Amiti and Weinstein (2018) extend it to granular bank shocks, and Degryse et al. (2019) recover shocks from market and sector cells so that single bank firms contribute as well. On the real side, Chodorow-Reich (2014) shows that firms attached to distressed banks cut employment in the United States, Greenstone et al. (2020) find small aggregate effects of local lending shocks in normal times, and Gutierrez et al. (2023) show that in Mexico a one standard deviation credit supply shock raises formal employment growth by about 1.4 percentage points.⁵ These designs share a constraint. Credit registries and social security records

⁵The response is not confined to marginal borrowers. Melcangi (2024) shows that even unconstrained firms hoard cash and shed employment in a credit crisis.

observe the formal sector by construction, so the outcome margin stops at the boundary of formality. By matching the universe of business loans to an economic census that counts informal establishments, we move the outcomes past that boundary and find that credit supply operates on the boundary itself. Credit expansions leave the total number of firms unchanged and recompose it instead, with more formal firms, faster formalization, and slower movement back to informality.⁶

The second literature studies informality and the policies that move it. The modern reading treats informal firms as a margin of development rather than a tax dodge alone, with [La Porta and Shleifer \(2014\)](#) and [Busso et al. \(2012\)](#) documenting their prevalence and productivity costs, [Ulyssea \(2018\)](#) modeling the choice between registering the firm and hiding workers in Brazil, and [Meghir et al. \(2015\)](#) embedding the sector choice in labor market search. The policy levers this literature evaluates are fiscal and administrative. Tax simplification in Brazil raised formality modestly ([Rocha et al., 2018](#)), enforcement raises it at a cost ([Almeida and Carneiro, 2012](#)), and the Mexican quasi experimental record on tax side reforms finds formality responses near zero ([Campos-Vazquez et al., 2020](#); [Bobba et al., 2021](#)).⁷ Relative to this work, the lever here is financial rather than fiscal, and the evidence is causal at the market level, so the paper documents that credit terms move the formalization margin that enacted tax reforms in the same country did not. The model matches this evidence in kind. Formality is a firm level status with entry, exit, and switching in both directions, disciplined by census transition rates, rather than an absorbing state.⁸

The third literature puts financial frictions into firm dynamics. The tradition runs from [Hopenhayn \(1992\)](#) through [Cooley and Quadrini \(2001\)](#) and [Gomes \(2001\)](#), with borrowing limits founded on limited enforcement in [Albuquerque and Hopenhayn \(2004\)](#) and [Clementi and Hopenhayn \(2006\)](#), development applications in [Arellano et al. \(2012\)](#), [Midrigan and Xu \(2014\)](#), and [Moll \(2014\)](#), and wealth constrained entry going back to [Evans and Jovanovic \(1989\)](#). Closest to this paper, [D’Erasmus and Boedo \(2012\)](#) and [Erosa et al. \(2023\)](#) study informality in economies with financial frictions, where the sector choice responds to financing conditions.⁹ we contribute on two fronts. The estimation disciplines the financial block with quasi experimental moments, stacking the Shift-Share

⁶The instrument follows the quasi experimental Shift-Share framework of [Borusyak et al. \(2021\)](#), with exposure robust inference from [Adão et al. \(2019\)](#) and diagnostics in the spirit of [Goldsmith-Pinkham et al. \(2020\)](#).

⁷Field experiments that subsidize registration find similarly small and often transitory effects, as in the Nicaraguan intervention studied by [Sutter et al. \(2017\)](#).

⁸[Adom \(2025\)](#) estimates a related transition model for Nigeria. The financial side here is richer, with participation costs, spreads, and collateral capacities by sector, and it is identified from quasi experimental variation rather than from the cross section alone.

⁹[Erosa et al. \(2023\)](#) model both the intensive and the extensive margins of evasion. Here the intensive margin is deliberately shut down so that the model’s sector indicator matches the census classification behind every empirical moment.

coefficients with the census cross section, so the incidence of an actual credit supply shock is an estimated object rather than a calibrated one. And the estimates locate the binding friction in a place this literature rarely prices, the fixed cost of participating in formal finance, which at twenty six wage units dwarfs the spread and collateral gaps. The three strands meet in one sentence. Formality is also a financing decision.

3 Data

The analysis combines four microdata sources and a public local labor market definition. A credit registry prices and counts every bank loan to firms, whether formal or informal. Social security records measure formal employment and wages. The economic census counts formal and informal establishments and follows their status over time. A financial survey measures leverage and borrowing terms for the structural estimation. All sources are merged at the level of local labor markets. The combination is the point. Credit, formality, and firm dynamics are observed in the same markets over the same twenty years.

The credit registry comes from the regulatory report R04C that every bank files with the banking commission (CNBV).¹⁰ It covers the universe of commercial bank loans to businesses at monthly frequency from 2005 to 2025, with the amount, the interest rate, the collateral attached, and the borrowing firm behind each credit. The registry spans 119 banks and 1.3 million borrowing firms. Matched to social security records, the panel covers 3.3 million formal and informal firms. Its central feature for this paper is that the borrower need not be formal. Banks lend against many things, and a social security registration is not a requirement, so the registry prices credit to informal firms directly. Loans are aggregated to bank and market cells by year to construct the exposure shares and the credit outcomes of Section 5.

The social security records (IMSS) cover the universe of registered employers and their workers. Wages are daily contribution base salaries, which include cash payments, bonuses, commissions, and payments in kind.¹¹ These records deliver the formal employment, wage bill, and mean wage outcomes.

The economic census (INEGI) enumerates establishments in fixed premises every five years, and the analysis uses the 2009, 2014, 2019, and 2024 waves.¹² An establishment is formal when it reports positive social security contributions and informal when it reports

¹⁰A public aggregate version is available on the commission’s website. The microdata are accessed through the EconLab environment of the Bank of Mexico, where all confidential sources reside on separated secure servers.

¹¹Wages are bottom coded at the minimum wage and top coded at twenty five times the Mexico City reference minimum. The unit of observation is a job rather than a person, so a worker holding two registered jobs contributes twice to employment counts.

¹²The census covers all localities with more than 2,500 inhabitants and all nonagricultural sectors. Activity without a fixed premise, such as street vending, is outside its frame. For employment context, the

workers and revenue but no contributions, the classification behind every formality outcome in the paper. Because censuses five years apart identify the same establishments, they deliver entry, exit, and the four formality transition cells, as well as value added by sector. The census is what closes the coverage gap of the matched registry data. Formal firms are observed whether or not they borrow, informal borrowers are observed in the registry, and informal firms without bank credit, invisible to any financial dataset, are counted here.

Markets are the 777 local labor markets that INEGI constructs from commuting flows, published in the Bank of Mexico’s market analysis dataset. Credit is observed in 705 of them, and the estimation sample of Section 5 keeps the 556 markets in which the instrument covers at least half of baseline credit, for 1,390 market and window observations across the three census windows. Finally, the national financing survey (ENAFIN 2024) provides borrowing incidence, approved credit lines, and loan pricing for the presets and targets of Section 9, with the measurement conventions collected in Appendix V.

4 Descriptive statistics

Table 1 crosses formality with credit holding in the 2024 census. Three facts organize the paper. First, informality is the sea the economy swims in. Of 5.5 million establishments, 88 percent are informal. Second, bank credit is scarce and skewed toward formality. Only 6.2 percent of firms hold a bank loan, and the incidence is 13.3 percent among formal firms against 5.2 percent among informal ones. Third, and least appreciated, informal firms are not outside the banking system. Because the informal sector is so large, informal establishments make up 74 percent of all firms with bank credit, 249,588 borrowers in 2024. A change in bank credit conditions therefore reaches informal firms directly, not only through spillovers from formal clients.

Table 1: Firms by formality and bank credit, 2024

	Formal	Informal	Total
Credit	87,788 (1.6%)	249,588 (4.6%)	337,376 (6.2%)
No Credit	570,277 (10.4%)	4,560,527 (83.4%)	5,130,804 (93.8%)
Total	658,065 (12.0%)	4,810,115 (88.0%)	5,468,180 (100%)

Source: Economic Census 2024.

Table 2 follows establishments across consecutive censuses. Formality is persistent but national labor force survey put informal employment at 54.5 percent of the workforce at the end of 2024.

far from absorbing. Reading shares off the counts, among formal establishments 60 percent remained formal between 2014 and 2019 and 51 percent between 2019 and 2024, while 12 and 22 percent respectively had become informal and about 28 percent had exited. Among informal establishments, between 55 and 61 percent remained informal, only 3.5 to 6.5 percent had formalized, and 35 to 38 percent had exited. Two readings of these numbers matter for what follows. In conditional terms, the formality border leaks mainly downward, since a formal establishment is two to six times more likely to turn informal than an informal one is to formalize. In absolute terms, however, the two flows are of comparable size, with 146,071 formalizations against 138,041 movements into informality in the latest window, because the informal base is so much larger. Both readings argue against treating formality as a one way ladder. The literature has concentrated on the climb into formality, yet the slide out of it is just as active a margin, and the model of Section 8 gives both directions a price.¹³

Table 2: Establishment transitions across censuses

		Destination		
		Formal	Informal	Exit
2014 Origin	Formal (%)	122,784 60.4	23,526 11.6	56,994 28.0
	Informal (%)	263,565 6.5	2,232,216 55.4	1,531,660 38.0
2019 Origin	Formal (%)	324,052 51.2	138,041 21.8	170,946 27.0
	Informal (%)	146,071 3.5	2,547,652 61.1	1,473,395 35.4

Note: Formal if Nb workers with social security is greater than zero; informal if zero.

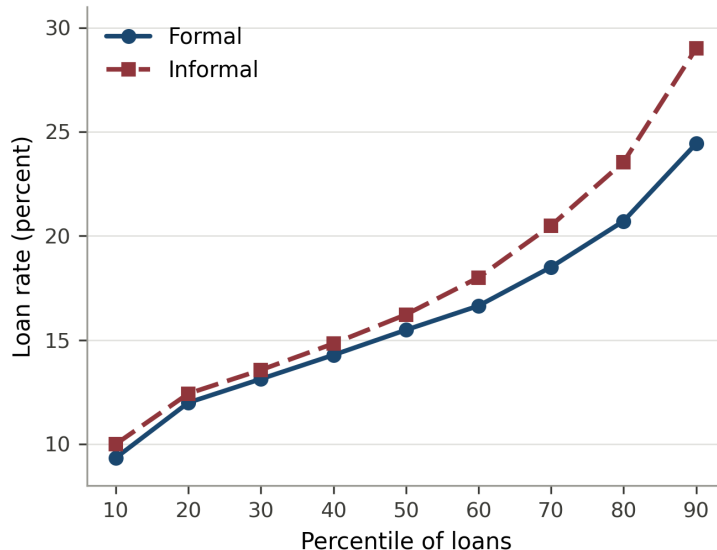
The financial terms behind these stocks differ sharply by sector. In the matched registry, where a borrower counts as informal if it has no IMSS employer registration in the loan year,¹⁴ the median rate on loans to informal firms is 16.2 percent against 15.5 percent for formal firms, a gap that stays between one and three percentage points across years, weighting schemes, and moments (Figure 2), and the survey data show informal firms operating with less capital per worker, investing less, and carrying far less bank debt per worker at every size and age (Appendix C). Collateral is thin, with the median market pledging about 14 centavos per peso lent. These gaps are the empirical faces of the model's

¹³Exit is measured as absence from the following census, so it bundles closure with death, and the informal exit rate exceeds the formal one in both windows, a ranking the estimation uses to preset the death rates. The transition counts come from the census panel described in Section 3.

¹⁴Figure 5 in Appendix A compares this classification with the census among firms holding bank credit: it matches the aggregate level of borrower informality, 76 percent against a census state mean of 81, but compresses the variation across states.

financial primitives, the sector spreads, the collateral capacities, and the participation costs that Section 9 reads off these same moments.

Figure 2: Loan pricing by formality in the credit registry



Percentiles of registry loan rates for formal and informal borrowers, unweighted. A borrower is informal if it has no IMSS employer registration in the loan year. Figure 5 in Appendix A compares this classification with the census.

Finally, the structure of banking relationships shapes the empirical design. Most borrowing firms use a single bank, 95.7 percent in 2009 and 85.2 percent in 2024, and no bank dominates the market for credit, with the largest banks’ lagged shares stable and the concentration index low and declining (Appendix C). The first fact motivates estimating the bank shocks within multibank firms in the baseline and recovering them from market and sector cells, which lets single bank firms contribute, in robustness. The second fact previews the shock level diagnostics of Section 6, where the design is shown to rest on many bank window cells rather than on any single institution.

5 Empirical strategy

5.1 Shift-Share identification with exogenous shocks

We are interested in estimating the impact of credit supply shocks on firm-level transitions (formality status, entry, and exit) and other outcomes. The problem is that OLS estimates are contaminated by reverse causality and omitted-variable bias: markets with more informal-to-formal transitions may demand more credit, and an unobserved technology shock may simultaneously raise firm entry and credit volumes.

A Shift-Share design with an exogenous shifter overcomes this endogeneity. Consider the ideal experiment where additional funding is randomly allocated across banks, raising the credit supply of the selected banks. Because the selection is random, it is uncorrelated with firms' demand for credit and, hence, with firm transitions and other outcomes. In this setting, identification requires, as an exclusion restriction, that the lottery affect outcomes only through credit in equilibrium.

For these reasons, we implement a Leave-One-Out Shift-Share Instrumental Variable (LOO-SSIV) approach, following [Amiti and Weinstein \(2018\)](#) and [Greenstone et al. \(2020\)](#). First, we decompose yearly bank-firm credit growth into demand and supply components. We exclude from the sample the market m that we are interested in (the Leave-One-Out step) to eliminate the mechanical bias that would arise from estimating $\hat{\gamma}_{bt}^{-m}$ on data that include market m 's own observations. Specifically, we estimate

$$\Delta \ln L_{bft} = \alpha_{ft}^{-m} + \gamma_{bt}^{-m} + \epsilon_{bft}^{-m}, \quad (1)$$

where Δ denotes the one-year log change, α_{ft}^{-m} is a firm-year fixed effect that absorbs credit demand, and γ_{bt}^{-m} is a bank-year fixed effect that recovers the bank-specific supply shock. The regression is weighted by the bank-firm pair's credit summed over the five preceding years, and the estimated bank-year effects are normalized to have zero credit-weighted mean within each year. Afterwards, we accumulate the yearly shocks over a five-year window, $\hat{\gamma}_{bt,5}^{-m} = \sum_{k=0}^4 \hat{\gamma}_{b,t-k}^{-m}$, matching the five-year changes of the outcome design below.¹⁵

Notice that the estimation relies on the yearly credit variation across banks within a firm, so that only firms borrowing from two or more banks in a given year contribute to the estimation¹⁶. In what follows, we maintain the standard assumption that the supply shock identified on multi-bank firms also applies to single-bank firms. Relaxing this assumption yields similar results, as shown in the robustness exercise in Section 6.3, where we replace firm-year fixed effects with market-sector-year fixed effects as in [Degryse et al. \(2019\)](#). This change allows single-bank firms to also contribute to identification at the cost of potentially absorbing demand shifters into the estimated supply shocks.

Second, we construct bank shares at baseline in each market,

$$s_{bm(t-5)} = \frac{L_{bm(t-5)}}{\sum_{b'} L_{b'm(t-5)}}, \quad (2)$$

¹⁵For banks without an estimated shock in some years of the window, the accumulation sums the available years. Because the yearly shocks have zero credit-weighted mean, this is equivalent to imputing the average shock in the missing years.

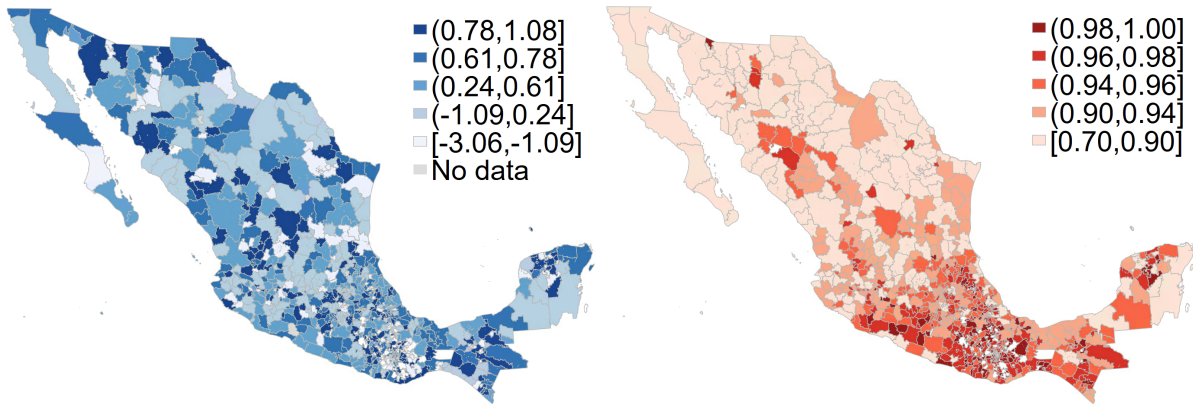
¹⁶From the pool of borrower firm-years, 13% hold loans from more than one bank and 87% borrow from a single bank. In terms of loan volume, 66% comes from multi-bank firms, while 34% comes from single-bank firms.

and define the Leave-One-Out Shift-Share instrument as

$$\hat{Z}_{mt}^{LOO} = \sum_{b \in \mathcal{B}_{mt}} s_{bm(t-5)} \hat{\gamma}_{bt,5}^{-m}, \quad (3)$$

where $\mathcal{B}_{mt}$ denotes the set of banks with at least one estimated shock in the window. Following the exogenous-shifter framework of [Borusyak et al. \(2021\)](#), identification requires i) many uncorrelated shocks with dispersed exposure and ii) that the shifters $\hat{\gamma}_{bt,5}^{-m}$ be as-good-as-randomly assigned across banks conditional on the included controls. The as-good-as-random requirement would fail if banks receiving larger supply shocks systematically lend in markets with atypical transition dynamics. For example, if an oil boom simultaneously raises firm entry in oil-intensive markets and lending by oil-specialized banks in those same markets. In this case, the estimated coefficient would partly capture sectoral demand rather than credit supply. We address these concerns by including controls that may correlate with both the shifter and the outcome (most directly, the market’s sectoral composition of employment and of the lending banks’ portfolios) and by conducting a battery of robustness checks (Section 6.3). The many-uncorrelated-shocks condition requires exposure to be dispersed across many effectively independent bank shocks, which is likely to hold given that the registry covers 119 banks and the effective number of bank shocks, i.e. the inverse Herfindahl index of exposure weights, is 16.5 pooled across windows (Table 3); additionally, Section 6.3 shows the estimates are robust to dropping the largest bank. Figure 3 displays the geography of the identifying variation. The instrument varies widely across markets, and its spatial distribution bears no visible relation to baseline informality, the raw form of the balance evidence in Section 6.3.

Figure 3: The identifying variation across local markets



The left panel maps the Shift-Share instrument across local labor markets. The right panel maps the baseline informality share. Gray areas have no credit data.

With the instrument in hand, the main estimating equation relates local credit growth to outcome variables at the market level:

$$\Delta_5 y_{mt} = \beta \Delta_5 \ln(C_{mt}) + X'_{m(t-5)} \Gamma + \lambda_t + \varepsilon_{mt}, \quad (4)$$

where $\Delta_5 y_{mt}$ is the five-year change in the outcome (such as the log number of firms by formality status, entry, wages, or value added); $\Delta_5 \ln(C_{mt})$ is the five-year log-change in credit in market m , instrumented by $\hat{Z}_{mt}^{LOO}$ standardized to unit variance; and λ_t captures the year fixed effects. The vector $X_{m(t-5)}$ contains the controls (used in every market level regression), which are: the sectoral composition of employment and of the lending banks' portfolios; the credit weighted average interest rate; the log numbers of banks and of loans; the credit share of the market's largest bank; and the sum of exposure shares, which is the share of credit granted by banks with a valid estimated supply shock, interacted with the year fixed effects. This last control accounts for the fact that exposure shares sum to less than one when some banks have no valid estimated supply shock, as prescribed by [Borusyak et al. \(2021\)](#) in the presence of incomplete shares. All controls vary at the market level and are fixed at their values at the start of each five year window. Regressions are weighted by the market's baseline credit $C_{m(t-5)}$.

Notice that the specification is estimated in five-year differences with updating exposure shares fixed at the start of each window, rather than in levels with market fixed effects. We choose this specification because, when shares update across windows, unit fixed effects do not isolate shock variation over time, whereas the first difference specification does, and the updating shares keep the first stage strong ([Borusyak et al., 2021](#)).

The corresponding first stage is

$$\Delta_5 \ln(C_{mt}) = \pi \hat{Z}_{mt}^{LOO} + X'_{m(t-5)} \Pi + \lambda_t + u_{mt}. \quad (5)$$

The corresponding map of realized credit growth is Figure 10 in Appendix D. The coefficient of interest is β , which is the causal elasticity of the outcome with respect to credit at the market level, identified by variation in credit induced by exogenous bank level supply shocks. Then, a one percent increase in credit implies an outcome change of $\hat{\beta}$ -percent if the outcome is in logs, or of $\hat{\beta}/100$ percentage point if the outcome is in shares. As [Adão et al. \(2019\)](#) point out, conventional standard errors over-reject in this setting because market residuals are correlated: markets with similar bank shares are exposed to the same shocks regardless of geographic distance. We follow their suggestion and report instead their exposure-robust standard errors, alongside standard errors clustered by market.

6 Results

6.1 Descriptive statistics of the shocks

Table 3 reports the [Borusyak et al. \(2021\)](#) diagnostics of our Shift-Share design. Although the regressions are estimated on 1,390 market-year observations, the identifying variation comes from the bank level supply shocks. So the table describes these shocks directly at the bank-window level. All weighted moments are computed using each bank’s share of baseline credit in the sample markets. The estimation draws on three five-year windows (2009–2014, 2014–2019, and 2019–2024), and the number of banks with an estimated shock rises from 26 in the first window to 61 in the last, tracking the deepening of the Mexican banking system over the period. Exposure is nonetheless concentrated: the effective sample size, $ESS = 1 / \sum_{b,t} w_{bt}^2$, is 16.5 pooled across windows, out of 129 bank-window cells from 79 distinct banks, and within the 2014–2019 window a single institution accounts for half of the exposure weight. Identification therefore rests on variation pooled across many bank-window cells rather than on any single institution or window: no single bank-window cell carries more than 17% of the pooled exposure weight, the Leave-One-Out construction removes each market’s own influence on its shocks, and Section 6.3 shows the estimates are robust to dropping the largest bank altogether, while the unweighted specification preserves every sign of the composition pattern, with precision concentrated where the credit is.

Table 3: Descriptive statistics of the standardized bank shocks

Window	Exposure weighted			Unweighted			Nb. banks	ESS	W
	Mean	Std. dev.	IQR	Mean	Std. dev.	IQR			
2009-2014	0.21	0.59	1.13	-0.21	0.61	0.70	26	8.38	0.18
2014-2019	0.14	1.53	3.05	0.79	1.05	1.83	42	3.39	0.51
2019-2024	-0.35	0.39	0.53	-0.62	0.54	0.42	61	7.93	0.29
Pooled	0.00	1.00	1.17	0.00	1.00	1.04	79	16.53	0.17

Notes: The table describes the accumulated five-year bank supply shocks $\hat{\gamma}_{bt,5}$, estimated on the full sample rather than the leave-one-out counterparts used by the instrument, whose only difference is the exclusion of one market at a time. All statistics are computed on the estimation sample of the main specification (1,390 market-years; three five-year windows ending in 2014, 2019, and 2024).

Within each block (exposure-weighted and unweighted), shocks are standardized to zero mean and unit variance over the estimation panel using the corresponding weights, so the pooled means and standard deviations equal 0 and 1 by construction. IQR is the difference between the 75th and 25th percentiles.

The weight w_{bt} is bank b 's share of baseline credit in each markets in window t , normalized to sum to one within each window for the window rows and across all bank-window cells for the pooled row. $ESS = 1/\sum w_{bt}^2$ is the effective sample size of the shocks (Borusyak et al., 2021). W denotes the largest normalized weight.

Nb. banks counts banks with an estimated shock in each window; the pooled row reports distinct banks (79, across 129 bank-window cells).

6.2 Main estimates

This section reports our main IV estimates. We follow the recent Shift-Share literature in reporting standard errors following Adão et al. (2019), which account for the cross-market correlation induced by common shocks to bank credit supply; significance stars are based on the corresponding p -values. We additionally report wild cluster bootstrap p -values in brackets (Cameron et al., 2008) as a nonparametric robustness check.

Table 4 reports the estimates of the main specification. The first-stage rows show that the Shift-Share shock, built to be orthogonal to local credit demand, strongly predicts credit in equilibrium: the coefficient on the standardized instrument is 1.10 (s.e. 0.06), and the Kleibergen–Paap F exceeds 310 in every column, far above conventional weak-instrument thresholds. Measured against the credit-weighted dispersion of the instrument in the estimation sample (a standard deviation of 0.62), a one-standard-deviation supply shock raises five-year credit growth by 0.68 log points, roughly a doubling of market credit over the window.

The second-stage estimates show that credit supply reshapes the composition of local firms, while leaving the total number of firms essentially unchanged. A one percent increase

in instrumented five-year credit growth raises the stock of formal firms by 0.05 percent, while the informal stock (-0.002) and the total count (0.004) do not move. At the mean market, where informal firms are nearly nine in ten establishments, a ten percent credit expansion adds about six formal firms and leaves the informal count unchanged. Aggregated over the 556 estimation markets, this amounts to roughly 3,300 additional formal firms. The mechanism behind this change is that formal entry responds strongly (0.067), informal entry does not (0.012 , n.s.), and exit rises across the board (0.049 total, 0.039 formal, 0.055 informal).

To put the formal-entry response in perspective, a supply-driven ten percent credit expansion adds about 2.9 formal entrants to the average market’s 451 per window, raising the five-year formal startup rate ($451/1,237 = 36.4\%$) by roughly 0.23 percentage points, holding the denominator fixed at the mean stock of 1,237 formal firms. This is about a third of the U.S. startup-rate decline documented by Pugsley and Şahin (2019), roughly 5 percentage points over 32 years, or 0.78 points per five-year window. A one-standard-deviation credit supply shock, which roughly doubles market credit, moves the formal startup rate by 1.7 percentage points, more than two U.S. quinquennia of decline reversed.

Exit rises for formal and informal firms alike, while entry responds only in the formal segment. Credit expansions thus raise market turnover with a formal tilt, replacing exiting incumbents, most of them informal, with formal entrants, a pattern in the spirit of the creative destruction of Schumpeter (1942). At the average market, a ten percent credit expansion is associated with about 15 (roughly 8,000 across the 546 estimation markets) additional exits (on a base of 3,144).

Wild cluster bootstrap p -values, reported in brackets, confirm the composition result under fully nonparametric market-level inference: the formal stock ($p = 0.021$), formal entry ($p = 0.049$), total exit ($p = 0.005$), and informal exit ($p = 0.002$) responses all survive, while the coefficients that are null under AKM inference remain null. The one divergence is formal exit, significant under AKM inference but not under the bootstrap ($p = 0.226$), we therefore read the exit response as driven by the informal segment.

Table 4: Shift-Share IV estimates of the effect of bank shocks on the number of firms

	Dependent variable: Five-year log change in the number of firms								
	Existing			Entering			Exiting		
	Total	Formal	Informal	Total	Formal	Informal	Total	Formal	Informal
$\widehat{\Delta_5 \ln(C)}$	0.004 (0.005) [0.797]	0.051*** (0.013) [0.021]	−0.002 (0.006) [0.910]	0.015 (0.009) [0.597]	0.067*** (0.018) [0.049]	0.012 (0.010) [0.652]	0.049*** (0.012) [0.005]	0.039*** (0.015) [0.226]	0.055*** (0.012) [0.002]
FS Coeff.	1.104 (0.063)	1.104 (0.063)	1.104 (0.063)	1.120 (0.060)	1.120 (0.060)	1.120 (0.060)	1.120 (0.060)	1.120 (0.060)	1.120 (0.060)
FS F	310.25	310.19	310.25	346.32	346.02	346.32	346.32	346.13	346.32
Mean	10,010	1,237	8,777	4,328	451	3,885	3,144	344	2,811
Obs.	1,390	1,385	1,390	992	975	992	992	961	992
Markets	556	552	556	546	536	546	546	529	546

Notes: Two-stage least squares estimates at the market (LLM) $\times$ five-year window level; the three windows end in 2014, 2019, and 2024, each relative to the previous Economic Census. Dependent variables are five-year log changes; markets with a zero count in a window drop from the corresponding column. The endogenous regressor is the five-year log change in market credit, instrumented by the leave-one-market-out shift-share credit-supply shock, standardized to unit variance. The Mean row reports the dependent variable’s level in the estimation sample. All columns include year fixed effects and the baseline (start-of-window) controls: market sector shares and credit-weighted sector shares of the lending banks’ portfolios; the credit-weighted average interest rate; log numbers of banks and of loans; the credit share of the market’s largest bank; and the coverage share (baseline credit granted by banks with a valid estimated supply shock), entered as a main effect and interacted with the year fixed effects. The sample is restricted to market-years with coverage of at least 0.5 (Section 6.3 varies this threshold). AKM standard errors following [Adão et al. \(2019\)](#) in parentheses; stars use the corresponding p -values. Wild cluster bootstrap p -values ([Cameron et al., 2008](#)), clustered at the market level, in brackets. The first-stage coefficient is reported with its market-clustered standard error; FS F is the Kleibergen–Paap rk Wald statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The change in the stock of formal firms need not come only from entry and exit: surviving firms can also switch formality status, as shown in the descriptive statistics section, Section 4. Table 5 closes this decomposition by estimating the response of the four origin→destination cells of surviving firms, and the pattern is uniform: more firms formalize, fewer abandon formality, more remain formal, and the count of informal stayers is flat. The informal-to-formal flow rises by 0.052 percent for a one percent increase in instrumented credit, the formal-to-informal flow falls by 0.039 percent, the number of firms remaining formal rises by 0.040 percent, and the informal-to-informal cell does not move (−0.007, n.s.). At the average market, a ten percent credit expansion adds about 1.5 formalizations per window (on a base of 310), prevents about one switch into informality (on 286), and adds about 1.8 formal stayers (on 481). Inference is strongest on the retention margin: the formal-to-formal response survives the wild bootstrap ($p = 0.032$), while the

two switching flows carry AKM significance but weaker bootstrap support ($p = 0.083$ and $p = 0.323$ respectively), the same pattern as formal exit above. Taken together with the entry and exit margins, the transition matrix points to a consistent reallocation toward formality among surviving firms, with the most conservative inference anchored by formal retention and, at the ten percent level, formalization.

Table 5: Shift-Share IV estimates of the effect of bank shocks on firm transitions

Dependent variable: Five-year log change in the number of firm transitions				
	Informal→Formal	Formal→Informal	Formal→Formal	Informal→Informal
$\widehat{\Delta_5 \ln(C)}$	0.052*** (0.018) [0.083]	-0.039*** (0.015) [0.323]	0.040*** (0.009) [0.032]	-0.007 (0.004) [0.699]
FS Coeff.	1.121 (0.060)	1.120 (0.060)	1.120 (0.060)	1.120 (0.060)
FS F	344.67	346.02	346.03	346.32
Mean	310	286	481	4,571
Obs.	953	954	966	992
Markets	520	525	533	546

Notes: Columns count surviving firms by formality status at the beginning and end of each five-year window (origin→destination), in logs. See Table 4 for specification and inference details; wild cluster bootstrap p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To trace where the additional credit goes, Table 7 decomposes the credit response into the number of borrowing firms, loans per borrower, and loan size. New borrowing firms account for 0.16 of the total elasticity; more loans per borrower, for 0.20; and larger loans, for 0.64 (the three coefficients sum to one by construction). So credit expansions mostly deepen existing lending relationships while also pulling new firms into the credit market. Table 6 then splits the borrowing margin by formality status: the number of firms with credit rises for formal and informal firms alike (0.041 and 0.035 per one percent of credit), with statistically indistinguishable responses, so the marginal borrower is drawn from both sectors. Notice that the borrowing-firm responses, in both tables, are significant under AKM inference but not under the wild bootstrap, so we read this margin as supportive of the access channel alongside the firm-margin results rather than as stand-alone evidence.

Table 6: Shift-Share IV estimates of the effect of bank shocks on the number of firms with credit

Dep. var.: Five-year log change, No. firms w/credit			
	Total	Formal	Informal
$\widehat{\Delta_5 \ln(C)}$	0.036*** (0.012) [0.530]	0.041** (0.018) [0.279]	0.035** (0.014) [0.727]
FS Coefficient	1.104 (0.063)	1.104 (0.063)	1.104 (0.063)
FS F	310.25	310.25	310.25
Mean	772	184	589
Obs.	1,390	1,390	1,390
Markets	556	556	556

Notes: See Table 4 for specification and inference details.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Decomposition of the credit response: extensive and intensive margins

	Dep. var.: Five-year log change in		
	Borrowing firms	Loans per borrower	Credit per loan
$\widehat{\Delta_5 \ln(C)}$	0.161*** (0.035) [0.197]	0.202*** (0.034) [<0.001]	0.637*** (0.044) [<0.001]
FS Coefficient	1.104 (0.063)	1.104 (0.063)	1.104 (0.063)
FS F	310.25	310.25	310.25
Mean	670	1.95	455
Obs.	1,390	1,390	1,390
Markets	556	556	556

Notes: Market credit satisfies $C = \text{borrowing firms} \times \text{loans per borrower} \times \text{credit per loan}$, so the three coefficients sum to the response of credit to itself, one, by construction ($0.161 + 0.202 + 0.637 = 1.00$). Outcomes are from the credit registry. Means in levels: the average market-year in the estimation sample has 670 borrowing firms with 1.95 loans per borrower. To scale the credit response itself we use the median market, whose baseline credit stock is 2.1 million USD: a ten percent credit supply shock corresponds to roughly 215 thousand USD of additional credit in the median market. See Table 4 for specification and inference details; wild cluster bootstrap p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Quantities expand, collateral terms ease, and the price of credit is essentially flat. Had the credit expansion been demand-driven, lending rates would have risen substantially and collateral terms would have tightened rather than eased; the rate response we find, while statistically robust, amounts to four basis points on a mean rate of 15 percent, too small to signal demand-driven growth. The two coefficients differ in robustness: the interest-rate response survives all three inference procedures (bootstrap $p = 0.030$), while the collateral response is significant under AKM and market-clustered inference but not under the wild bootstrap ($p = 0.196$). We therefore emphasize the qualitative pattern, easier non-price terms, over the collateral point estimate itself. The model section (Section 8) returns to the question of which lending term, interest rate or collateral, transmits bank shocks.

Table 8: Shift-Share IV estimates of the effect of bank shocks on interest rate and collateral

	Collateral ratio (log change)	Interest rate (pp change)
$\widehat{\Delta_5 \ln(C)}$	-1.857^{***} (0.322) [0.196]	0.422^{***} (0.102) [0.030]
FS Coefficient	1.120 (0.060)	1.104 (0.063)
FS F	345.53	310.25
Mean	4.33	15.00
Obs.	816	1,390
Markets	451	556

Notes: The collateral ratio is the credit-weighted collateral-to-loan ratio (in logs; mean 4.33). The interest rate is the credit-weighted average lending rate in percentage points (mean 15.0), so its coefficient is a percentage-point change per log point of credit. The collateral sample covers the market-years with collateral information. See Table 4 for specification and inference details; wild cluster bootstrap p -values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

We now turn to the labor-market and production effects of these credit supply shocks. Table 9 shows that credit expansions raise formal employment: IMSS-registered workers, formal by definition, increase by 0.019 percent per one percent of credit, about 57 workers at the average market (on a mean of 30,695) for a ten percent expansion, and roughly 400 for a one-standard-deviation shock that doubles market credit. The formal-employment response stands out against Mexico’s policy record: evaluations of enacted tax-side reforms, the halving of the border VAT (Campos-Vazquez et al., 2020) and revenue-neutral payroll-tax reforms (Bobbà et al., 2021), find formality responses close to zero, whereas credit supply moves the formal worker margin. The total formal wage bill rises accordingly (0.023), while the mean daily wage falls slightly (-0.012 , about 23 centavos on a mean of 204 MXN per day), consistent with marginal hires earning slightly below-average wages; the wage-bill response is approximately the sum of the employment and wage responses. Value added moves with the composition results: average and total value added rise marginally (0.027 and 0.030, significant at the ten percent level), in line with reallocation toward formal firms rather than dilution of average productivity by marginal entrants. The split by formality sharpens the picture: informal-sector value added falls (-0.052 , significant at five percent under AKM inference) while formal-sector value added does not move (0.025, n.s.), so the

modest aggregate gain masks a reallocation of production toward formal firms.

Three caveats bound this table. First, none of the seven coefficients survives the wild cluster bootstrap (bracketed p -values between 0.14 and 0.69), so we read the labor and production responses as a directionally consistent complement to the firm-margin results, estimated with less power, rather than as stand-alone findings. Second, the value-added responses are sensitive to market-specific trends: when the dependent variable’s own lagged five-year change is added as a control, the aggregate and formal value-added coefficients reverse sign, while the informal value-added decline strengthens (Appendix Table 19); the value-added result we rely on is therefore the reallocation of production away from the informal sector, which is robust. Third, the design cannot speak to informal employment: worker counts covering informal workers exist only in the 2019 and 2024 census waves, leaving a single five-year window on which the instrument is weak (first-stage F of 2.6); Appendix B, Table 17, reports those estimates for completeness.

Table 9: Shift-Share IV estimates of the effect of bank shocks on workers, wagebill, and value added

	$\Delta_5 \log$ workers	$\Delta_5 \log$ total wagebill	$\Delta_5 \log$ mean wage	$\Delta_5 \log$ avg. VA	$\Delta_5 \log$ total VA	$\Delta_5 \log$ VA formal	$\Delta_5 \log$ VA informal
$\widehat{\Delta_5 \ln(C)}$	0.019*** (0.005) [0.177]	0.023** (0.009) [0.324]	−0.012*** (0.004) [0.139]	0.027* (0.014) [0.402]	0.030* (0.016) [0.377]	0.025 (0.031) [0.692]	−0.052** (0.022) [0.619]
FS Coeff.	1.104 (0.063)	1.104 (0.063)	1.104 (0.063)	1.104 (0.063)	1.104 (0.063)	1.104 (0.063)	1.104 (0.063)
FS F	310.21	310.21	310.21	310.25	310.25	310.25	310.25
Mean	30,695	12,567,111	204	1,057	22,276,545	16,637,337	5,639,208
Obs.	1,382	1,382	1,382	1,390	1,390	1,390	1,390
Markets	552	552	552	556	556	556	556

Notes: See Table 4 for specification and inference details; wild cluster bootstrap p -values in brackets. The three labor-market columns are from IMSS records and therefore cover formal (registered) employment: “workers” is the market total of IMSS-registered workers, “total wagebill” the market total of daily contribution-base wages, and “mean wage” the market average daily wage. The four value-added columns are from the Economic Census and cover all firms: “avg. VA” is average per-firm value added, “total VA” aggregate value added, and the last two columns split total value added by the census formality classifier, firms with positive income and workers but zero social security contributions. Means are in the levels of each variable: the mean daily wage is 204 MXN (≈ 10.7 USD); the mean daily wage bill is 12.6 million MXN (≈ 0.66 million USD); value added is recorded in thousands of current pesos, so average per-firm value added is ≈ 1.06 million MXN ($\approx 56,000$ USD) and total market value added ≈ 22.3 billion MXN (≈ 1.2 billion USD). All conversions at 19 MXN/USD. Census worker counts, available only in the 2019 and 2024 waves, are reported in Appendix B, Table 17; the instrument is weak on that single-window subsample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Overall, these results point towards three lessons. First, bank credit supply is a causal driver of firm dynamics in Mexican local economies, but its imprint is on composition rather than scale: credit expansions leave the total number of firms essentially unchanged while reshaping who operates, with more formal entrants, more exits of incumbents, and more survivors moving into and remaining in formality. Second, the transformation is real rather than nominal: formal employment and the formal wage bill expand, and value added reallocates away from the informal sector, so the formality gained carries production and jobs with it, reallocating resources. Third, although banks lend to formal and informal borrowers alike, the effects concentrate on marginal firms at the border of formality. This pattern suggests that credit does not merely relax a uniform financing constraint, and points instead to credit interacting with the formality status itself, the mechanism that Section 8 formalizes as a fixed cost of formal financial participation. The margins carrying these conclusions, the formality composition, entry and exit, and retention responses, are also the ones that survive every inference procedure we use, so the section’s core message rests on its most robust estimates. For credit policy the implication is immediate: bank credit supply moves the formalization margin that enacted tax-side reforms have struggled to move.

6.3 Robustness checks

This section examines the robustness of our results. We proceed in three steps. First, we test the identifying assumption directly. We check that the exposure shares are balanced on predetermined market characteristics, we control for lagged outcome changes, and we run placebo regressions on pre-period outcomes. Second, we vary the construction of the instrument. We drop the Leave-One-Out correction, renormalize by coverage, move the coverage threshold from zero to 0.8, and rebuild the instrument without the largest lender. Third, we vary the population and the estimator. We exclude the three metropolitan areas, drop the credit weights, compare the IV estimates with their OLS and reduced-form counterparts, and re-estimate everything at the bank level with re-identified shocks. All tables are in the Appendix. Two facts survive every exercise: credit supply shifts the composition of firms toward formality without changing their number and credit expansions deepen existing lending relationships.

A central identifying assumption of our design is that the bank credit supply shocks aggregated by the Shift-Share instrument are as-good-as-randomly assigned conditional on the controls in our main specification. Following the balance tests suggested by [Borusyak et al. \(2021\)](#) and [Goldsmith-Pinkham et al. \(2020\)](#), we regress each of the seven predetermined market controls of the main specification on the Leave-One-Out Shift-Share instrument, conditional on year fixed effects and the coverage control and weighted by baseline market credit. Table 18 in Appendix E reports the results with each control measured five and ten

years before the baseline period, that is, ten and fifteen years before the outcome wave. None of the fourteen coefficients is statistically distinguishable from zero at the ten percent level and the point estimates are economically small. As a complementary dynamic check, Table 19 in Appendix F re-estimates the main specifications adding the dependent variable’s own lagged five-year change as a control. The core results are essentially unchanged, with the formal-firm response at 0.048 against 0.051 in the baseline; the exceptions are the aggregate and formal value-added responses, which are not robust to this control and on which we therefore do not lean, while the value-added result we emphasize, the reallocation of production away from the informal sector, strengthens under it.

As a model-free check on our identification, Table 20 in Appendix G reports, for every outcome, the naive OLS regression on credit growth and the reduced-form regression on the standardized Leave-One-Out instrument, with the same controls, weights, and sample restrictions as the main IV specification and standard errors clustered at the market level. The reduced form confirms that the identifying correlation is present in the raw data, with significance concentrated in the same margins the paper emphasizes, and the OLS-IV comparison shows why instrumenting matters: the naive interest-rate correlation is negative and insignificant while the instrumented response is positive and precise, and the naive value-added correlations are strongly positive while the instrumented ones are not, the sign patterns expected when demand shocks dominate observational credit variation.

The credit-weighted regression places disproportionate weight on the three largest metropolitan areas (Mexico City, Monterrey, and Guadalajara). To address the concern that the main estimates reflect dynamics specific to these urban banking centers, Table 21 in Appendix H re-estimates every specification excluding them: only nine of the 1,390 market-year observations, but roughly two-thirds (69 percent) of the regression weight. The composition core is essentially unchanged, with the formal-firm response at 0.049 against 0.051 and formal entry identical at 0.067, and several margins strengthen outside the metros (the employment response rises to 0.027 and the informal value-added decline deepens to -0.116). Two results soften and are disclosed as such: the informal-to-formal switching flow loses significance, with the composition shift operating instead through formal retention, fewer switches into informality, and a declining count of informal stayers, and the mean-wage decline disappears, indicating that the wage-composition effect of marginal hires is a metropolitan phenomenon. Dropping less than one percent of the observations but two-thirds of the regression weight therefore leaves the formalization core intact.

The credit weights in our main specification imply that the elasticities in Tables 4–9 are credit-weighted averages of market-specific responses, in which markets with larger base-period credit, notably the three metropolitan areas, carry proportionally more weight. Table 22 in Appendix I therefore re-estimates the main specification with every market-year contributing equally, which shifts the estimand toward the typical market, a market

with roughly \$2.1 million (USD) of credit outstanding at baseline, on estimation samples identical to the weighted ones. The first stage remains strong and the composition pattern keeps its signs, with the formal-firm stock at 0.064 against the credit-weighted 0.051, the collateral and interest-rate responses growing in magnitude (-2.27 and 0.94), and the informal value-added decline persisting (-0.305); the credit-deepening decomposition again sums exactly to one and tilts further toward credit per loan (0.783), the one margin that remains significant under exposure-robust inference ($p = 0.048$). What changes is precision: exposure-robust standard errors are roughly two to seven times the clustered ones, the employment and wage-bill responses lose significance, and the mean-wage dip flips to marginally positive, so we read the unweighted results as a statement about where the identifying power of the design resides, in the markets carrying the credit, rather than as a contradiction.

As a direct test of the identifying assumption that our Shift-Share instrument is uncorrelated with pre-existing dynamics in local outcomes, Table 23 in Appendix J reports placebo IV estimates in which the dependent variable is the five-year log change of each outcome before the shock window, $\Delta_5 Y_{t-5} = \ln Y_{t-5} - \ln Y_{t-10}$, with the endogenous regressor, instrument, controls, weights, and sample condition unchanged. The answer is margin-specific and disclosed in full. Firm counts display a positive pre-period correlation that is common to the two sectors rather than differential (formal 0.017, informal 0.019), whereas the estimated effects load entirely on formal firms, and the formal-firm effect is essentially unchanged under the lagged-outcome control; the level outcomes, wages, value added, and loan sizes, attract significant negative placebo coefficients (-0.165 for average value added), which we read as mean reversion in bank lending: shocks were directed disproportionately toward markets whose financial and real conditions had been deteriorating, so the contemporaneous wage and value-added estimates partly capture recoveries and we do not lean on them. The entry, exit, formality-transition, and collateral placebos are uninformative, because the required further lag exists in a single census window and the first stage collapses (a Kleibergen–Paap F of about 2.6), and we report them for transparency rather than as evidence in either direction.

A natural question is whether our findings depend on the Leave-One-Out construction of the Shift-Share instrument. Table 24 in Appendix K re-estimates every outcome with the conventional construction, in which the bank shocks $\hat{\gamma}_{bt}$ are estimated on the full sample including each market’s own loans, so that $\hat{Z}_{mt} = \sum_b s_{bm(t-5)} \hat{\gamma}_{bt}$ assigns every market the same shock value for a given bank; the sample, controls, weights, and inference are otherwise identical to the main specification. As [Borusyak et al. \(2021\)](#) emphasize, this construction suffers from own-observation overlap: each market’s own credit dynamics enter its instrument, which pulls the IV estimate toward its OLS counterpart. Qualitatively nothing changes, and the modest deviations move exactly as the overlap predicts, toward

the attenuated OLS counterparts: the formal-firm elasticity falls from 0.051 to 0.036 and the interest-rate response from 0.42 to 0.33, while the conventional first stage is mechanically tighter. We therefore retain the Leave-One-Out instrument as our preferred specification: the comparison shows that the refinement, far from generating our results, if anything strengthens the composition margins.

As a complementary identification check, Table 25 in Appendix L moves the analysis to the shock level. The credit supply shocks are re-identified with the cell-fixed-effects approach of Degryse et al. (2019), an extension of the within-firm framework of Khwaja and Mian (2008) in which credit demand is absorbed at the market-sector ($\text{LLM} \times \text{industry}$) cell rather than within multi-bank firms. Both multi-bank and single-bank firms now contribute to the shock estimates, a consequential expansion in Mexico, where single-bank firms dominate local credit markets; the unit of observation is the bank-year, outcomes and market credit growth are aggregated across the markets each bank serves using baseline exposure shares, and the shocks, units, aggregation, controls, weights, and samples all differ from the market-level design, so nothing mechanical links the two sets of estimates: if our results reflected a particular scheme for purging demand, the own-market overlap, or fragile specification choices, this regression would be free to disagree.

It does not disagree; it reproduces the composition of the response outcome by outcome: the total stock of firms in a bank’s markets does not respond while the formal stock rises (0.036, bootstrap $p < 0.001$, against 0.051 at the market level), turnover again runs through informal exit, the formality transitions match sign by sign, the credit-deepening decomposition again attributes the bulk of the response to credit per loan (0.716 against 0.637), collateral falls, the interest rate rises, and the fragile value-added responses are absent, consistent with our decision not to lean on them. That a design with different shocks, different units, different aggregation, and different inference reproduces the composition of the response, and not merely its direction, is among the strongest pieces of evidence for the credit supply interpretation of our results.

Our main analysis restricts the estimation sample to market-years for which the Leave-One-Out instrument is constructed from at least half of the local base-period banking exposure. Table 26 in Appendix M re-estimates every outcome at the two extremes: a threshold of zero, which keeps every market-year with a constructible instrument (1,520 observations in 568 markets), and a threshold of 0.8, which keeps only market-years where the instrument pools the great majority of local banking exposure (1,215 observations in 541 markets). The coverage requirement behaves like the measurement restriction it is: the Kleibergen-Paap statistic rises steeply with the threshold, from about 120 to between 284 and 321, while the substantive conclusions do not move, with the formal-firm elasticity in a narrow band around the main estimate (0.042 at zero, 0.051 in the main specification, 0.045 at 0.8), formal entry at 0.058 at both extremes, the credit-deepening decomposition

summing exactly to one and dominated by credit per loan at every threshold, and the interest-rate response virtually unchanged; the changes are confined to margins we already treat cautiously. The threshold matters for the precision of the instrument, not for any substantive conclusion of the paper.

A potential concern with our Leave-One-Out instrument is that its magnitude scales mechanically with coverage: in a market where only half of the base-period banks have an identifiable shock, the instrument is a coverage-weighted sum that is mechanically half as large as it would otherwise be. Table 27 in Appendix N re-estimates the main specification with the renormalized instrument $\tilde{z}_{mt} = z_{mt}/\text{coverage}_{mt}$, which converts the coverage-weighted sum into a coverage-weighted average, standardized as in the main specification and estimated on the identical sample. The estimates are nearly indistinguishable from the main ones, with the formal-firm elasticity at 0.052 against 0.051, the credit-deepening decomposition again summing exactly to one, and a very strong first stage throughout (Kleibergen–Paap F above 335 in every family); the coverage-driven scaling of the instrument drives no result in the paper.

A remaining concentration concern is that the identifying variation might come disproportionately from a single dominant lender: the shock-level diagnostics show an effective sample of roughly sixteen independent bank-window shocks, with the largest carrying 17 percent of the weight. Table 28 in Appendix O therefore rebuilds the instrument from scratch excluding the bank with the largest share of aggregate baseline credit, removing both its shocks and its exposure shares, and re-estimates the main specification on the identical sample. The cost appears exactly where it should, in the first stage, which falls to a Kleibergen–Paap statistic of about 32, while no conclusion changes: the formal stock rises by the same magnitude as in the main specification (0.048 against 0.051, with a bootstrap p -value that rises to 0.17 from 0.02), turnover still runs through total and informal exits, the formality transitions keep their signs with formalization and informalization if anything larger (0.093 and -0.078), the credit-deepening decomposition again sums exactly to one, the collateral response strengthens, and the interest-rate response keeps its sign. A harsher variant purges the lender from every input to the design, rebuilding shocks and shares with no information from it at all: the first stage falls to an F of 10.3, while the formal share response, at 0.123, keeps its sign and magnitude with a wide interval that contains the baseline (-0.014 to 0.260 ; AKM $p = 0.015$, bootstrap $p = 0.176$); the same purge applied to the next largest lender leaves the design intact, with a first-stage F of 98 and a formal response of 0.051. What the largest lender contributes is precision, not the result.

The checks in this section converge on a single conclusion: the two facts at the center of the paper are not the product of any single methodological choice. The composition margins survive the balance tests, lagged-outcome controls, and placebo scrutiny, in which the only pre-trends are common to the two sectors or attached to level outcomes we already

treat cautiously; they survive replacing the Leave-One-Out construction with the conventional one, renormalizing by coverage, moving the coverage threshold from zero to 0.8, and rebuilding the instrument without the largest lender; and they are reproduced outcome by outcome by a bank-level design with re-identified shocks. Population and weights matter only for precision: the identifying power of the data resides where the credit is, and the employment, wage, and value-added responses, the margins that depend on specification, are the ones we lean on nowhere. Bank credit supply changes the composition of firm activity in Mexican local economies: it formalizes the surviving margin, deepens existing lending relationships, and does not expand the number of firms.

7 A toy model of formality and finance

This section develops a two-period version of the full model of Section 8, used for estimation. Its purpose is threefold. First, it delivers closed-form decision rules and transition thresholds for every margin observed in the data: entry into the informal or the formal sector, exit from either sector, and switching in both directions. Second, it signs, in closed form, the response of every threshold and of every transition probability to the two financial primitives that the credit supply shocks of Section 5 move: the cost of external finance and the collateral capacity of each sector. Third, it produces the composition results that discipline the interpretation of the Shift-Share estimates. A credit expansion operates through marginal firms. It moves the formality margins of entrants and survivors, raises the turnover of the marginal cohort it admits, and lowers average productivity within that cohort, while the formality margin removes the best firms from the informal pool. This combination is the estimated pattern of Section 6: formal entry and formalization rise, de-formalization falls, exit rises, and informal value added declines. The general model of Section 8 adds sector-specific productivity processes, a size-dependent cost of informality on the labor margin, a fixed cost of borrowing, a formality taste shock, and an infinite horizon; every mechanism there is already visible here.

7.1 Environment

Time lasts two periods, $t \in \{1, 2\}$. A firm is run by a risk-neutral entrepreneur who consumes her dividends and discounts period 2 at factor $\beta \in (0, 1)$. The economy is populated by (i) incumbents, firms that begin period 1 attached to a sector $s \in \{i, f\}$ (informal or formal) with installed capital $k_1 > 0$, and (ii) potential entrants, agents endowed with financial wealth $k_0 \geq 0$ who may create a firm that operates in period 2. The output good is the numéraire and the wage w is exogenous to the firm (in the quantitative model w is the local-labor-market wage, normalized to 1; the partial-equilibrium treatment mirrors the within-market identification of the empirical design).

An active firm in sector s with capital k and labor ℓ produces

$$y = \theta k^\alpha \ell^\nu, \quad \alpha, \nu > 0, \quad \alpha + \nu < 1. \quad (6)$$

Log productivity combines a permanent component v , drawn at birth from $\ln v \sim N(0, \sigma^2)$ and known to the firm, and transitory shocks:

$$\ln \theta_1 = \ln v + \varepsilon_1, \quad \ln \theta_2 = \rho \ln \theta_1 + (1 - \rho) \ln v + \varepsilon_2, \quad \varepsilon_t \stackrel{iid}{\sim} N(0, \sigma_\varepsilon^2), \quad (7)$$

with $\rho \in (0, 1)$. In the general model $(\rho, \sigma_\varepsilon)$ are sector-specific; here they are common so that sector choice is exclusively driven by the financial frictions.

A formal firm pays a proportional tax $\tau_y \in [0, 1)$ on revenue and a payroll tax $\tau_\ell \geq 0$ on its wage bill. An informal firm evades both. To isolate the financial mechanism, the two-period model shuts down the size-dependent audit technology of the general model (formally, $\eta_i \rightarrow \infty$ in the notation of Section 8), so the only advantages of formality are financial and the only advantages of informality are fiscal and its lower fixed costs. Operating in sector s in period 2 also requires a per-period fixed cost $c_s \geq 0$, with $c_f > c_i$ (registration upkeep, accounting, visibility).

All within-period payments (wages, taxes) are settled out of contemporaneous revenue. Investment, in contrast, must be paid one period ahead of the revenue it generates, so a firm whose desired capital exceeds its internal funds must raise external funds $a \geq 0$. External finance is intermediated at a sector-specific cost: raising a costs the firm a fee of the form

$$\Phi_s(a) = \phi_s a, \quad \phi_i > \phi_f \geq 0, \quad (8)$$

paid on top of repayment of principal within the period. The parameter ϕ_s is the spread between the marginal cost of external funds and the risk-free rate in sector s ; informality raises it because informal firms cannot pledge audited accounts, registered assets, or formal payroll as information collateral (Section 4 documents the corresponding gap in loan pricing). In addition, external funds are limited by a collateral constraint on the capital being installed,

$$a \leq b_s k', \quad 0 \leq b_i < b_f < 1, \quad (9)$$

where k' is next-period capital: lenders finance at most a fraction b_s of the installed capital of a sector- s borrower, and formal status raises pledgeability. The pair (ϕ_s, b_s) summarizes the financial conditions faced by sector s ; the empirical counterpart of a bank credit supply shock is a change in this pair, and all comparative statics below are taken with respect to it. The general model adds a fixed cost of borrowing, $\Phi_s(a) = \phi_{0s} \mathbf{1}_{a>0} + \phi_s a$ (Remark P.4); the linear case makes every formula below explicit.

Regarding costs, creating a firm in sector s costs c_s , with $c_f > c_i$ (registration, licenses, documentation). An informal incumbent that formalizes pays $c_{if} \geq 0$; a formal incumbent that de-formalizes pays $c_{fi} \geq 0$. A firm operating in sector s in period 2 is destroyed with exogenous probability $\lambda_s \in [0, 1)$ before producing, in which case the owner recovers only the undepreciated capital; the ranking $\lambda_i \geq \lambda_f$ (formal status insures against expropriation, eviction, and enforcement shocks) is allowed but not imposed. Capital depreciates at rate $\delta \in (0, 1)$ and is resold at its undepreciated value.

Timing mirrors the recursive model and is summarized in Appendix Figure 11: sector choices for period $t + 1$ are made at the end of period t , before the period- $(t + 1)$ innovation is observed. After period-1 production, taxes, and fixed costs, incumbents observe (θ_1, v) and choose $d \in \{\text{exit}, i, f\}$, while potential entrants, knowing (v, k_0) but not ε_2 , choose $d \in \{\text{stay out}, i, f\}$; a continuing or entering firm pays the relevant entry or switching fee, chooses next-period capital k_2 , and borrows $a = \max\{0, k_2 - n\}$ at fee $\Phi_{s'}(a)$ subject to $a \leq b_{s'}k_2$, where n denotes internal funds at that stage. In period 2 death shocks realize, survivors produce, pay taxes and the fixed cost, and sell undepreciated capital.

At the transition stage the resources available inside the firm are the period-1 operating profit plus the resale value of undepreciated capital, net of the fixed cost and of any switching or entry fee:

$$n = \begin{cases} \pi s(\theta_1, k_1) - c_s + (1 - \delta)k_1 - c_{ss'}\mathbf{1}_{\{s' \neq s\}}, & \text{incumbent of sector } s, \\ k_0 - c^e s', & \text{entrant into } s', \end{cases} \quad (10)$$

with $c_{ss} \equiv 0$. An exiting incumbent simply consumes $n^{\text{exit}} = \pi s(\theta_1, k_1) - c_s + (1 - \delta)k_1$; a non-entrant consumes k_0 . Throughout “financial conditions” means the four-vector $(\phi_i, \phi_f, b_i, b_f)$.

7.2 Profits, investment, and financing

Conditional on operating with state (θ, k) , optimal labor choice yields the separable operating -profit function $\pi_s(\theta, k) = \Omega_s x k^\gamma$, where $x \equiv \theta^{1/(1-\nu)}$ is the profitability index, $\gamma \equiv \alpha/(1 - \nu) \in (0, 1)$, and Ω_s collects technology, the wage, and the taxes effectively paid in sector s ; the tax wedge $T \equiv \Omega_f/\Omega_i = (1 - \tau_y)^{1/(1-\nu)}(1 + \tau_\ell)^{-\nu/(1-\nu)} < 1$ summarizes the fiscal cost of formality per unit of x , and labor demand is increasing in θ and k and decreasing in the wage and in own-sector taxes. Profits are linear in x , so all expectations below reduce to means of a log-normal variable: conditional on the transition-stage information set, the expected profitability index $\hat{x} \equiv E[x_2 \mid \mathcal{I}_1]$, strictly increasing in θ_1 and in v (for entrants, a function of v alone), is the single sufficient statistic for expected productivity in every decision below. Formal statements are in Appendix P, and all proofs are collected in Appendix Q.

Consider a firm that has chosen to operate in sector s in period 2 (the sector subscript is momentarily suppressed) with internal funds $n \geq 0$ and expected index $\hat{x}$. Its investment problem at the transition stage is

$$G_s(\hat{x}, n) \equiv \max_{k \geq 0} -k - \Phi_s(a) + \beta \left[(1 - \lambda s) \Omega_s \hat{x} k^\gamma + (1 - \delta)k \right], \quad (11)$$

with $a = \max\{0, k - n\}$, $a \leq b_s k$, and that the value of operating in s is $U_s(\hat{x}, n) = n + G_s(\hat{x}, n) - \beta(1 - \delta_s)c_s$. Thus, the firm consumes n minus the resources it puts into the firm, and expects profit net of the fixed cost if it survives, plus the capital resale value in either survival state. Now, define the internal and external user costs of capital and the two notional targets

$$R^0 \equiv 1 - \beta(1 - \delta), \quad R_s \equiv R^0 + \phi_s, \quad \bar{k}_s(\hat{x}) = \left[\frac{\gamma M_s \hat{x}}{R^0} \right]^{\frac{1}{1-\gamma}}, \quad \tilde{k}_s(\hat{x}) = \left[\frac{\gamma M_s \hat{x}}{R_s} \right]^{\frac{1}{1-\gamma}}, \quad (12)$$

with $M_s \equiv \beta(1 - \lambda s)\Omega_s$. Notice that since $\phi_s > 0$ and $\tilde{k}_s < \bar{k}_s$, a borrower's optimal scale is compressed relative to a self-financed firm's.

The optimal policy is a four-region hierarchy of finance, $k_s^*(\hat{x}, n) = \bar{k}_s(\hat{x})$ for unconstrained firms (R1), n for firms self-financed at the kink (R2), $\tilde{k}_s(\hat{x})$ under interior borrowing (R3), and $n/(1 - b_s)$ where the collateral constraint binds (R4), with borrowing $a_s^* = \max\{0, k_s^* - n\}$ strictly positive exactly in R3 and R4; Appendix P states the full policy and its derivatives.

Three of its properties connect to the data: investment-cash-flow sensitivity is positive only where finance binds and is largest where collateral binds; the spread ϕ_s acts on the intensive margin of interior borrowers while collateral capacity b_s acts on the most severely constrained, typically young, low- n firms, so the two instruments of a credit expansion reach different parts of the firm distribution; and borrowing incidence rises with expected productivity and falls with internal funds.

The value of operating inherits the four branches, with $\Psi_s(\hat{x}; R) \equiv (1 - \gamma) \gamma^{\gamma/(1-\gamma)} (M_s \hat{x})^{1/(1-\gamma)} R^{-\gamma/(1-\gamma)}$ the value on the power branches, and the shadow price of a peso inside the firm, $q_s \equiv 1 + \partial G_s / \partial n$, exceeds one for constrained firms and jumps above $1 + \phi_s$ when collateral binds, the object behind the role of entrant wealth in the quantitative model.

The analytical workhorse of the paper is the envelope pair $\partial U_s / \partial \phi_s = -a_s^*(\hat{x}, n) \leq 0$ and $\partial U_s / \partial b_s = \mu_s k_s^* \geq 0$, with μ_s the multiplier on the collateral constraint and zero cross-derivatives, $\partial U_s / \partial \phi_{s'} = \partial U_s / \partial b_{s'} = 0$ for $s' \neq s$: the value of choosing a sector responds to its own financial conditions in proportion to equilibrium borrowing, to its own collateral capacity in proportion to the shadow value of pledgeability, and to the other sector's terms not at all, with both own-sector responses strict exactly for borrowers. In the general

model the financing fee carries an additional fixed component $\phi_{0s} \mathbf{1}\{a > 0\}$; conditional on borrowing the fixed cost is sunk, so all policies and envelope derivatives are unchanged, and the borrowing-participation cutoff it creates is the moment that identifies the fixed cost at estimation (Appendix P).

7.3 Sector choice: a single-crossing formality margin

Define the origin state of a firm as $o \in \{\text{entrant}, i\text{-incumbent}, f\text{-incumbent}\}$ and its internal funds available before paying the origin-specific fee of each destination as n_o . Thus, the destination s leaves internal funds $n_{o,s} = n_o - \kappa_{os}$ inside the firm, where $\kappa_{os} \in \{c_s^e, c_{if}, c_{fi}, 0\}$ per (10). Then, the firm's formality decision compares

$$\Delta_o(\hat{x}, n_o) \equiv U_f(\hat{x}, n_o - \kappa_{of}) - U_i(\hat{x}, n_o - \kappa_{oi}). \quad (13)$$

Notice that the empirically relevant region for this comparison is R3 for both destinations. This is growing firms whose desired capital exceeds its internal funds. The statement below is for that region R3, and Section Q verifies that the threshold structure extends piecewise to R2/R4 (and characterizes the cash-rich case in Remark P.7). On origin $\times$ destination region R3 $\times$ R3, the following expression holds

$$\Delta_o(\hat{x}, n_o) = A \hat{x}^{\frac{1}{1-\gamma}} - B n_o - C_o, \quad (14)$$

where

$$A \equiv (1 - \gamma) \gamma^{\frac{\gamma}{1-\gamma}} \left[M_f^{\frac{1}{1-\gamma}} R_f^{-\frac{\gamma}{1-\gamma}} - M_i^{\frac{1}{1-\gamma}} R_i^{-\frac{\gamma}{1-\gamma}} \right], \quad B \equiv \phi_i - \phi_f > 0, \quad (15)$$

$$C_o \equiv (1 + \phi_f) \kappa_{of} - (1 + \phi_i) \kappa_{oi} + \beta[(1 - \lambda_f)c_f - (1 - \lambda_i)c_i]. \quad (16)$$

Assumption 1 (Finance-driven formality). *Define $T_\lambda \equiv T \cdot \frac{1 - \lambda_f}{1 - \lambda_i}$ as the tax wedge adjusted by the survival advantage of formality. Then*

$$\left(\frac{R_f}{R_i} \right)^\gamma < T_\lambda. \quad (17)$$

Assumption 1 states that, for a firm that has to borrow, the financing advantage of formality ($\left(\frac{R_f}{R_i} \right)^\gamma < 1$) outweighs its fiscal cost net of the survival advantage ($T_\lambda \leq 1$). This assumption is equivalent to $A > 0$, and it is the exact sense in which formality is, in this model, access to capital. Absent a financing or survival advantage ($R_f = R_i$, $\lambda_f = \lambda_i$), $T < 1$ implies $A < 0$ and no borrower would ever formalize.

Assumption 2 (Interior thresholds and cost ordering). (a) $C_o + B n_o > 0$, means that the fixed and transition costs of formality are not dominated by its switching benefits. (b) The switching and entry fees satisfy $c_{if}, c_{fi} > 0$, $0 \leq c_i^e \leq c_f^e$, and

$$(1 + \phi_f) c_f^e - (1 + \phi_i) c_i^e < (1 + \phi_f) c_{if}, \quad (\text{A2-i})$$

$$(1 + \phi_f) c_f^e + (1 + \phi_i) c_{fi} > (1 + \phi_i) c_i^e. \quad (\text{A2-ii})$$

Conditions (A2-i) and (A2-ii) are exactly equivalent to a cost ordering that places the entrant formality threshold strictly between the two incumbent thresholds, with the formal incumbent facing the lowest cost of formality and the informal incumbent the highest: (A2-i) says that entering formal is cheaper than entering informal and immediately formalizing, and (A2-ii) says that entering informally is cheaper than entering formally and immediately switching down, which holds automatically whenever the switch-down fee is at least the informal entry fee, and in particular whenever informal entry is nearly free, the empirically relevant case (Appendix P states the cost-ordering lemma).

Proposition 7.1 (Formality threshold). *Under Assumptions 1–2, for each origin o and internal funds n_o , $\Delta_o(\cdot, n_o)$ is strictly increasing and strictly convex in $\hat{x}$ and crosses zero exactly once, at*

$$\hat{x}_o^F(n_o) = \left[\frac{B n_o + C_o}{A} \right]^{1-\gamma}. \quad (18)$$

Thus, the firm chooses the formal sector if and only if $\hat{x} \geq \hat{x}_o^F(n_o)$. The threshold is strictly increasing in internal funds, so

$$\frac{\partial \hat{x}_o^F}{\partial n_o} = (1 - \gamma) \frac{B}{A} \left[\frac{B n_o + C_o}{A} \right]^{-\gamma} > 0, \quad (19)$$

this is, conditional on expected productivity, firms with more internal funds are less likely to formalize, because self-finance substitutes for the financial advantage of formality.

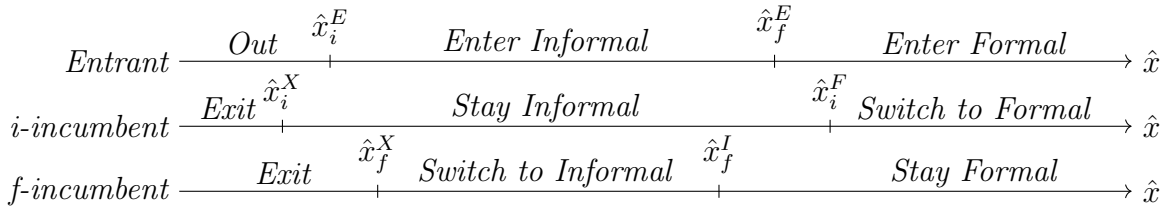
Proposition 7.1 formalizes the sorting logic of the model. The firms that select into formality are those whose growth needs are large relative to their internal resources, i.e. high $\hat{x}$, low n . These are precisely the firms for which a bank credit supply shock is decisive, which is why the Shift-Share elasticities of Section 6 are informative about (ϕ_s, b_s) .

For unconstrained, cash-rich firms the finance advantage is irrelevant, and with taxes as the only non-financial difference a sufficiently wealthy firm prefers informality; a strong enough survival advantage of formality ($T_\lambda \geq 1$) or the size-dependent audit cost of the general model removes this artifact, and the sizable mass of formal-to-informal transitions in the data suggests that for many firms the net non-financial advantage of formality is indeed weak (Appendix P).

7.4 The full transition map and its comparative statics

Each exit and entry margin reduces to one comparison, continuation in the informal sector (the relevant alternative at low $\hat{x}$, by the ordering below) against consuming the outside value, and each is characterized by a condition of the form $G_s(\hat{x}, \tilde{n}) = m$, where $\tilde{n}$ are the funds carried into the continuation and m collects the operating and transition costs. Solving branch by branch of the financing regions yields a single master threshold whose four instances are exit from either sector and entry into either sector; Appendix P states it and Appendix S develops the complete taxonomy. Two of its properties matter below: a firm without internal funds exits or fails to enter at any productivity (financial exit), and a constrained firm whose quasi-rents on internal funds cover the relevant costs never exits. The thresholds are ordered as follows.

Proposition 7.2 (Ordered thresholds). *Under Assumptions 1–2 and $c_i > 0$, for each level of internal funds the policy of every origin state is a partition of the $\hat{x}$ line, such that $\hat{x}_i^X < \hat{x}_i^E < \hat{x}_f^X < \hat{x}_f^I < \hat{x}_f^E < \hat{x}_i^F$,*



Proposition 7.3 (Comparative statics of all thresholds). *On the borrowing regions, with $D_o \equiv Bn_o + C_o > 0$ and a_s^* evaluated at the relevant margin, the thresholds respond to financial conditions as follows:*

	$\phi_f \uparrow$	$\phi_i \uparrow$	$b_f \uparrow$	$b_i \uparrow$
Formality margins $\hat{x}_e^F, \hat{x}_i^F, \hat{x}_f^F$	$\frac{(1-\gamma)a_f^*}{D_o} > 0$	$-\frac{(1-\gamma)a_i^*}{D_o} < 0$	≤ 0	≥ 0
Informal entry/exit $\hat{x}_i^E, \hat{x}_i^X$	0	$\frac{a_i^*}{\Psi_i'} > 0$	0	≤ 0
Formal exit $\hat{x}_f^X$	0	$\frac{a_i^*}{\Psi_i'} > 0$	0	≤ 0

Note: entries assume the threshold ordering of Proposition 7.2, so the formal incumbent's relevant alternative at its exit margin is the switch-down option; off that ordering the formal-exit row picks up ≥ 0 (ϕ_f) and ≤ 0 (b_f) terms through the value of staying formal. The first row reports $\partial \ln(\text{threshold}) / \partial (\text{parameter})$; the second and third rows report level derivatives, with a_i^* evaluated at the margin's funds (n for informal exit/entry, $n - c_{fi}$ for formal exit).

Proposition 7.3 carries three observations. First, cheaper or more abundant formal credit (a fall in ϕ_f or a rise in b_f) lowers every formality threshold. This produces more

formal entry, more informal-to-formal switching, and less formal-to-informal switching. It leaves the formal exit margin unchanged, because the exiting formal firm's relevant alternative is the switch-down option, and formal-side terms do not enter it. This is the pattern of the IV estimates: formal entry and formalization rise, and de-formalization falls. Second, cheaper or more abundant informal credit (a fall in ϕ_i or a rise in b_i) lowers the informal entry, informal exit, and formal exit thresholds. It would raise informal entry, lower exit from either sector, reduce formalization, and raise de-formalization. The estimates show none of these movements, informal entry is flat and exit rises, so the data assign the shock a predominantly formal-side incidence. The structural estimation of Section 9 agrees: the informal-side loading of the shock is small but positive, about two thirds of the formal loading. Third, all effects are zero for unconstrained firms. They are proportional to equilibrium borrowing at the margin for constrained firms, and hence concentrated on firms with high growth needs relative to internal funds.

A remark on what this implies for the general quantitative model. The exact zeros in the table are exclusion restrictions delivered by the two-period structure. However, with an infinite horizon, continuation values embed the option to switch sectors again later, so a formal-side parameter leaks into an informal continuation value at second order (and conversely). The general model therefore preserves the table's sign pattern and its near-diagonal loading, each margin responds first and foremost to its own sector's terms, in proportion to marginal borrowing, but not the literal zeros. Section 9 verifies exactly this at the estimated parameters, by reporting the full Jacobian of moments with respect to parameters: the quantitative counterpart of this table.

7.5 From thresholds to transition rates

Integrating over the period-1 population translates the thresholds into transition probabilities, the objects the Shift-Share design estimates: each transition probability is a normal probability of the standardized gap between log productivity and the operative threshold, so the response of any transition rate to financial conditions is the density of firms at the margin times the threshold shift of the comparative statics above, scaled by borrowing at the relevant margin. Appendix P derives the semi-elasticity of every transition rate in closed form; self-financed firms contribute exactly zero, and aggregate responses are incidence-weighted integrals that concentrate among small, young, low-net-worth firms near their thresholds. The payoff for the empirical strategy is the incidence algebra of the switching margins.

Corollary 7.4 (Incidence and the switching margins). *Consider a credit supply expansion that improves both sectors' borrowing terms: spreads fall by the absolute amounts $\varepsilon_s \equiv \chi_s \phi_s \geq 0$, i.e. $d\phi_s = -\varepsilon_s$, and/or collateral capacities rise, $db_s \geq 0$. The conditions below*

are stated for the spread components; a collateral improvement adds $\mu_s k_s^* db_s$ to its own sector's side of each inequality. Then:

(i) Entry and exit. Under Proposition 7.2, the binding entry margin and both exit margins, $\hat{x}_i^E$, $\hat{x}_i^X$, $\hat{x}_f^X$, are informal-continuation comparisons, so they respond only to informal-side terms: all three fall when $\varepsilon_i > 0$ or $db_i > 0$, and are unchanged under a purely formal-side expansion. Hence total entry rises and exit falls at both incumbent margins whenever the expansion reaches the informal side; the formal/informal composition of entry is governed by $\hat{x}_e^F$, which obeys the same condition as the switching margins of (ii).

(ii) Switching up. $\mathbb{P}(i \rightarrow f)$ rises if and only if

$$\varepsilon_f a_f^* > \varepsilon_i a_i^* \quad \text{at the } i \rightarrow f \text{ margin.} \quad (20)$$

Because the switch fee travels with the formal option, $a_f^* - a_i^* = \tilde{k}_f - \tilde{k}_i + c_{if} > 0$ under Assumption 1; hence equal absolute improvements ($\varepsilon_f = \varepsilon_i$) always raise informal-to-formal switching. Under equal proportional improvements ($\chi_f = \chi_i$), (20) reads $\phi_f a_f^* > \phi_i a_i^*$: the borrowing-scale gap must outweigh the spread gap. The same condition, with the a^* 's evaluated at the entrant's funds ($a_s^* = \tilde{k}_s - (k_0 - c_s^e)$), governs the entrant margin $\hat{x}_e^F$: there $a_f^* - a_i^* = \tilde{k}_f - \tilde{k}_i + c_f^e - c_i^e > 0$, so equal absolute improvements also raise the formal share of entrants.

(iii) Switching down. $\mathbb{P}(f \rightarrow i)$ rises if and only if

$$\varepsilon_i a_i^* > \varepsilon_f a_f^* \quad \text{at the } f \rightarrow i \text{ margin,} \quad (21)$$

where now the fee travels with the informal option, $a_i^* - a_f^* = \tilde{k}_i - \tilde{k}_f + c_{fi}$. Both switching margins can therefore rise together only if the expansion has a sufficiently informal-side component (ε_i , or $b_i \uparrow$).

This incidence algebra is the toy model's central payoff for the empirical strategy, and the estimates of Section 6 discipline it. Formalization rises and de-formalization falls; by (ii) and (iii) both movements require the same inequality, that the formal-side improvement dominates at each switching margin, $\varepsilon_f a_f^* > \varepsilon_i a_i^*$, while the size of the de-formalization decline is informative about the informal side, because a larger informal-side improvement attenuates it. The entry and exit block points the same way: informal entry is flat and exit rises, so the informal-side threshold effects of (i) are absent from the data, and the exit increase instead reflects the turnover of marginal cohorts, the subject of the next subsection. The estimated switching pattern is thereby converted into a statement about which sector's borrowing terms the shock reached, and this is the restriction taken to the structural estimation: the first-stage credit response and the collateral response pin

the formal-side incidence, the two switching responses discipline the informal-side loading, which the data keep small, and the interest rate is kept as a measurement equation rather than an estimation target.

7.6 Composition: marginal entrants, turnover, and average productivity

The last set of results concerns the composition of the response. This is where the model meets the value-added estimates of Section 6: informal value added falls, formal value added does not move, and the average across firms is at most marginally positive. Allow period-2 firms a shutdown option: after observing θ_2 , an active firm operates iff variable profit covers the fixed cost, $\Omega_s x_2 k_2^\gamma \geq c_s$; otherwise it stays idle for the period (it still resells its capital). Shutdown is the two-period analogue of endogenous exit for a young cohort. For an entering cohort into sector i (the same algebra applies to a formal entering cohort, the margin the estimated shock moves most), entry occurs iff $v \geq \bar{v}(k_0)$, where $\bar{v}$ solves $\hat{x}_e(\bar{v}) = \hat{x}_i^E(k_0)$, and:

$$\mathbb{P}(\text{shutdown of an entrant with } (v, k_2)) : \pi^X(v) = \Phi\left(\frac{(1 - \nu) \ln(c_i / (\Omega_i k_2^\gamma)) - \ln v}{\sigma_\varepsilon}\right), \quad (22)$$

which is strictly decreasing in v (directly, and indirectly through k_2).

Proposition 7.5 (Composition of a credit expansion). *Let financial conditions improve for the informal sector ($\phi_i \downarrow$ or $b_i \uparrow$), so that $\bar{v}$ falls for every constrained k_0 . Then, for the entering cohort:*

- (i) Entry rises: *the mass of entrants $\int \mathbb{P}(v \geq \bar{v}(k_0)) dF(k_0)$ increases.*
- (ii) Average entrant productivity falls: *for $\ln v \sim N(0, \sigma_v^2)$, $\mathbb{E}[v \mid v \geq \bar{v}]$ is strictly increasing in $\bar{v}$; hence a lower threshold mechanically lowers the mean (and every quantile) of entrant productivity, and with it average value added per entrant.*
- (iii) Cohort turnover rises: *the cohort shutdown rate $\mathbb{E}[\pi^X(v) \mid v \geq \bar{v}]$ is strictly decreasing in $\bar{v}$; the marginal entrants admitted by the expansion fail at higher rates than infra-marginal ones, so measured exit rises with a lag even as incumbent exit thresholds fall.*

Part (iii) of the proposition isolates the composition channel: for the collateral-constrained entrants where the margin lives, capital is pinned by wealth and collateral, so a pure spread reduction leaves their capital unchanged and the composition effect is the entire response, while a collateral relaxation deepens their capital and partially offsets it. The proposition thus reproduces, inside the two-period model, the composition signature of Section 6:

the expansion admits marginal, lower- v entrants and raises their subsequent turnover, the model counterpart of the estimated exit response, while on the informal side the formality threshold falls, the most productive informal firms leave the pool, and informal average value added declines. The two forces lower average productivity within the informal sector, the response that is robust in the data, and leave the economy-wide average roughly flat, exactly as estimated: the model reads the value-added results as reallocation plus selection, not as productivity gains within infra-marginal firms.

7.7 The model-implied Shift-Share estimand

The toy model, finally, makes precise what the Leave-One-Out Shift-Share coefficient identifies through the lens of the model: to first order, the model counterpart of the IV estimand is the ratio of two equilibrium responses to the same financial-conditions shift, a density-weighted sum of threshold shifts in the numerator over a denominator that mixes intensive loan deepening by continuing borrowers with loans created and swapped at the margins. Appendix P presents the generic closed form and Appendix R works out the informality-mass case in detail. Three lessons carry to the quantitative model. First, the IV coefficient is large where many firms sit near their thresholds, which rationalizes why the gross-flow elasticities exceed the net stock response, flows add up threshold crossings while the stock nets them out, and why the responses concentrate on the formal side, the sector whose borrowing terms the shock improves. Second, both blocks of the ratio are built from parameters estimated off cross-sectional moments, so matching the observed first stage disciplines the financing block directly. Third, the ratio is invariant to the scale of the shock, so only the incidence of the credit supply shock, not its size, needs to be estimated; the informal collateral loading is set to zero in the baseline because bank credit supply shocks do not shift informal collateral technology, and the estimation pins the remaining loadings from the stacked quantity, collateral, and switching responses, reserving the rest for validation.

8 The quantitative model

This section embeds the mechanisms of Section 7 in a stationary firm dynamics model in the tradition of Hopenhayn (1992), with two operating regimes as in Ulyssea (2018) and costly external finance as in Gomes (2001). Relative to the toy period model it adds an infinite horizon with persistent productivity, a cost of informality that grows with firm size on the labor margin, sector specific productivity dynamics, a fixed cost of borrowing, and a taste shock on the sector choice. These additions discipline the size distribution, the age profiles, and the relative turnover of the two sectors. Every comparative static that drives the counterfactuals, however, is inherited from the closed forms of Section 7.

8.1 Environment

Time is discrete and infinite, and a period is a year. A local labor market contains a continuum of production sites. Each date a measure m_0 of potential entrants is born. A potential entrant draws a permanent productivity type v , with $\ln v \sim N(0, \sigma_v^2)$, and initial wealth k_0 , with $\ln k_0 \sim N(\mu_{k_0}, \sigma_{k_0}^2)$, independent of v . An incumbent is described by its sector $s \in \{i, f\}$, its type v , its lagged productivity, and its installed capital k . An incumbent in sector s dies at the beginning of each period with probability λ_s , in which case the owner recovers the undepreciated capital $(1 - \delta)k$. Owners are risk neutral, consume dividends, and discount at β . The output good is the numeraire. The wage w is common to both sectors and exogenous to the firm. In the baseline it is normalized to one and held fixed, which mirrors the within market identification of the empirical design.

Production is $y = \theta k^\alpha \ell^\nu$ with $\alpha + \nu < 1$, where $\ell = 1 + \ell^h$ adds the owner's inelastic unit to hired labor $\ell^h \geq 0$. Productivity has three layers with distinct jobs,

$$\ln \theta_t = \ln v + z_t + u_t, \quad z_t = \rho_s z_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_{\varepsilon,s}^2), \quad u_t \sim N(-\frac{1}{2}\sigma_{u,s}^2, \sigma_{u,s}^2). \quad (23)$$

The permanent type v generates selection at entry and the long tails of the age and size profiles. The persistent component z carries the slow mean reversion that the census autocovariances identify. The transitory component u is real. The firm observes it before its exit, switching, input, and borrowing decisions, so it generates annual turnover, and because it is independent over time it adds no state variable.¹⁷ The processes are sector specific, while σ_v is common because v is drawn before any sector is chosen.

A formal firm pays the revenue tax τ_y and hires on registered contracts, so its labor cost is $(1 + \tau_\ell) w \ell^h$. An informal firm evades both taxes but its payroll is exposed to detection. Hiring ℓ^h costs

$$\mathcal{C}_i(\ell^h) = w \ell^h \left(1 + \frac{\ell^h}{\eta_i}\right), \quad \eta_i > 0, \quad (24)$$

an expected penalty schedule that rises with size in the spirit of [Ulyssea \(2018\)](#). Informality therefore saves τ_ℓ per worker at small scale, and the saving erodes as the firm grows, which restores formality as the optimal regime for large firms even absent financial motives. Static profits are $\pi_s(\theta, k) = \max_{\ell^h \geq 0} (1 - \tau_y^s) \theta k^\alpha (1 + \ell^h)^\nu - \mathcal{C}_s(\ell^h)$ with $\tau_y^f = \tau_y$ and $\tau_y^i = 0$. The corner $\ell^h = 0$ is the self employment margin that accounts for the mass of owner only firms in the census.¹⁸

¹⁷Measured productivity adds classical measurement error on top of (23). The error lives in the data and never in decisions. Section 9 shows how the census autocovariances separate the layers and where the split between real transitory variation and error comes from.

¹⁸The model is silent on registered firms that underreport part of their payroll, the intensive margin of informality in [Ulyssea \(2018\)](#). That margin responds to enforcement and labor cost policy rather than to credit supply, and its omission buys an exact mapping between the model's sector indicator and the census classification used in every empirical moment.

Investment is financed one period ahead of the revenue it generates. A firm entering the next period in sector s with target capital k' and internal funds n raises external funds $a = \max\{0, k' - n\}$ at cost

$$\Phi_s(a) = \phi_{0s} \mathbf{1}\{a > 0\} + \phi_s a, \quad a \leq b_s k'. \quad (25)$$

The three financial primitives have distinct empirical counterparts. The slope ϕ_s is the spread of marginal loan pricing over the risk free rate. The fixed component ϕ_{0s} collects application, documentation, and relationship costs, and it is what keeps borrowing incidence far below one even when spreads are moderate. The cap b_s limits pledgeability. No sector ranking is imposed. A bank credit supply shock is a shift of (ϕ_s, b_s) , and its incidence across sectors is estimated, not assumed. Entry into sector s costs c_s^e , switching costs are c_{if} and c_{fi} , operating in sector s costs c_s per period, and exit is free and yields the firm's internal funds. Finally, the sector comparison at the transition stage carries an idiosyncratic taste draw. Each option receives an independent extreme value shock with scale σ_c , so sector choices are smooth in the value gaps and the model generates gross switching flows in both directions at any parameter vector.¹⁹

8.2 Recursive formulation

The within period timing replicates Figure 11. Death shocks realize first, then θ_t is observed, then production and taxes and the operating cost settle, and then the firm chooses next period's status and, if continuing, capital and borrowing. Sector status is therefore chosen before the next innovation is observed, which makes lagged productivity the relevant state for transitions. Let $W_s(v, \theta_{-1}, k)$ denote the value of beginning a period committed to sector s ,

$$W_s(v, \theta_{-1}, k) = \lambda_s(1 - \delta)k + (1 - \lambda_s) \mathbb{E}_{\theta|\theta_{-1}, v}^s [V_s(v, \theta, k)], \quad (26)$$

where the post production value solves

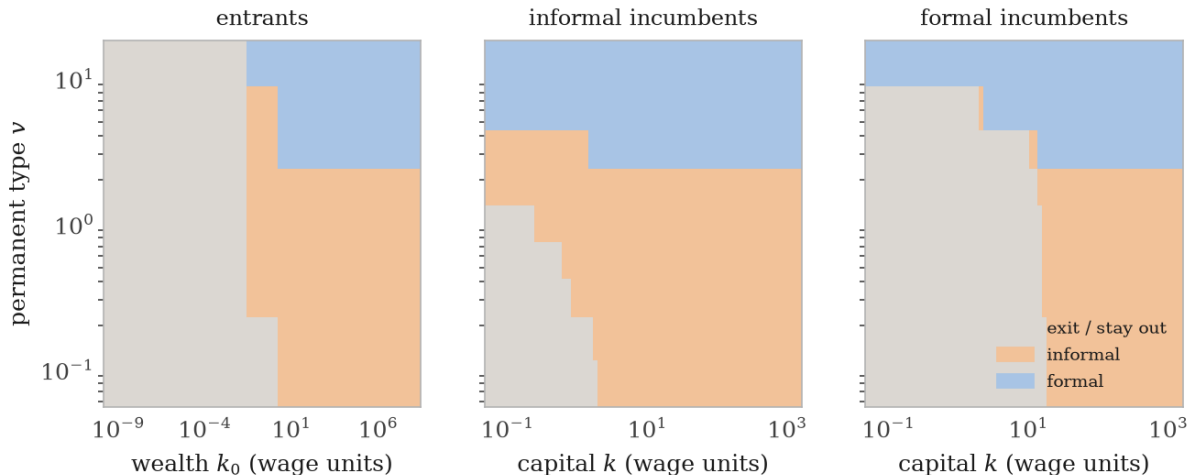
$$V_s(v, \theta, k) = \max \left\{ n_s^{\text{exit}}, \max_{s'} \max_{k' \geq 0} n_{ss'} - k' - \Phi_{s'}(a) + \beta W_{s'}(v, \theta, k') \right\}, \quad (27)$$

subject to $a = \max\{0, k' - n_{ss'}\}$ and $a \leq b_{s'} k'$, with internal funds $n_s^{\text{exit}} = \pi_s(\theta, k) - c_s + (1 - \delta)k$ and $n_{ss'} = n_s^{\text{exit}} - c_{ss'} \mathbf{1}\{s' \neq s\}$. A potential entrant solves the same continuation problem with funds $k_0 - c_s^e$ and first period productivity centered on v , against the outside value k_0 . The taste draws enter the discrete comparison in (27) additively. Standard arguments deliver existence and uniqueness of the values and measurable policies for the sector decision, capital, borrowing, and labor.

¹⁹At $\sigma_c = 0$ the choice collapses to the deterministic maximum of Section 7. The taste draw stands in for heterogeneity the state vector omits, such as owner preferences over dealing with authorities.

Because continuation values increase in θ and the exit value does not, the policies inherit the threshold structure of Section 7 conditional on (v, k) , now smoothed by the taste scale. Two forces shape the boundaries. The finance force makes formality attractive for firms whose growth needs are large relative to internal funds. The audit force in (24) makes formality attractive at large scale regardless of finance. Both are estimated, and Figure 4 displays their combined effect at the estimated parameters.

Figure 4: Decision rules at the estimated parameters



The panels partition the state space into exit, informal, and formal regions for each origin status. The ordering of Proposition 7.2 holds at the estimates.

8.3 Stationary equilibrium

The policies induce a transition kernel on the state space. Incumbents die, draw productivity, produce, and transit according to the decision rules, while entrants add mass according to the entry policy. A stationary equilibrium given w is an invariant measure μ together with optimal values and policies such that the entrant inflow balances exogenous and endogenous outflows sector by sector. Aggregates are integrals against μ , in particular the formal share of firms, credit $C = \int a d\mu$, employment, value added, and the flow masses of entry, exit, and switching. The flow identity

$$N_{s,t+1} = (1 - \lambda_s) \text{Stay}_{s,t} + (1 - \lambda_{s'}) S_{s',t} + E_{s,t} \quad (28)$$

is the model counterpart of the margin decomposition estimated in Section 6, with the same five year definitions. One implication of stationarity matters for estimation. The entrant share of the stock is pinned by the exit flows through the balance condition, so it serves as a consistency check rather than as a free moment.

8.4 The model counterpart of the Shift-Share experiment

The empirical design identifies the response of market outcomes to equilibrium credit induced by bank credit supply shocks, using Leave-One-Out exposure across markets. The model counterpart replicates the design rather than the estimand alone. The economy is a collection of markets in the stationary equilibrium of the estimated model, identical except for their exposure ω_m to a common credit supply shifter. At date zero an unanticipated permanent shock of intensity $\zeta_m = \omega_m g$ moves market m 's financing schedules to

$$\phi_s(\zeta_m) = (1 - \chi_s \zeta_m) \phi_s, \quad b_f(\zeta_m) = (1 + \kappa_f \zeta_m) b_f, \quad (29)$$

where $(\chi_i, \chi_f, \kappa_f)$ is the incidence vector, the object that says how much of a marginal peso of bank funding reaches each sector's price and quantity margins. The baseline holds $\kappa_i = 0$, since bank credit does not shift the collateral technology available to informal lending, and this restriction is relaxed in robustness. Policies are resolved, the distribution is iterated forward five years, the horizon of the outcome window, and every empirical outcome is recorded along the way. The estimator of Section 5 is then run on the simulated market panel. For small shocks each coefficient approaches

$$\beta_Y^{\text{model}} = \frac{\ln Y_5(\zeta) - \ln Y_5(0)}{\ln C_5(\zeta) - \ln C_5(0)}, \quad (30)$$

which is invariant to the scale of the shock because numerator and denominator are both proportional to ζ to first order, exactly as in (52). In practice the shifter g is set so that the simulated first stage reproduces the empirical response of five year credit to a one standard deviation bank shock, and the invariance of the coefficients to that choice is verified numerically. Each coefficient is a density weighted aggregate of the threshold responses of Section 7, concentrated on constrained marginal firms. Section 9 stacks a subset of these coefficients with the census moments in estimation and reserves the rest, together with the untargeted cross section, for validation. Appendix T discusses the modeling choices, the fixed wage, within period debt, safe loans, and measured financial primitives, and what each implies for the counterfactuals.

9 Estimation

The parameters divide into three groups. Institutional parameters are read off legislation, administrative ratios, and market prices. A second group is estimated outside the model by closed form inversions of census and survey moments. The remaining thirteen parameters, the operating, entry, and switching costs, the borrowing participation costs, the entrant wealth distribution, the taste scale, and the incidence of the credit supply shock, are estimated jointly by minimum distance on twenty three moments, ten from the census and

financial cross sections and thirteen quasi experimental coefficients from Section 6. The system is overidentified with ten degrees of freedom. Identification rests on the comparative statics of Section 7, each parameter has a leading moment, and the sensitivity measure of Andrews et al. (2017) reports how far that assignment holds at the estimates. Validation then uses what estimation never touched, the untargeted responses and the untargeted cross section.

9.1 Measurement and preset parameters

Every moment is defined once, in the microdata language of its source, and the model reproduces the measurement process rather than an idealized object: exit and switching are five year outcomes with death and exit indistinguishable, employment includes owners and unpaid workers, borrowing is a flow of credit raised within the year, and all monetary objects are multiples of the census median annual wage of 77.5 thousand pesos. The preset parameters are read off legislation, administrative ratios, and market prices: taxes from effective rates, technology from census cost shares, the productivity process from census autocovariances at horizons free of measurement error, death rates from the flat tail of exit by size, the spreads directly from registry pricing (the medians over CETES 28 by borrower IMSS registration, $\varsigma_i = 0.068$ and $\varsigma_f = 0.050$), and the collateral capacities from the ninetieth percentile of leverage among borrowers, with the formal read capped at 0.90. Appendix U reports the full preset table and the measurement choices behind it, and Appendix V collects the conventions line by line.

9.2 Estimated parameters, moments, and the criterion

The estimated vector is $\theta = (c_i, c_f, c_i^e, c_f^e, \sigma_c, c_{fi}, \phi_{0i}, \phi_{0f}, \mu_{k_0}, \sigma_{k_0}, \chi_f, \kappa_f, \chi_i)$.²⁰ The moment vector stacks ten cross sectional moments, the five year exit rates by sector, the two switching rates, the formal share of entrants, borrowing incidence by sector, the mean and ninetieth percentile of informal entrant capital, and the relative size of informal entrants, with thirteen coefficients from the Shift-Share design, the stocks by sector, entry in total and by sector, exit by sector, the two switching responses, average and total value added, collateral, and the first stage.²¹ Census moments enter in relative deviations and coefficients

²⁰The switching cost c_{if} is held at the floor the deterministic version of the model pushed it to. In that version the data ask the formalization rate to be carried by an ever smaller fee, which is the signature of a missing smoothing device rather than of a large fee, so the taste scale σ_c carries this margin and c_{if} stays frozen. The informal collateral loading κ_i is held at zero, since bank credit supply does not shift the collateral technology available to informal lending (Section 8.4).

²¹The interest rate response is deliberately excluded. The observed book rate mixes repricing with near complete turnover of the loan stock, so the model treats it as a measurement equation rather than a target, at 0.42 percentage points per log point of credit.

in units of their AKM standard errors, so the criterion is a diagonally weighted minimum distance. The estimator solves

$$\hat{\theta} = \arg \min_{\theta} \sum_j \left(\frac{m_j(\theta) - \hat{m}_j}{\omega_j} \right)^2, \quad (31)$$

with ω_j the moment's scale, by a damped Gauss Newton routine with a numerical Jacobian, restarted from multiple initial vectors. Model moments are computed exactly from the stationary distribution and the fifth power of the transition kernel, so the criterion is smooth and free of simulation noise. The shock experiment inside the criterion moves only (ϕ_i, ϕ_f, b_f) , which is the exclusion restriction of the design. At the estimates the criterion equals 203.7 against ten degrees of freedom, and its composition summarizes what the model can and cannot do: about 55 units are the disclosed first stage level gap, the model moving market credit by 0.024 log points per unit bank shock against the 0.15 targeted; about 147 come from the six dynamic coefficients the model underfits, collateral coverage, both switching margins, the two exit margins, and formal entry; and less than 2 come from the ten census level moments, each matched within one standard error.²²

9.3 Identification

Each parameter has a leading moment, and the pairing follows a monotone comparative static of Section 7. Table 10 lists the assignments. The operating costs govern the exit thresholds, so the two exit rates read (c_i, c_f) . The formal entry fee governs the entrant formality cutoff, so the formal share of entrants reads c_f^e , while the informal entry fee governs selection into activity, so the size of informal entrants relative to incumbents reads c_i^e . The two switching rates discipline the taste scale and the switching fee toward informality. The participation costs govern who borrows at all, so borrowing incidence by sector reads (ϕ_{0i}, ϕ_{0f}) . Constrained entrants install capital proportional to wealth, so the location and spread of entrant capital read $(\mu_{k_0}, \sigma_{k_0})$. On the incidence side, the first stage reads χ_f , the collateral response reads κ_f , and the two switching responses read χ_i , the margin that Section 7 shows is informative because a formal side improvement raises formalization and lowers its reverse while informal side incidence pushes both the other way. The remaining coefficients over identify the system. Two exhibits report this logic at the estimates rather than assume it. The Jacobian heat map in Figure 13 in Appendix U shows strong own responses exactly where the estimates are precise, the operating and participation costs, the wealth block, and the taste scale, alongside one global column: the entrant wealth dispersion

²²The weighting is diagonal, so this statistic is a fit summary rather than a formally sized test. The full moment covariance, and with it a sized test and covariance robust standard errors, awaits a market level bootstrap. Reported standard errors are delta method sandwich formulas conditional on the presets and on the incidence estimates.

σ_{k0} moves most moments, because it governs how many firms sit against their financing constraints. The sensitivity matrix of Andrews et al. (2017) in Appendix Figure 12 shows the mirror image and is candid about the weak corner: the small entry and switching fees have near zero own elasticities, so they are the loosely identified parameters of the system, exactly where the standard errors of Table 11 are large.

Table 10: Leading moments

Parameter	Leading moment
Operating costs c_i, c_f	five year exit rate, by sector
Entry cost, informal c_i^e	entrant size relative to incumbents
Entry cost, formal c_f^e	formal share of entrants
Taste scale σ_c and switching cost c_{fi}	five year switching rates
Participation costs ϕ_{0i}, ϕ_{0f}	borrowing incidence, by sector
Entrant wealth μ_{k0}, σ_{k0}	entrant capital, mean and ninetieth percentile
Incidence χ_f	first stage credit response
Incidence κ_f	collateral response
Incidence χ_i	switching responses

9.4 Parameter estimates

Table 11 reports the estimates. The picture they paint is a cost wall. Operating formally costs 15.2 wage units a year against 0.85 informally, about 62 thousand dollars against 3.5 thousand, a factor of eighteen. Borrowing formally carries a participation cost of 26.1 wage units against 0.40 informally, about 107 thousand dollars against 1.6 thousand, a factor of about sixty five. The entry fees themselves are small, about 60 and 320 dollars, so the wall is not at the door but in the recurring cost of operating and of dealing with the formal financial system. Against this wall stands the entrant wealth distribution. The median entrant starts with about 810 dollars, three orders of magnitude below the formal operating bill, with enormous dispersion, so only firms that expect scale, or that draw exceptional wealth, find the formal package worth paying for. This is the quantitative content of the sorting result of Proposition 7.1. Appendix Figure 17 displays the estimates in money units.

The incidence estimates put $\chi_f = 0.073$ with a mild size gradient ($\kappa_f = 0.009$), and a small but positive informal side loading, $\chi_i = 0.048$: bank credit reaches informal firms directly, at roughly two thirds of the formal pass through, rather than not at all. The switching pair is what forces it there. In the data formalization rises by 0.052 and its reverse falls by 0.039 per log point of credit, and in the model both movements require the formal side improvement to dominate at the margin, while any appreciable informal side

loading would attenuate the second and eventually reverse the first. A bank credit supply expansion, through the lens of the model, is a formal sector event that reaches informal firms by making formality cheaper to finance, not by financing informality.

Table 11: Estimated parameters

Parameter	Estimate	(s.e.)	Dollar equivalent
Operating cost, informal c_i	0.85	(0.13)	3,462
Operating cost, formal c_f	15.24	(0.37)	62,162
Entry cost, informal c_i^e	0.015	(0.76)	62
Entry cost, formal c_f^e	0.078	(0.46)	319
Taste scale σ_c	1.91	(0.46)	—
Switching cost $c_{fi}^\dagger$	0.024	at bound	98
Participation cost, informal ϕ_{0i}	0.40	(0.34)	1,626
Participation cost, formal ϕ_{0f}	26.14	(2.16)	106,624
Entrant wealth location μ_{k_0}	-1.61	(0.10)	—
Entrant wealth dispersion σ_{k_0}	4.09	(0.08)	—
Incidence χ_f	0.073	—	—
Incidence κ_f	0.009	—	—
Incidence χ_i	0.048	—	—

Estimates in wage units (77,500 pesos); the dollar column converts the cost parameters at 19 pesos per dollar. Delta method standard errors from the core Jacobian at the estimates, conditional on the presets and on the incidence estimates; the incidence loadings carry no standard errors. The wall parameters are tightly estimated while the small entry and switching fees are not, consistent with their near zero own elasticities in Figure 13 in Appendix U. [†]At the boundary of its admissible range: the moment equation holds as an inequality and no standard error is reported. The criterion at these estimates is 203.7 on ten degrees of freedom. The entry costs and the informal loading, at their bounds in earlier drafts, are now interior.

9.5 Fit

Appendix Figure 14 compares the targeted cross sectional moments with the model. The exit rates, the formal share of entrants, and the entrant capital moments are matched closely. The model overshoots formalization and informal borrowing incidence and undershoots the reverse switching rate. The common thread of the misses is that the financial parameters vary by formality status and not by firm, so the model cannot simultaneously give informal firms rare borrowing and frequent switching.

Table 12: Model fit: targeted coefficients and untargeted validation

	Data	(s.e.)	Model
<i>Panel A: targeted quasi-experimental coefficients, per log point of market credit</i>			
First stage (five-year $\Delta \ln C$ per unit shock)	0.150	(0.017)	0.024
Formal firms (stock)	0.051	(0.013)	0.074
Informal firms (stock)	−0.002	(0.006)	−0.001
Entry, total	0.015	(0.009)	0.011
Entry, formal	0.067	(0.018)	0.150
Entry, informal	0.012	(0.010)	0.001
Exit, formal	0.039	(0.015)	−0.004
Exit, informal	0.055	(0.012)	−0.019
Switching, informal to formal	0.052	(0.018)	0.143
Switching, formal to informal	−0.039	(0.015)	−0.097
Average value added	0.027	(0.014)	0.014
Total value added	0.030	(0.016)	0.027
Collateral coverage	−1.857	(0.322)	0.016
<i>Panel B: untargeted sectoral value added responses</i>			
Formal value added	0.025	(0.031)	0.029
Informal value added	−0.052	(0.022)	−0.082

Note: Data columns reproduce the Shift-Share estimates with cluster-robust standard errors in parentheses; Panel B reports the untargeted sectoral value added responses with AKM standard errors. Model columns are computed at the reported point. The first-stage level miss (0.024 against 0.150) is disclosed in the text; because the coefficients are invariant to the scale of the shock, the counterfactual and validation experiments set the scale to reproduce the empirical first stage.

Table 12 compares the thirteen targeted coefficients with their model counterparts, and Appendix Figure 15 displays them against their confidence bands. The model reproduces the composition core and the value added responses. The formal stock response is 0.074 against 0.051, within two standard errors; the informal stock, informal entry, and total entry responses are small in the model as in the data; and the value added pair carries the right signs, 0.027 against 0.030 for the total and 0.014 against 0.027 for the average, both within one standard error. The dynamic margins are where the model falls short, in three ways it is useful to separate. First, the reallocation responses overshoot: formal entry is 0.150 against 0.067, and the switching pair is 0.143 and −0.097 against 0.052 and −0.039, so the model concentrates too much of the adjustment in firms that move between sectors. Second, the exit responses are slightly negative in the model against 0.04 and 0.06 in the data: with a fixed wage and no competition for inputs an expansion displaces no incumbents, so the empirical exit response points to a general equilibrium turnover channel the model omits. Third, the collateral response is near zero with the wrong sign against

−1.86 in the data, because the cap is a hard technological limit here while in the data collateral coverage is a negotiated loan term that loosens when credit expands. And the first stage level falls short of its target, 0.024 against 0.15, a miss we disclose rather than absorb, which cautions against reading the incidence loadings as complete pass through. Because the Shift-Share coefficients are invariant to the scale of the shock, the validation experiment of Section 8.4 sets that scale to reproduce the empirical first stage, so the level miss affects the interpretation of the loadings and nothing downstream.²³

9.6 Validation

The sharpest validation uses responses the estimation never saw. The model, estimated without the sectoral value added responses, predicts that a credit expansion moves value added toward formal firms, raising it in the formal sector and lowering it in the informal one. The corresponding untargeted coefficients in the data are 0.025 and −0.052, and the informal decline is the value added result that survives every robustness exercise of Section 6.3. Panel B of Table 12 displays the comparison, and Appendix Figure 16 the graphical version. Both model predictions lie inside the 95 percent confidence bands of the untargeted estimates.

The untargeted cross section points the same way. The switching gradients by size line up with the census, with movements toward informality concentrated among the smallest firms and movements toward formality rising with size, and the stationary formal share of firms comes out near one fifth against roughly one eighth in the census, with nothing in the estimation targeting it. We read the validation as supporting the mechanism the counterfactuals exercise, selection into formality governed by financing costs, while the fit misses delimit what the model should not be used for, the level of aggregate credit and the turnover of incumbents.

10 Counterfactuals

The estimation delivers a specific diagnosis. Formality in Mexico is priced as a package, a recurring operating bill of about 62 thousand dollars a year and a financial participation cost of about 107 thousand, against 3.5 and 1.6 thousand in informality, while the median entrant holds about 810 dollars. The counterfactuals ask which part of that package matters, by equalizing one primitive at a time across sectors, re solving the stationary equilibrium at the estimated parameters, and comparing steady states. The wage stays fixed throughout, so the numbers are the model analogue of the within market evidence rather

²³These misses are disclosed rather than absorbed elsewhere.

than economy wide general equilibrium predictions.²⁴

Table 13: Counterfactual policy menu

Experiment	formal share Δ (pp)	ln total VA Δ (%)	ln average VA Δ (%)	ln credit Δ (%)	ln entry Δ (%)
Grant formal access, $\phi_{0f} \rightarrow \phi_{0i}$	+3.72	+5.08	+3.80	+30.59	+0.15
Remove informal access, $\phi_{0i} \rightarrow \phi_{0f}$	+2.69	-4.37	+23.19	-13.95	-24.54
Equal taxes	+0.89	-1.29	+3.01	+1.45	-2.10
Equal spreads, $\phi_i \rightarrow \phi_f$	+0.37	+0.41	+0.16	+3.42	+0.13
Equal collateral caps, $b_i \rightarrow b_f$	-0.08	+0.31	-1.27	+0.93	+1.20
All financing equalized	+2.73	-3.33	+22.13	-12.07	-23.63

Table 13 contains the paper’s policy result in six rows. Figure 18 in Appendix U displays the menu graphically. The margins a planner might reach for first do almost nothing. Equalizing lending spreads moves the formal share by less than half a point, and equalizing collateral caps it essentially does not move it at all, because at the estimated parameters few informal firms sit against those constraints once the participation cost has already kept them out of the credit market. Equalizing taxes formalizes modestly, less than one point of the firm distribution, which is consistent with the quasi experimental record on tax based formalization programs in Mexico. The action is in the fixed cost of formal financial participation, and it can be removed from either side, with opposite consequences for output.

Granting formal financial access at informal cost formalizes 3.7 points of the firm distribution and grows the economy at the same time. Credit expands by 31 percent, total value added rises by 5.1 percent, and average value added rises with it, because the firms that formalize are exactly the constrained high expectation firms of Proposition 7.1, which then borrow and scale up. Entry barely moves, so this is reallocation and deepening rather than a change in the number of firms, the same composition signature the Shift-Share estimates document. Removing informal financial access instead formalizes 2.7 points by destruction. Credit contracts by 14 percent, entry collapses by 25 percent, and total value added falls by 4.4 percent, while average value added jumps by 23 percent because the survivors are a selected elite. The two roads reach similar formal shares and opposite economies. Formalization is therefore the wrong summary statistic for this class of policies, and the output consequences separate financial inclusion from financial repression.²⁵

²⁴Holding the informal financial primitives fixed under a large formalization likely understates the financial channel, since information and enforceability would improve with formalization itself, the direction Brooks and DAVIS (2020) document. The fixed wage omits displacement effects, which is the same omission behind the model’s missed exit responses.

²⁵The equal taxes row uses the effective rates of Section 9. Comparing the first and third rows, the

The mechanism ties the menu back to the evidence. The estimated incidence of the observed bank shocks is formal sided, so history has been running a small version of the first row, and the model reproduces its footprint, more formal firms and formal entry, faster formalization, slower movement back to informality, deeper loans to existing borrowers, and value added shifting toward the formal sector. The first row scales that footprint by removing the participation wall entirely rather than shaving the spread. Reading the menu together with the estimates, the binding financial friction behind informality in these data is not the price of credit and not pledgeability. It is the cost of entering the formal financial system at all.

11 Conclusion

This paper matched the universe of Mexican bank loans to the economic censuses and asked whether formality responds to financial conditions. It does. Bank credit supply expansions leave the number of firms unchanged and recompose it toward formality, with more formal firms and formal entrants, faster formalization, slower slippage back into informality, and production reallocating away from the informal sector. The credit behind these movements flows mainly into deeper loans for existing borrowers, on easier collateral rather than lower prices. The pattern survives eleven robustness exercises and reappears, sign by sign, in a bank level design built on independently identified shocks.

The structural half of the paper explains why the lever works. Formality in Mexico is priced as a package whose binding component is financial, an operating bill seventeen times the informal one and a cost of entering formal borrowing nearly seventy times the informal one, facing entrants whose median wealth is three orders of magnitude smaller. Credit supply shocks arrive on the formal side of that wall, which is what the estimated incidence says and what the opposite movements of the two switching margins independently require. The firms that respond are the marginal ones, productive enough to want scale and poor enough to need finance, and formality is how they buy access to it.

The policy content follows directly. Two roads formalize in the model, and they are not equivalent. Extending formal borrowing terms to informal firms formalizes about four times more than equalizing taxes and raises output while doing so. Suppressing informal credit formalizes almost as much and burns output, entry, and the informal margin it squeezes. Formalization alone is therefore the wrong summary statistic for policy, and the output consequences separate financial inclusion from financial repression. The finding that

access lever formalizes about four times more than the tax lever and is the only experiment in the menu that formalizes and raises output together. A full welfare statement would require weights on tax revenue, on the owners displaced under repression, and on the insurance value of the social security package, and we do not take that step here. The ranking of the two access experiments by output does not depend on those weights.

enacted tax reforms moved the formality margin little, while credit terms move it, relocates the formalization debate from the tax code to the financial system.

The limits of the current analysis mark the next steps on the research agenda. The design is a within country comparison across markets, so aggregate general equilibrium adjustments, in wages, in bank competition, and in enforcement, are outside it, and the model inherits those omissions, which is where its fit misses concentrate. Collateral is modeled as a technological cap rather than a negotiated term, welfare statements would require weights on tax revenue and on displaced owners that the paper does not impose, and the informal financial primitives are held fixed in counterfactuals that would plausibly improve them, so the financial channel is likely understated. What stands without qualification is the empirical fact the paper set out to establish, and what stands in outline is the mechanism the model selects for it, marginal firms weighing a fixed cost of formal financial participation that their wealth cannot clear. Formality is also a financing decision, and credit supply is a policy margin for it.

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A Firm transitions under alternative definitions of informality

Table 14: Establishment transition across censuses with the intensive margin

		Destination			
		Formal	Informal	Mix	Exit
2014 Origin	Formal	94,053 (54.0%)	20,182 (11.6%)	10,990 (6.3%)	48,971 (28.1%)
	Informal	111,773 (2.8%)	2,232,216 (55.4%)	151,792 (3.8%)	1,531,660 (38.0%)
	Mix	10,522 (36.1%)	3,344 (11.5%)	7,219 (24.8%)	8,023 (27.6%)
2019 Origin	Formal	197,196 (51.8%)	54,785 (14.4%)	20,666 (5.4%)	108,369 (28.4%)
	Informal	84,059 (2.0%)	2,547,652 (61.1%)	62,012 (1.5%)	1,473,395 (35.4%)
	Mix	48,735 (19.3%)	83,256 (33.0%)	57,455 (22.8%)	62,577 (24.8%)

Note: Percentages are calculated within each origin category. Formal: 100% of workers have social security (SS); Informal: 0%; Mix: strictly between 0% and 100%.

Table 15: Establishment transition across censuses with social security number definition

		Destination		
		Formal	Informal	Exit
2019 Origin	Formal	84,146 (33.4%)	73,597 (29.2%)	93,991 (37.3%)
	Informal	142,684 (3.8%)	2,261,091 (60.5%)	1,331,140 (35.6%)

Note: Percentages are calculated within each origin category. Informality is defined as an establishment without social security (IMSS) number. The table excludes 813,508 observations (17%) with missing responses to the question about the social security (IMSS) registration number.

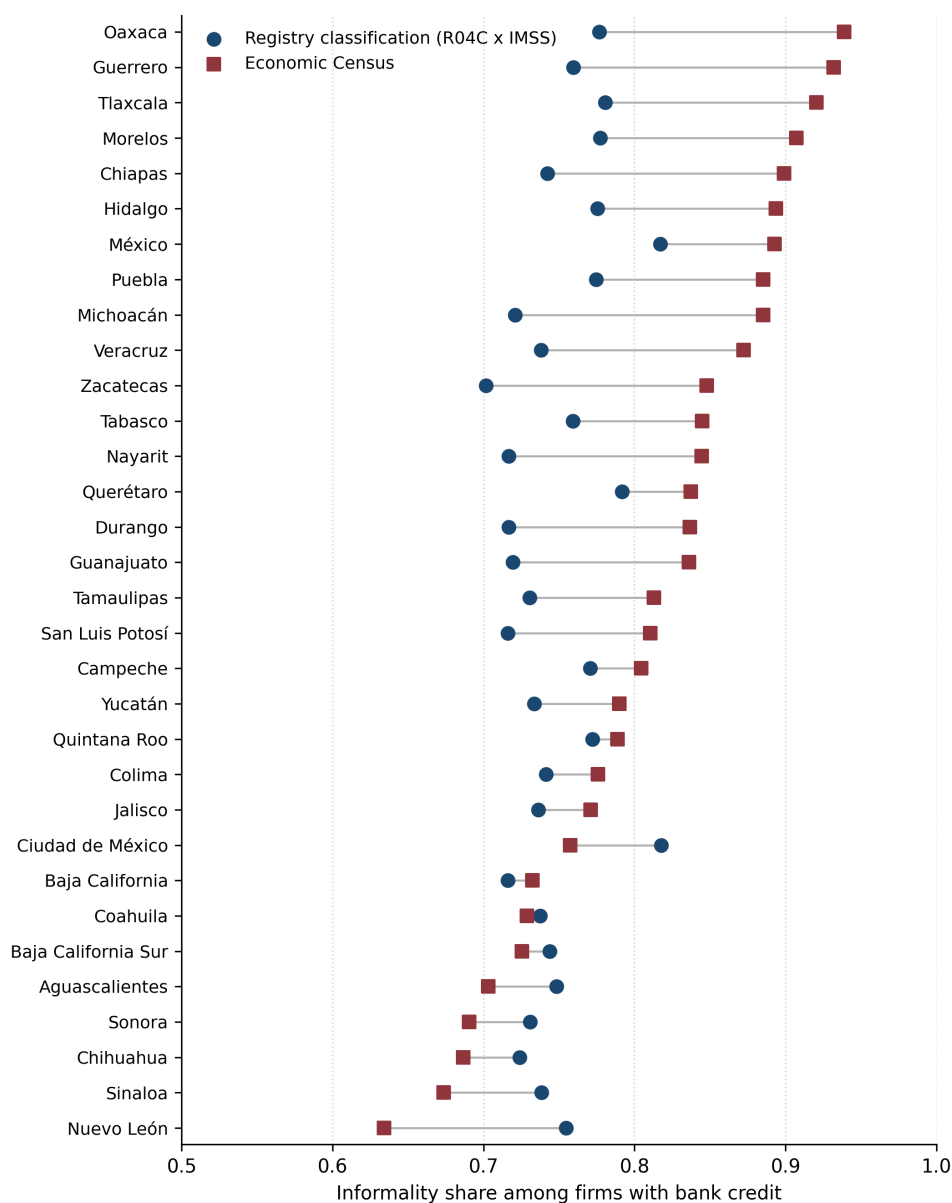
Table 16: Establishment transition across censuses with tax administrative number definition

		Destination		
		Formal	Informal	Exit
2019 Origin	Formal	749,911	284,018	475,078
		(49.7%)	(18.8%)	(31.5%)
	Informal	424,885	1,439,976	1,162,103
		(14.0%)	(47.6%)	(38.4%)

Note: Percentages are calculated within each origin category. Informality is defined as an establishment without tax (RFC) number. The table excludes 264,186 observations (5%) with missing responses to the question about the tax (RFC) number.

The census series in Figure 5 uses the census IMSS registration question, the closest counterpart to the administrative match; the establishment counts and transition tables elsewhere in the paper instead classify an establishment as informal when it reports zero social security payments while employing workers and earning income, and the two measures should not be read as the same object.

Figure 5: Registry classification of borrower informality against the census



Share of firms with bank credit classified as informal, by state, sorted by the census value. Registry: firms borrowing in the 2018 R04C whose RFC has no IMSS employer registration (474,237 firms). Census: establishments not registered with IMSS (question h424a) among those reporting bank credit, 2019 census. The classification, used only for loan pricing, gets the level of borrower informality approximately right (76 percent overall against a census state mean of 81) but compresses the cross-state gradient: it overstates informality where the census puts it lowest (Nuevo León, Sinaloa, Chihuahua, Sonora) and understates it where the census puts it highest (Oaxaca, Guerrero, Tlaxcala, Morelos).

B Census workers count

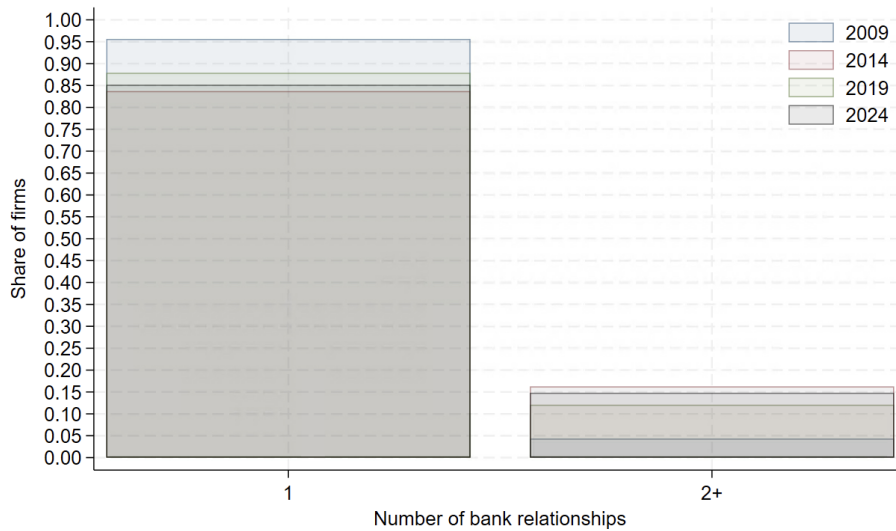
Table 17: Census worker counts

	Total	Formal	Informal
$\widehat{\Delta_5 \ln(C)}$	0.033 (0.075)	-0.058 (0.168)	0.065 (0.189)
FS Coeff.	-2.523 (1.569)	-2.522 (1.569)	-2.523 (1.569)
FS F	2.59	2.58	2.59
Obs.	517	494	517
Markets	517	494	517

Notes: Census worker counts exist only in the 2019 and 2024 waves, leaving a single five-year window (a cross-section of markets, so clusters equal observations). The first stage on this subsample is weak ($F = 2.6$), so these estimates are reported for completeness only and are not informative. AKM standard errors in parentheses.

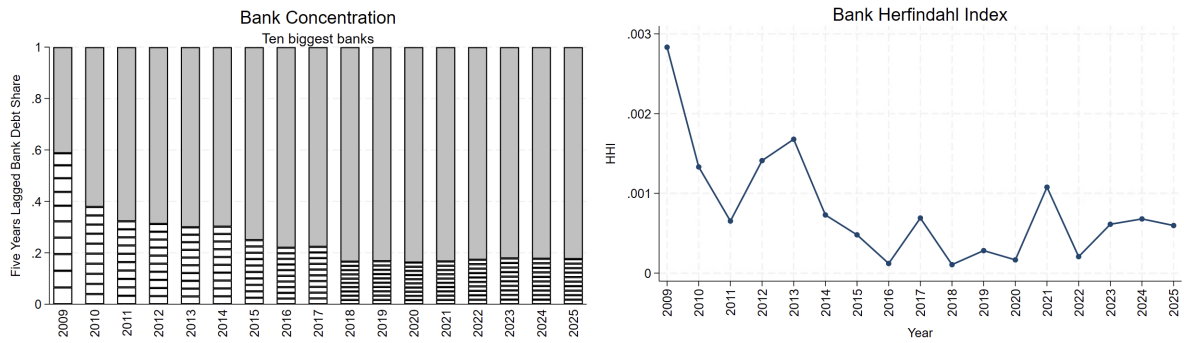
C Additional descriptive figures

Figure 6: Number of banking relationships per firm



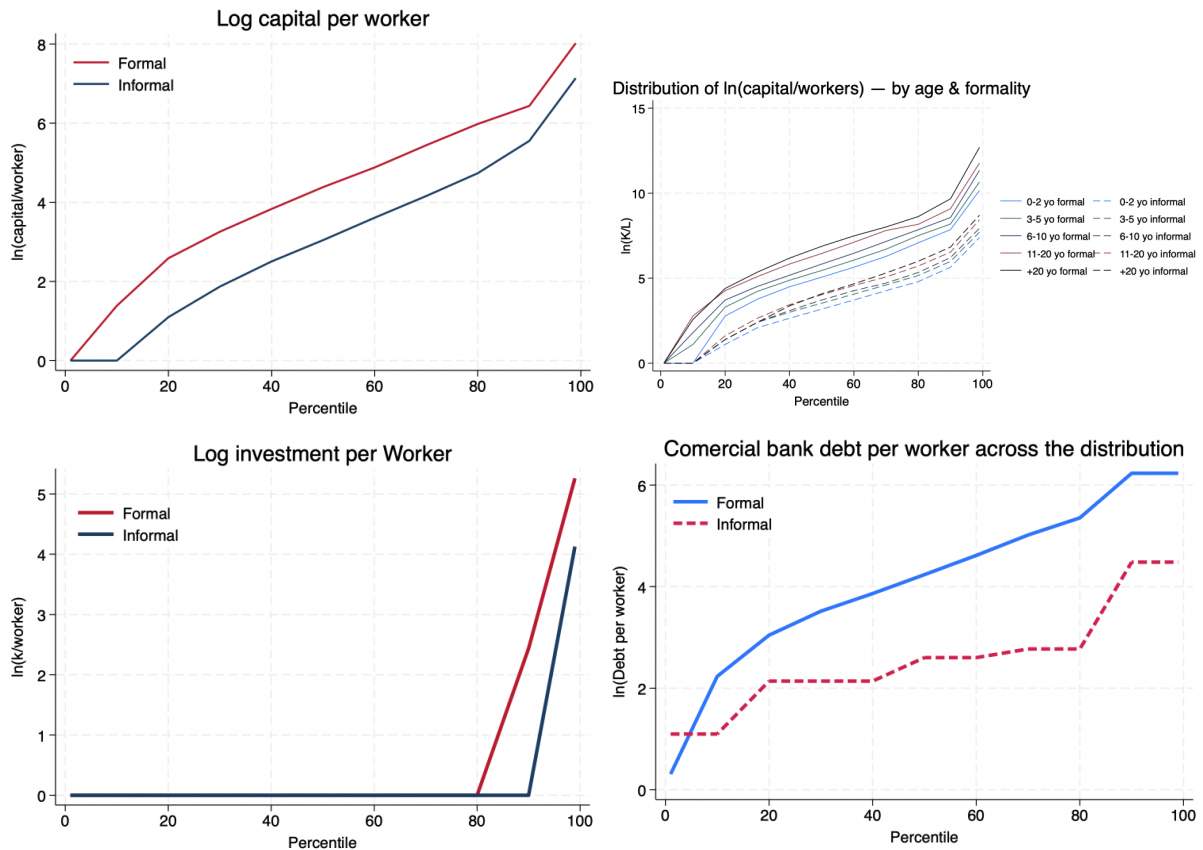
Share of borrowing firms by number of bank relationships in 2009, 2014, 2019, and 2024.

Figure 7: Bank concentration



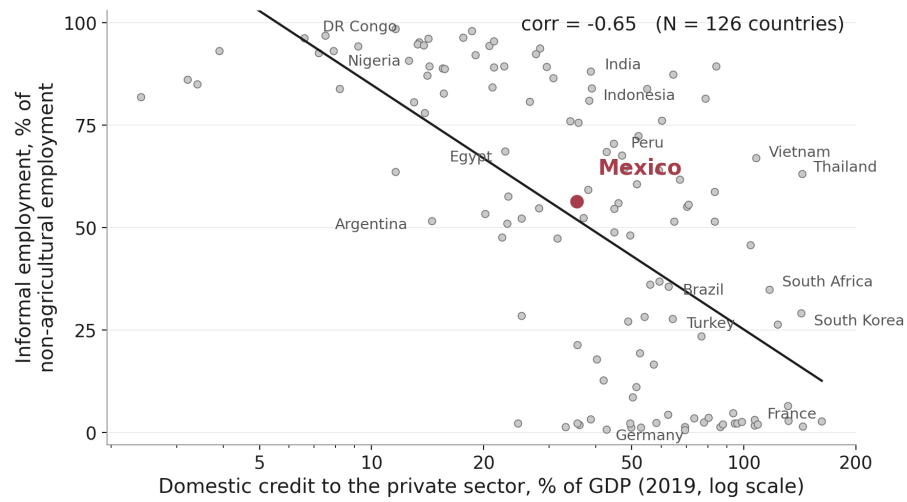
The left panel shows the five year lagged credit shares of the ten largest banks, the predetermined exposure shares of the instrument. The right panel shows the concentration index over time.

Figure 8: Capital, investment, and debt by formality



Census capital per worker by size and by age, census investment per worker, and survey bank debt per worker, formal against informal firms.

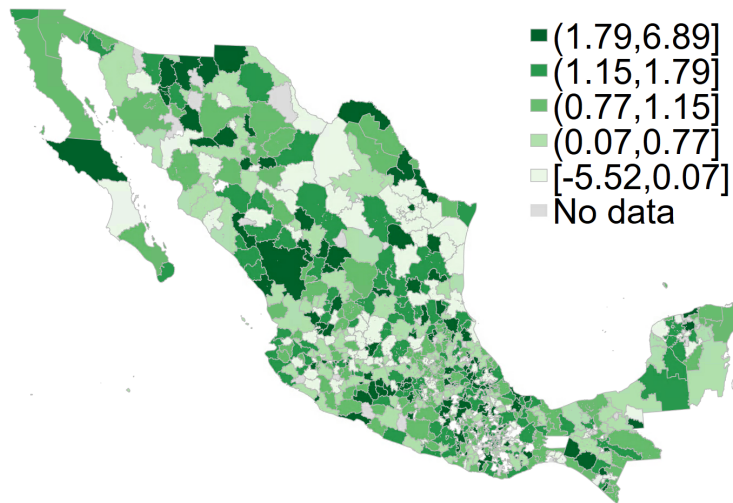
Figure 9: Informality and credit across countries



Sources: ILO SDG 8.3.1 via Our World in Data; World Bank WDI (2019).

D Map of credit growth

Figure 10: Five year credit growth across markets



Realized five year log change in bank credit by local labor market, the outcome side of the first stage.

E Balance test

Table 18: Balance test: Shift-Share instrument and predetermined market characteristics

<i>Predetermined variable</i>	Lagged 5 years		Lagged 10 years	
	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Market employment share, sector 1	0.040	(0.026)	0.038	(0.027)
Market employment share, sector 2	−0.007	(0.011)	−0.011	(0.012)
Bank-portfolio sector share, sector 1	−0.007	(0.031)	0.028	(0.053)
Bank-portfolio sector share, sector 2	0.010	(0.034)	−0.057	(0.061)
Credit-weighted interest rate	0.330	(0.258)	0.760	(0.809)
Log number of banks in market	−0.040	(0.072)	−0.072	(0.069)
Log number of credits in market	0.206	(0.312)	−0.038	(0.316)
Observations (range)	1,200–1,356		766–934	
Markets (range)	521–545		457–515	

Notes: Each cell reports the coefficient and market-clustered standard error from a separate regression of the indicated predetermined market characteristic on the standardized leave-one-market-out credit-supply instrument, conditional on year fixed effects and the coverage share interacted with year fixed effects, weighted by baseline market credit. Lags are relative to the baseline period, so the two panels measure each characteristic ten and fifteen years before the outcome wave. Sample restricted to market-years with instrument coverage ≥ 0.5 . None of the fourteen coefficients is statistically distinguishable from zero at the 10% level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F Pre-trend test

Table 19: Pre-trend test: main estimates with the lagged outcome change as a control

<i>Outcome (five-year log change)</i>	Main specification		With pretrend control	
	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Formal firms	0.051***	(0.013)	0.048***	(0.013)
Total firms	0.004	(0.005)	0.003	(0.005)
Informal firms	−0.002	(0.006)	−0.002	(0.006)
Workers (IMSS)	0.019***	(0.005)	0.018***	(0.006)
Total wage bill	0.023**	(0.009)	0.020**	(0.010)
Mean daily wage	−0.012***	(0.004)	−0.008**	(0.003)
Interest rate (pp)	0.422***	(0.102)	0.408***	(0.101)
Borrowing firms	0.161***	(0.035)	0.061***	(0.014)
Loans per borrower	0.202***	(0.034)	0.194***	(0.037)
Average VA	0.027*	(0.014)	−0.036*	(0.020)
Total VA	0.030*	(0.016)	−0.050**	(0.021)
VA formal	0.025	(0.031)	−0.114***	(0.030)
VA informal	−0.052**	(0.022)	−0.074***	(0.016)
Obs. (range)	1,382–1,390		929–979	
Markets (range)	552–556		525–541	

Notes: The pretrend columns re-estimate the main specification adding the dependent variable's own lagged five-year change as a control; the sample shrinks because the lagged change requires data two windows back. AKM standard errors in parentheses, with stars from the corresponding p -values; the main-specification columns repeat the corresponding entries of the results tables for comparison. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

G Reduced-form estimates

Table 20: Reduced-form: Shift-Share instrument and local labor market outcomes

<i>Dependent variable</i>	OLS on credit growth		Reduced form on $\hat{Z}$		<i>N</i>
	$\hat{\beta}$	SE	$\hat{\beta}$	SE	
Existing firms — Total	0.006	(0.006)	0.004	(0.012)	1,390
Existing firms — Formal	0.029**	(0.013)	0.056***	(0.016)	1,385
Existing firms — Informal	0.003	(0.006)	−0.002	(0.013)	1,390
Entry — Total	0.014	(0.011)	0.016	(0.022)	992
Entry — Formal	0.038**	(0.018)	0.075**	(0.030)	975
Entry — Informal	0.012	(0.012)	0.013	(0.023)	992
Exit — Total	0.035***	(0.007)	0.054***	(0.011)	992
Exit — Formal	0.028**	(0.012)	0.044*	(0.025)	961
Exit — Informal	0.039***	(0.008)	0.061***	(0.012)	992
Informal→Formal	0.039**	(0.018)	0.058**	(0.027)	953
Formal→Informal	−0.015	(0.014)	−0.044*	(0.026)	954
Formal→Formal	0.032**	(0.015)	0.045***	(0.016)	966
Informal→Informal	0.003	(0.007)	−0.007	(0.013)	992
Total	0.009	(0.021)	0.040	(0.037)	1,390
Formal	0.031	(0.026)	0.046	(0.031)	1,390
Informal	0.017	(0.024)	0.039	(0.051)	1,390
Workers (IMSS)	0.017**	(0.007)	0.021*	(0.013)	1,382
Total wage bill	0.014	(0.011)	0.026	(0.019)	1,382
Mean daily wage	−0.012***	(0.003)	−0.013*	(0.007)	1,382
Average VA	0.041***	(0.016)	0.030	(0.029)	1,390
Total VA	0.046***	(0.018)	0.033	(0.030)	1,390
VA formal	0.042	(0.027)	0.028	(0.047)	1,390
VA informal	0.003	(0.032)	−0.057	(0.063)	1,390
Borrowing firms	0.111**	(0.056)	0.178*	(0.099)	1,390
Loans per borrower	0.107***	(0.029)	0.223***	(0.039)	1,390
Credit per loan	0.782***	(0.066)	0.703***	(0.095)	1,390
Collateral ratio	−1.771***	(0.495)	−2.080**	(0.946)	816
Interest rate (pp)	−0.045	(0.124)	0.466***	(0.169)	1,390

Notes: Each row reports two separate regressions with the same controls, weights, and sample as the main specification: OLS of the outcome on the five-year log change in market credit, and the reduced form of the outcome on the standardized leave-one-market-out instrument. Standard errors clustered at the market (LLM) level. Because the model is exactly identified, each reduced-form coefficient equals the product of the corresponding first-stage and IV coefficients. Census worker counts (single-window subsample) are reported in Appendix B. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

H Excluding the three largest metropolitan areas

Table 21: Shift-Share IV estimates excluding the three largest metropolitan areas

<i>Dependent variable</i>	$\widehat{\Delta_5 \ln(C)}$	FS <i>F</i>	<i>N</i>
Existing firms — Total	−0.003 (0.006)	91.66	1,381
Existing firms — Formal	0.049*** (0.014)	91.65	1,376
Existing firms — Informal	−0.009 (0.007)	91.66	1,381
Entry — Total	0.018 (0.014)	119.35	986
Entry — Formal	0.067*** (0.020)	119.33	969
Entry — Informal	0.015 (0.015)	119.35	986
Exit — Total	0.041** (0.018)	119.35	986
Exit — Formal	0.067*** (0.018)	119.27	955
Exit — Informal	0.037** (0.018)	119.35	986
Informal→Formal	0.014 (0.020)	118.67	947
Formal→Informal	−0.048** (0.023)	119.25	948
Formal→Formal	0.055*** (0.009)	119.28	960
Informal→Informal	−0.025*** (0.006)	119.35	986
Total	0.113*** (0.021)	91.66	1,381
Formal	0.064*** (0.021)	91.66	1,381
Informal	0.142*** (0.025)	91.66	1,381
Workers (IMSS)	0.027*** (0.008)	91.66	1,373
Total wage bill	0.035*** (0.013)	91.66	1,373
Mean daily wage	0.001 (0.004)	91.66	1,373
Average VA	0.019 (0.020)	91.66	1,381
Total VA	0.015 (0.018)	91.66	1,381
VA formal	0.026 (0.037)	91.66	1,381
VA informal	−0.116*** (0.033)	91.66	1,381
Borrowing firms	0.157*** (0.031)	91.66	1,381
Loans per borrower	0.285*** (0.062)	91.66	1,381
Credit per loan	0.558*** (0.083)	91.66	1,381
Collateral ratio	−0.370* (0.218)	118.90	810
Interest rate (pp)	0.470*** (0.144)	91.66	1,381

Notes: Same specification and inference as the main tables (AKM standard errors in parentheses, stars from the corresponding *p*-values; FS *F* is the Kleibergen–Paap statistic), excluding Mexico City, Monterrey, and Guadalajara. These markets are nine of the 1,390 baseline market-year observations but carry 69 percent of the regression weight. Wild cluster bootstrap *p*-values are not recomputed for robustness specifications. * *p* < 0.10, ** *p* < 0.05, *** *p* < 0.01.

I Unweighted Shift-Share IV estimates

Table 22: Unweighted Shift-Share IV estimates

<i>Dependent variable</i>	$\widehat{\Delta_5 \ln(C)}$	AKM s.e.	First-stage F	N
Log existing firms — Total	0.022 (0.015)	[0.029]	34.86	1,390
Log existing firms — Formal	0.064* (0.039)	[0.080]	40.97	1,385
Log existing firms — Informal	0.021 (0.016)	[0.032]	34.86	1,390
Log firm entry — Total	0.067** (0.033)	[0.089]	26.04	992
Log firm entry — Formal	0.090 (0.060)	[0.167]	38.41	975
Log firm entry — Informal	0.066* (0.034)	[0.095]	26.04	992
Log firm exit — Total	0.017 (0.034)	[0.120]	26.04	992
Log firm exit — Formal	−0.048 (0.095)	[0.246]	30.07	961
Log firm exit — Informal	0.027 (0.035)	[0.143]	26.04	992
Formal–Formal	0.102 (0.065)	[0.248]	29.55	966
Informal–Formal	0.053 (0.060)	[0.327]	51.97	953
Formal–Informal	0.064 (0.065)	[0.450]	39.31	954
Informal–Informal	−0.006 (0.019)	[0.072]	26.04	992
Firms with credit — Total	0.148 (0.126)	[0.305]	34.86	1,390
Firms with credit — Formal	0.160* (0.095)	[0.183]	34.86	1,390
Firms with credit — Informal	0.099 (0.152)	[0.250]	34.86	1,390
Formal workers (IMSS)	−0.041 (0.056)	[0.171]	34.86	1,382
Formal wage bill (IMSS)	−0.026 (0.061)	[0.234]	34.86	1,382
Mean daily wage (IMSS)	0.028* (0.015)	[0.026]	34.86	1,382
Avg. value added	−0.137* (0.075)	[0.137]	34.86	1,390
Total value added	−0.096 (0.106)	[0.242]	34.86	1,390
Formal value added	0.558 (0.385)	[1.199]	34.86	1,390
Informal value added	−0.305*** (0.101)	[0.328]	34.86	1,390
Log borrowers	0.176** (0.079)	[0.299]	34.86	1,390
Log loans per borrower	0.041 (0.060)	[0.427]	34.86	1,390
Log credit per loan	0.783*** (0.077)	[0.396]	34.86	1,390
Log collateral	−2.265*** (0.604)	[1.961]	43.77	816
Interest rate (pp)	0.935** (0.425)	[2.561]	34.86	1,390

Notes: Two-stage least squares estimates of the main specification without the baseline-credit weights, so that each market-year contributes equally to the regression. All other specification details (controls, fixed effects, instrument, sample definition) follow Table 4, and the estimation samples are identical to the corresponding weighted specifications. Standard errors clustered at the market level in parentheses; exposure-robust standard errors following [Adão et al. \(2019\)](#) in brackets. Stars are based on the clustered standard errors, as in the main tables. Under exposure-robust inference, credit per loan is the only outcome that remains significant at the 5% level ($p = 0.048$). The credit-deepening decomposition again sums to one ($0.176 + 0.041 + 0.783$). First-stage F is the Kleibergen–Paap rk Wald statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

J Shift-Share IV Placebo test

Table 23: Placebo tests: Shift-Share IV on pre-period (five-year) outcome changes

<i>Pre-period outcome</i> ($\Delta_5 Y_{t-5}$, <i>log</i>)	$\widehat{\Delta_5 \ln(C)}$	AKM s.e.	First-stage F	N
Log existing firms — Total	0.016*** (0.006)	[0.005]***	346.22	979
Log existing firms — Formal	0.017 (0.013)	[0.008]**	346.15	975
Log existing firms — Informal	0.019** (0.009)	[0.006]***	346.22	979
Log firm entry — Total	0.030 (0.050)	[0.106]	2.59	512
Log firm entry — Formal	−0.065 (0.099)	[0.242]	2.59	506
Log firm entry — Informal	0.041 (0.050)	[0.094]	2.59	512
Log firm exit — Total	−0.038 (0.051)	[0.110]	2.59	512
Log firm exit — Formal	0.122 (0.116)	[0.294]	2.59	493
Log firm exit — Informal	−0.070 (0.063)	[0.102]	2.59	512
Formal–Formal	−0.015 (0.042)	[0.066]	2.26	486
Informal–Formal	−0.093 (0.111)	[0.171]	2.59	494
Formal–Informal	0.060 (0.077)	[0.169]	2.59	483
Informal–Informal	−0.040 (0.042)	[0.103]	2.59	512
Firms with credit — Total	0.078 (0.083)	[0.058]	346.22	979
Firms with credit — Formal	0.091 (0.058)	[0.037]**	346.22	979
Firms with credit — Informal	0.003 (0.159)	[0.121]	346.22	979
Formal workers (IMSS)	−0.011 (0.009)	[0.006]*	346.20	966
Formal wage bill (IMSS)	−0.025** (0.012)	[0.005]***	346.20	966
Mean daily wage (IMSS)	−0.015 (0.010)	[0.006]***	346.20	966
Avg. value added	−0.165*** (0.038)	[0.014]***	346.22	979
Total value added	−0.148*** (0.041)	[0.013]***	346.22	979
Formal value added	−0.175*** (0.063)	[0.020]***	346.22	979
Informal value added	−0.037 (0.037)	[0.019]*	346.22	979
Log borrowers	−0.471 (0.288)	[0.169]***	358.24	929
Log loans per borrower	−0.179*** (0.056)	[0.041]***	358.24	929
Log credit per loan	−0.711*** (0.102)	[0.071]***	358.24	929
Log collateral	4.851 (2.990)	[5.394]	2.60	400
Interest rate (pp)	−0.181 (0.114)	[0.069]***	358.24	929

Notes: Each row reports a placebo IV estimate in which the dependent variable is the five-year log change in the indicated outcome *before* the credit-supply shock window, $\Delta_5 Y_{t-5} = \ln Y_{t-5} - \ln Y_{t-10}$. The endogenous regressor, instrument, controls, baseline-credit weights, and sample condition follow the main specification (Table 4); the sample additionally requires a non-missing outcome ten years prior. Standard errors clustered at the market level in parentheses; exposure-robust standard errors following [Adão et al. \(2019\)](#) in brackets. Stars on coefficients are based on the clustered standard errors and stars on brackets on the exposure-robust ones. First-stage F is the Kleibergen–Paap rk Wald statistic on each restricted sample. The entry, exit, transition, and collateral outcomes require lagged flows that exist in a single census window, so their samples halve and the first stage collapses; we report these rows for transparency only. The census-based worker outcomes cannot be estimated in the pre-period at all and are omitted. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

K Conventional (Non-Leave-One-Out) IV estimates

Table 24: Shift-Share IV estimates using the conventional (Non-Leave-One-Out) instrument

<i>Dependent variable</i>	$\widehat{\Delta_5 \ln(C)}$	AKM s.e.	Boot. p	First-stage F	N
Log existing firms — Total	0.001 (0.010)	[0.005]	0.960	264.47	1,390
Log existing firms — Formal	0.036** (0.013)	[0.008]***	0.067	264.42	1,385
Log existing firms — Informal	−0.002 (0.012)	[0.006]	0.901	264.47	1,390
Log firm entry — Total	0.007 (0.019)	[0.009]	0.781	272.77	992
Log firm entry — Formal	0.043* (0.024)	[0.010]***	0.120	272.58	975
Log firm entry — Informal	0.006 (0.020)	[0.010]	0.808	272.77	992
Log firm exit — Total	0.049*** (0.010)	[0.009]***	0.002	272.77	992
Log firm exit — Formal	0.023 (0.021)	[0.018]	0.442	272.58	961
Log firm exit — Informal	0.060*** (0.011)	[0.008]***	0.000	272.77	992
Formal–Formal	0.027** (0.014)	[0.004]***	0.145	272.52	966
Informal–Formal	0.050** (0.024)	[0.020]**	0.102	271.75	953
Formal–Informal	−0.044* (0.025)	[0.013]***	0.279	272.53	954
Informal–Informal	−0.003 (0.012)	[0.003]	0.865	272.77	992
Firms with credit — Total	0.025 (0.029)	[0.007]***	0.568	264.47	1,390
Firms with credit — Formal	0.018 (0.026)	[0.011]*	0.594	264.47	1,390
Firms with credit — Informal	0.031 (0.039)	[0.009]***	0.617	264.47	1,390
Formal workers (IMSS)	0.015 (0.010)	[0.004]***	0.219	264.44	1,382
Formal wage bill (IMSS)	0.014 (0.015)	[0.007]**	0.523	264.44	1,382
Mean daily wage (IMSS)	−0.010 (0.006)	[0.003]***	0.269	264.44	1,382
Avg. value added	0.014 (0.025)	[0.016]	0.645	264.47	1,390
Total value added	0.014 (0.025)	[0.016]	0.613	264.47	1,390
Formal value added	−0.004 (0.036)	[0.028]	0.921	264.47	1,390
Informal value added	−0.030 (0.050)	[0.012]**	0.684	264.47	1,390
Log borrowers	0.101 (0.074)	[0.028]***	0.379	264.47	1,390
Log loans per borrower	0.220*** (0.030)	[0.024]***	0.000	264.47	1,390
Log credit per loan	0.679*** (0.066)	[0.025]***	0.000	264.47	1,390
Log collateral	−2.441*** (0.829)	[0.432]***	0.085	271.17	816
Interest rate (pp)	0.330** (0.143)	[0.064]***	0.043	264.47	1,390

Notes: Two-stage least squares estimates using the conventional (non-leave-one-out) shift-share instrument, in which the bank shocks are estimated on the full sample including each market's own loans, standardized to unit variance. Specification, controls, baseline-credit weights, and sample condition follow Table 4; the estimation samples are identical to the corresponding leave-one-out regressions. Standard errors clustered at the market level in parentheses, with stars on coefficients based on them; exposure-robust standard errors following [Adão et al. \(2019\)](#) in brackets, with stars on brackets based on exposure-robust p -values; wild cluster bootstrap p -values in the third column. The credit-deepening decomposition sums to one (0.101 + 0.220 + 0.679). First-stage F is the Kleibergen–Paap rk Wald statistic; the tighter first stage relative to the leave-one-out design reflects the mechanical own-market overlap in the conventional shocks. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

L Shift-Share IV results at the bank level

Table 25: Bank level IV estimates: Khwaja-Mian-style identification at the bank-year

<i>Dependent variable</i>	$\widehat{\Delta_5 \ln(C)}$	Boot. p	First-stage F	N
Log existing firms — Total	−0.004 (0.005)	0.562	34.91	131
Log existing firms — Formal	0.036*** (0.008)	0.000	34.91	131
Log existing firms — Informal	−0.009* (0.005)	0.186	34.91	131
Log firm entry — Total	−0.001 (0.008)	0.962	30.40	105
Log firm entry — Formal	0.045*** (0.011)	0.000	30.40	105
Log firm entry — Informal	−0.004 (0.009)	0.716	30.40	105
Log firm exit — Total	0.058*** (0.011)	0.001	30.40	105
Log firm exit — Formal	0.020 (0.016)	0.326	30.40	105
Log firm exit — Informal	0.070*** (0.010)	0.001	30.40	105
Formal–Formal	0.021*** (0.008)	0.040	30.40	105
Informal–Formal	0.068*** (0.013)	–	30.40	105
Formal–Informal	−0.077*** (0.015)	0.003	30.40	105
Informal–Informal	−0.003 (0.004)	0.704	30.40	105
Firms with credit — Total	−0.003 (0.011)	0.839	34.91	131
Firms with credit — Formal	0.003 (0.016)	0.885	34.91	131
Firms with credit — Informal	0.017 (0.010)	0.164	34.91	131
Formal workers (IMSS)	0.009 (0.007)	0.348	34.91	131
Formal wage bill (IMSS)	0.008 (0.012)	0.654	34.91	131
Mean daily wage (IMSS)	−0.015*** (0.004)	0.001	34.91	131
Avg. value added	0.022 (0.015)	0.261	34.91	131
Total value added	0.018 (0.017)	0.450	34.91	131
Formal value added	−0.006 (0.037)	0.906	34.91	131
Informal value added	−0.013 (0.017)	0.468	34.91	131
Log borrowers	0.083** (0.032)	0.046	34.91	131
Log loans per borrower	0.209*** (0.046)	0.002	34.91	131
Log credit per loan	0.716*** (0.035)	0.000	34.91	131
Log collateral	−2.417*** (0.331)	0.000	30.40	66
Interest rate (pp)	0.299*** (0.091)	0.024	37.14	151

Notes: Two-stage least squares estimates at the bank–census–year level. The unit of observation is a bank-year; market outcomes are aggregated to the bank level using the bank’s baseline market exposure shares. The endogenous regressor is the bank-level aggregate of five-year market credit growth, instrumented by the bank’s credit-supply shock identified from market-sector cell fixed effects following Degryse et al. (2019), an extension of Khwaja and Mian (2008). Controls are census-year fixed effects and the bank-level analogs of the two sector-share controls; estimation is weighted by the bank’s baseline credit. The sample keeps bank-years in which the aggregated outcome covers at least half of the bank’s baseline credit, so the sample varies across outcomes. Standard errors clustered at the bank level in parentheses, with stars based on them; wild cluster bootstrap p -values in the second column where computed. The credit-deepening decomposition sums to approximately one (0.083 + 0.209 + 0.716). First-stage F is the Kleibergen–Paap rk Wald statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

M Shift-Share IV with different coverage thresholds

Table 26: Sensitivity to instrument coverage threshold

<i>Dependent variable</i>	Coverage ≥ 0				Coverage ≥ 0.8			
	$\hat{\beta}$ (s.e.)	F	N	M	$\hat{\beta}$ (s.e.)	F	N	M
Log existing firms — Total	−0.000 (0.007)	124.21	1,520	568	0.003 (0.010)	284.48	1,215	541
Log existing firms — Formal	0.042*** (0.013)	124.22	1,514	564	0.045*** (0.014)	284.42	1,210	537
Log existing firms — Informal	−0.005 (0.008)	124.21	1,520	568	−0.001 (0.012)	284.48	1,215	541
Log firm entry — Total	0.009 (0.013)	121.56	1,092	568	0.012 (0.018)	321.21	859	525
Log firm entry — Formal	0.058*** (0.020)	121.54	1,071	555	0.058** (0.023)	320.85	843	514
Log firm entry — Informal	0.006 (0.013)	121.56	1,091	568	0.010 (0.019)	321.21	859	525
Log firm exit — Total	0.040*** (0.012)	121.56	1,092	568	0.047*** (0.011)	321.21	859	525
Log firm exit — Formal	0.017 (0.022)	121.69	1,049	549	0.032* (0.018)	321.03	831	508
Log firm exit — Informal	0.047*** (0.012)	121.56	1,092	568	0.056*** (0.012)	321.21	859	525
Formal–Formal	0.018 (0.014)	122.28	1,051	553	0.034*** (0.008)	320.91	834	511
Informal–Formal	0.058*** (0.022)	120.08	1,042	540	0.051** (0.023)	319.38	824	498
Formal–Informal	−0.060*** (0.019)	121.73	1,040	541	−0.034 (0.024)	320.90	826	503
Informal–Informal	−0.008 (0.008)	121.56	1,091	568	−0.005 (0.010)	321.21	859	525
<i>Panel C: Firms with credit</i>								
Firms with credit — Total	0.016 (0.024)	124.21	1,520	568	0.026 (0.019)	284.48	1,215	541
Firms with credit — Formal	0.029 (0.022)	124.21	1,520	568	0.031 (0.021)	284.48	1,215	541
Firms with credit — Informal	0.012 (0.033)	124.21	1,520	568	0.024 (0.025)	284.48	1,215	541
Formal workers (IMSS)	0.005 (0.011)	124.20	1,511	563	0.015** (0.007)	284.44	1,208	538
Formal wage bill (IMSS)	0.009 (0.014)	124.20	1,511	563	0.016 (0.010)	284.44	1,208	538
Mean daily wage (IMSS)	−0.018*** (0.007)	124.20	1,511	563	−0.011* (0.006)	284.44	1,208	538
Avg. value added	0.028 (0.021)	124.21	1,520	568	0.022 (0.022)	284.48	1,215	541
Total value added	0.027 (0.021)	124.21	1,520	568	0.025 (0.022)	284.48	1,215	541
Formal value added	0.006 (0.036)	124.21	1,520	568	0.011 (0.028)	284.48	1,215	541
Informal value added	−0.019 (0.040)	124.21	1,520	568	−0.033 (0.036)	284.48	1,215	541
Log borrowers	0.110* (0.060)	124.21	1,520	568	0.135** (0.059)	284.48	1,215	541
Log loans per borrower	0.165*** (0.035)	124.21	1,520	568	0.216*** (0.032)	284.48	1,215	541
Log credit per loan	0.725*** (0.058)	124.21	1,520	568	0.649*** (0.052)	284.48	1,215	541
Log collateral	−1.793*** (0.592)	124.59	868	464	−2.167*** (0.535)	319.97	713	433
Interest rate (pp)	0.434*** (0.144)	124.21	1,520	568	0.423*** (0.142)	284.48	1,215	541

Notes: Two-stage least squares estimates re-estimating the main specification at alternative thresholds on instrument coverage. The left block includes all market-years with a non-missing instrument ($\text{cov_z_5} \geq 0$); the right block restricts to market-years for which the leave-one-market-out shift-share instrument pools at least 80% of the local base-period banking exposure ($\text{cov_z_5} \geq 0.8$). The headline specification in Tables 4–9 uses the intermediate threshold $\text{cov_z_5} \geq 0.5$; all other specification details are unchanged. F denotes the first-stage Kleibergen–Paap rk Wald statistic; N denotes observations; M denotes markets. Standard errors clustered at the market level in parentheses, with stars based on them; exposure-robust standard errors following [Adão et al. \(2019\)](#) are similar or smaller in these samples and deliver the same or sharper significance, so the clustered inference reported here is the more conservative choice. The credit-deepening decomposition sums to one in both blocks ($0.110 + 0.165 + 0.725$ and $0.135 + 0.216 + 0.649$). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

N Shift-Share IV with instrument renormalization

Table 27: Robustness to instrument renormalization

<i>Dependent variable</i>	$\widehat{\Delta_5 \ln(C)}$	AKM s.e.	First-stage F	N
Log existing firms — Total	0.004 (0.010)	[0.006]	335.43	1,390
Log existing firms — Formal	0.052*** (0.016)	[0.016]***	335.36	1,385
Log existing firms — Informal	−0.002 (0.011)	[0.007]	335.43	1,390
Log firm entry — Total	0.016 (0.020)	[0.011]	372.47	992
Log firm entry — Formal	0.070** (0.028)	[0.022]***	372.13	975
Log firm entry — Informal	0.013 (0.020)	[0.012]	372.47	992
Log firm exit — Total	0.049*** (0.010)	[0.014]***	372.47	992
Log firm exit — Formal	0.042* (0.023)	[0.018]**	372.27	961
Log firm exit — Informal	0.054*** (0.011)	[0.014]***	372.47	992
Formal–Formal	0.042*** (0.016)	[0.011]***	372.15	966
Informal–Formal	0.049* (0.026)	[0.022]**	371.02	953
Formal–Informal	−0.039* (0.023)	[0.018]**	372.12	954
Informal–Informal	−0.008 (0.011)	[0.005]	372.47	992
Firms with credit — Total	0.042 (0.036)	[0.014]***	335.43	1,390
Firms with credit — Formal	0.045 (0.029)	[0.023]**	335.43	1,390
Firms with credit — Informal	0.043 (0.049)	[0.017]**	335.43	1,390
Formal workers (IMSS)	0.021* (0.012)	[0.007]***	335.38	1,382
Formal wage bill (IMSS)	0.026 (0.018)	[0.012]**	335.38	1,382
Mean daily wage (IMSS)	−0.012* (0.006)	[0.005]**	335.38	1,382
Avg. value added	0.030 (0.027)	[0.017]*	335.43	1,390
Total value added	0.033 (0.028)	[0.020]*	335.43	1,390
Formal value added	0.030 (0.044)	[0.038]	335.43	1,390
Informal value added	−0.060 (0.061)	[0.028]**	335.43	1,390
Log borrowers	0.174* (0.095)	[0.043]***	335.43	1,390
Log loans per borrower	0.203*** (0.031)	[0.043]***	335.43	1,390
Log credit per loan	0.623*** (0.087)	[0.055]***	335.43	1,390
Log collateral	−1.734** (0.876)	[0.413]***	371.35	816
Interest rate (pp)	0.441*** (0.162)	[0.127]***	335.43	1,390

Notes: Two-stage least squares estimates re-estimating the main specification with the renormalized shift-share instrument $\tilde{z}_{mt} = z_{mt}/\text{cov_z_5}_{mt}$, which converts the instrument from a coverage-weighted sum of bank shocks into a coverage-weighted average, standardized as in the main specification. The estimation sample, controls, weights, and inference are identical to Table 4. Standard errors clustered at the market level in parentheses, with stars based on them; exposure-robust standard errors following [Adão et al. \(2019\)](#) in brackets, with stars on brackets based on exposure-robust p -values. The credit-deepening decomposition sums to one (0.174 + 0.203 + 0.623). First-stage F is the Kleibergen–Paap rk Wald statistic. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

O Shift-Share IV estimates excluding the biggest bank

Table 28: Shift-Share IV estimates excluding the biggest bank

<i>Dependent variable</i>	$\widehat{\Delta_5 \ln(C)}$	AKM s.e.	Boot. p	First-stage F	N
Log existing firms — Total	0.000 (0.011)	[0.006]	0.993	32.17	1,390
Log existing firms — Formal	0.048* (0.026)	[0.010]***	0.173	32.17	1,385
Log existing firms — Informal	−0.006 (0.013)	[0.008]	0.741	32.17	1,390
Log firm entry — Total	0.004 (0.021)	[0.011]	0.871	31.54	992
Log firm entry — Formal	0.058 (0.038)	[0.012]***	0.248	31.54	975
Log firm entry — Informal	0.001 (0.022)	[0.013]	0.960	31.54	992
Log firm exit — Total	0.046*** (0.014)	[0.016]***	0.015	31.54	992
Log firm exit — Formal	−0.000 (0.026)	[0.017]	0.989	31.53	961
Log firm exit — Informal	0.061*** (0.015)	[0.016]***	0.007	31.54	992
Formal–Formal	0.030* (0.017)	[0.009]***	0.165	31.53	966
Informal–Formal	0.093** (0.042)	[0.021]***	0.116	31.73	953
Formal–Informal	−0.078** (0.038)	[0.025]***	0.185	31.53	954
Informal–Informal	0.000 (0.013)	[0.004]	1.000	31.54	992
Firms with credit — Total	0.000 (0.033)	[0.013]	0.993	32.17	1,390
Firms with credit — Formal	0.005 (0.038)	[0.021]	0.908	32.17	1,390
Firms with credit — Informal	0.018 (0.037)	[0.017]	0.718	32.17	1,390
Formal workers (IMSS)	0.013 (0.011)	[0.007]*	0.297	32.17	1,382
Formal wage bill (IMSS)	0.008 (0.016)	[0.012]	0.633	32.17	1,382
Mean daily wage (IMSS)	−0.010* (0.006)	[0.005]*	0.158	32.17	1,382
Avg. value added	−0.004 (0.030)	[0.021]	0.901	32.17	1,390
Total value added	−0.005 (0.029)	[0.025]	0.854	32.17	1,390
Formal value added	−0.039 (0.041)	[0.045]	0.404	32.17	1,390
Informal value added	−0.031 (0.050)	[0.024]	0.586	32.17	1,390
Log borrowers	0.093 (0.069)	[0.032]***	0.266	32.17	1,390
Log loans per borrower	0.246*** (0.045)	[0.056]***	0.001	32.17	1,390
Log credit per loan	0.660*** (0.073)	[0.048]***	0.004	32.17	1,390
Log collateral	−2.855*** (0.810)	[0.275]***	0.029	31.45	816
Interest rate (pp)	0.335** (0.164)	[0.118]***	0.055	32.17	1,390

Notes: Two-stage least squares estimates re-estimating the main specification with the shift-share instrument reconstructed to exclude the bank holding the largest share of aggregate baseline credit: both its shocks and its exposure shares are removed, and the resulting instrument is standardized as in the main specification. The estimation sample, controls, weights, and sample condition are identical to Table 4. Standard errors clustered at the market level in parentheses, with stars based on them; exposure-robust standard errors following [Adão et al. \(2019\)](#) in brackets, with stars on brackets based on exposure-robust p -values; wild cluster bootstrap p -values in the third column. The credit-deepening decomposition sums to one (0.093 + 0.246 + 0.660). First-stage F is the Kleibergen–Paap rk Wald statistic; its decline relative to the main specification reflects the excluded bank’s substantial contribution to the identifying variation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

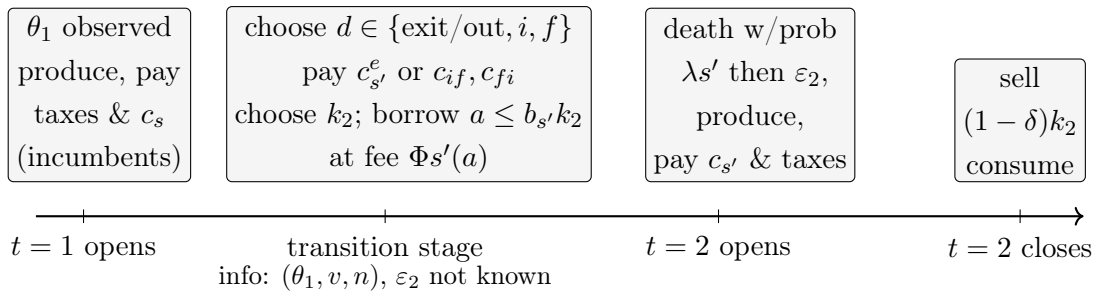
P The two-period model: derivations and additional results

This appendix collects the formal material behind the two-period model of the main text: the timing, the static and dynamic value functions and their envelope properties, the cost ordering behind the threshold structure and the cash-rich region, the master threshold and its four instances, the transition semi-elasticities, and the closed form of the model-implied estimand. Statements keep the notation of the main text, and proofs are in Appendix Q.

P.1 Timing

Figure 11 summarizes the timing, which mirrors the recursive model: sector choices for period $t+1$ are made at the end of period t , before the period- $(t+1)$ innovation ε_{t+1} is observed. In period 1, incumbents observe (θ_1, v) , hire labor at wage w , produce, settle taxes, and pay the fixed cost c_s . Then each incumbent chooses $d \in \{\text{exit}, i, f\}$ and, if continuing in sector s' , next-period capital k_2 , borrowing $a = \max\{0, k_2 - n\}$ at fee $\Phi s'(a)$ subject to $a \leq b_{s'} k_2$, where n denotes internal funds at this stage (defined in (10) below). Potential entrants, knowing (v, k_0) but not ε_2 , choose $d \in \{\text{stay out}, i, f\}$, pay $c_{s'}^e$ if entering, and choose k_2 under the same financial technology. Dividends (possibly negative, interpreted as the owner's outside funds absorbed at the financing fee's shadow price) are consumed. Finally, in period 2, death shocks realize (probability $\lambda_{s'}$). A destroyed firm's owner keeps $(1 - \delta)k_2$. Surviving firms observe θ_2 , hire labor, produce, pay taxes and the fixed cost $c_{s'}$, then sell undepreciated capital $(1 - \delta)k_2$. The world ends.

Figure 11: Timing of the two-period model.



P.2 Static problem: labor demand and the profit function

Let τ_y^s and τ_ℓ^s denote the taxes effectively paid in sector s : $\tau_y^f = \tau_y$, $\tau_\ell^f = \tau_\ell$, $\tau_y^i = \tau_\ell^i = 0$. Conditional on operating with state (θ, k) , the firm solves $\max_{\ell \geq 0} (1 - \tau_y^s) \theta k^\alpha \ell^\nu - (1 + \tau_\ell^s) w \ell$; substituting the optimal ℓ back into the objective yields the operating-profit function

$$\pi_s(\theta, k) = (1 - \nu) (1 - \tau_y^s) \theta k^\alpha \left[\frac{\nu (1 - \tau_y^s) \theta k^\alpha}{(1 + \tau_\ell^s) w} \right]^{\frac{\nu}{1-\nu}},$$

which the next lemma rewrites in the separable form used throughout.

Lemma P.1 (Labor demand and profits). *Optimal labor and the operating-profit function are*

$$\ell_s^*(\theta, k) = \left[\frac{\nu (1 - \tau_y^s) \theta k^\alpha}{(1 + \tau_\ell^s) w} \right]^{\frac{1}{1-\nu}}, \quad \pi_s(\theta, k) = \Omega_s x k^\gamma, \quad (32)$$

where

$$x \equiv \theta^{\frac{1}{1-\nu}}, \quad \gamma \equiv \frac{\alpha}{1-\nu} \in (0, 1), \quad \Omega_s \equiv (1 - \nu) (1 - \tau_y^s)^{\frac{1}{1-\nu}} \left[\frac{\nu}{(1 + \tau_\ell^s) w} \right]^{\frac{\nu}{1-\nu}}. \quad (33)$$

Labor demand is strictly increasing in θ and k ($\partial \ell^* / \partial \theta > 0$, $\partial \ell^* / \partial k > 0$) and strictly decreasing in the wage and in own-sector taxes ($\partial \ell^* / \partial w < 0$, $\partial \ell_f^* / \partial \tau_y < 0$, $\partial \ell_f^* / \partial \tau_\ell < 0$). The profit function is strictly increasing and strictly concave in k ($\gamma < 1$), and the tax wedge

$$T \equiv \frac{\Omega_f}{\Omega_i} = (1 - \tau_y)^{\frac{1}{1-\nu}} (1 + \tau_\ell)^{-\frac{\nu}{1-\nu}} < 1 \quad (34)$$

summarizes the fiscal cost of formality per unit of the profitability index x .

All proofs are collected in appendix Section Q. The index $x = \theta^{1/(1-\nu)}$ is the natural unit of firm heterogeneity: profits are linear in x , so all expectations below reduce to means of a log-normal random variable. Conditional on the transition-stage information set, define the expected profitability index as

$$\hat{x} \equiv \mathbb{E}[x_2 \mid \mathcal{I}_1] = \exp \left\{ \frac{\rho \ln \theta_1 + (1 - \rho) \ln v}{1 - \nu} + \frac{\sigma_\varepsilon^2}{2(1 - \nu)^2} \right\} \quad (\text{incumbents}), \quad (35)$$

and $\hat{x}_e(v) = \exp \{ \ln v / (1 - \nu) + \sigma_\varepsilon^2 / (2(1 - \nu)^2) \}$ for entrants (whose period-2 productivity is $\ln \theta_2 = \ln v + \varepsilon_2$). $\hat{x}$ is strictly increasing in θ_1 and in v ; it is the single sufficient statistic for expected productivity in every decision below.

P.3 Investment and financing: formal statements

Lemma P.2 (Capital and borrowing policy). *The solution to (11) is*

$$k_s^*(\hat{x}, n) = \begin{cases} \bar{k}_s(\hat{x}), & n \geq \bar{k}_s(\hat{x}) & (R1: \text{unconstrained}) \\ n, & \tilde{k}_s(\hat{x}) \leq n < \bar{k}_s(\hat{x}) & (R2: \text{self-financed at the kink}) \\ \tilde{k}_s(\hat{x}), & (1 - b_s) \tilde{k}_s(\hat{x}) \leq n < \tilde{k}_s(\hat{x}) & (R3: \text{interior borrowing}) \\ \frac{n}{1 - b_s}, & 0 \leq n < (1 - b_s) \tilde{k}_s(\hat{x}) & (R4: \text{collateral-constrained}) \end{cases} \quad (36)$$

with borrowing $a_s^* = \max\{0, k_s^* - n\}$, strictly positive exactly in R3-R4. This policy is continuous, weakly increasing in $\hat{x}$ and in n , and satisfies the following derivatives

$$\begin{aligned} \frac{\partial k_s^*}{\partial \phi_s} &= -\frac{\tilde{k}_s}{(1-\gamma)R_s} < 0 \text{ in } R3, = 0 \text{ ow}; & \frac{\partial k_s^*}{\partial b_s} &= \frac{n}{(1-b_s)^2} > 0 \text{ in } R4, = 0 \text{ ow}; \\ \frac{\partial k_s^*}{\partial n} &= 1 \text{ in } R2, \frac{1}{1-b_s} > 1 \text{ in } R4, = 0 \text{ ow}; & \frac{\partial k_s^*}{\partial \hat{x}} &> 0 \text{ in } R1, R3, = 0 \text{ ow}. \end{aligned} \quad (37)$$

Three observations connect Lemma P.2 to the data. First, investment-cash-flow sensitivity ($\partial k^*/\partial n > 0$) is positive only for constrained firms (R2, R4), and is largest exactly where the collateral constraint binds (R4). This is the firm-level footprint of the credit channel that the Shift-Share design picks up at the market level. Second, the marginal cost of external funds ϕ_s acts on the intensive margin of constrained borrowers (R3), while collateral capacity b_s acts on the most severely constrained firms (R4), typically young, low- n entrants. Therefore, the two instruments of a credit expansion reach different parts of the firm distribution. Third, borrowing incidence $\mathbb{P}(a^* > 0)$ rises with expected productivity and falls with internal funds. The Remark P.4 below turns this margin into the identification argument for the fixed cost of borrowing in Section 9. Finally, notice that this four-region finance-restriction policy is a generalization of the hierarchy of finance with three endogenous regimes of Bond and Meghir (1994).

Lemma P.3 (Value of operating). Define $\Psi_s(\hat{x}; R) \equiv (1-\gamma)\gamma^{\frac{\gamma}{1-\gamma}}(M_s\hat{x})^{\frac{1}{1-\gamma}}R^{-\frac{\gamma}{1-\gamma}}$, which is strictly increasing and strictly convex in $\hat{x}$ and strictly decreasing in R . Then

$$G_s(\hat{x}, n) = \begin{cases} \Psi_s(\hat{x}; R^0), & R1 \\ M_s\hat{x}n^\gamma - R^0n, & R2 \\ \phi_s n + \Psi_s(\hat{x}; R_s), & R3 \\ \phi_s n + M_s\hat{x}\left(\frac{n}{1-b_s}\right)^\gamma - R_s\frac{n}{1-b_s}, & R4, \end{cases} \quad (38)$$

where G_s is continuous, strictly increasing and strictly convex in $\hat{x}$, and concave and non-decreasing in n . Therefore, the marginal value of internal funds, $q_s \equiv 1 + \partial G_s/\partial n$, satisfies

$$q_s = \begin{cases} 1, & R1 \\ 1 + \gamma M_s\hat{x}n^{\gamma-1} - R^0 \in (1, 1 + \phi_s], & R2 \\ 1 + \phi_s, & R3 \\ 1 + \phi_s + \frac{\gamma M_s\hat{x}k_s^*\gamma^{-1} - R_s}{1-b_s} > 1 + \phi_s, & R4, \end{cases} \quad (39)$$

The step function q_s is the shadow price of a peso inside the firm. It exceeds one for constrained firms and jumps above $1 + \phi_s$ when collateral binds; it is the model object behind the role of entrant wealth $(\mu_{k_0}, \sigma_{k_0})$ in the quantitative model.

Remark P.4 (Borrowing participation and the fixed cost of borrowing). *In the general model the financing fee has an additional fixed component, $\Phi_s(a) = \phi_{0s}\mathbf{1}_{a>0} + \phi_s a$. However, all results in this section extend directly because conditional on borrowing, the fixed cost is sunk. Thus, all policies and envelope derivatives are unchanged, and thresholds are unchanged up to the bookkeeping $m \rightarrow m + \phi_{0s}$ and $C_o \rightarrow C_o + \phi_{0f} - \phi_{0i}$ on branches whose continuation choice is to borrow. Define the borrowing participation premium as the surplus gain from capital expansion using external funds over staying at the kink in capital without external funding,*

$$\pi_s^B(\hat{x}, n) \equiv G_s^{R3/R4}(\hat{x}, n) - [M_s \hat{x} n^\gamma - R^0 n],$$

which equals zero at the R2/R3 boundary and is strictly increasing in $\hat{x}$ beyond it (its $\hat{x}$ -derivative is $M_s(k_s^{}^\gamma - n^\gamma) > 0$ whenever $k_s^* > n$). With $\phi_{0s} > 0$ the firm borrows iff $\pi_s^B \geq \phi_{0s}$, so the R2/R3 boundary shifts to the unique root of $\pi_s^B = \phi_{0s}$, which is a borrowing-participation cutoff in $\hat{x}$ given n . The population mass beyond the cutoff, i.e. the borrowing incidence, is strictly decreasing in ϕ_{0s} given (ϕ_s, b_s) , which is the moment that identifies the fixed cost in Section 9; the intensive-margin objects (a^* , the thresholds, their derivatives) in the toy model are unchanged in the quantitative model, conditional on participation.*

The next lemma is the analytical workhorse of the paper. By the envelope theorem, the value of choosing a sector responds to i) its financial conditions in proportion to equilibrium borrowing and ii) to collateral capacity in proportion to the shadow value of pledgeability.

Lemma P.5 (Value of financial conditions). *Since G_s , and hence U_s , are differentiable with respect to (ϕ_s, b_s) , except at region boundaries, we can derive the following derivatives*

$$\frac{\partial U_s}{\partial \phi_s} = -a_s^*(\hat{x}, n) \leq 0, \quad \frac{\partial U_s}{\partial b_s} = \mu_s k_s^* = \mathbf{1}_{R4} \left[\gamma M_s \hat{x} k_s^{*\gamma-1} - r_s \right] \frac{n}{(1-b_s)^2} \geq 0, \quad (40)$$

where $\mu_s \geq 0$ is the multiplier on (9). Notice that both inequalities are strict exactly when the firm borrows (respectively, when the collateral constraint binds). Finally, observe that the cross-derivatives are zero: $\partial U_s / \partial \phi_{s'} = \partial U_s / \partial b_{s'} = 0$ for $s' \neq s$.

P.4 Cost ordering and the cash-rich region

Notice that conditions (A2-i)-(A2-ii) are exactly equivalent (by Lemma P.6 below) to the cost ordering $C_{f\text{-inc}} < C_{\text{entrant}} < C_{i\text{-inc}}$ that places the entrant formality threshold between the two incumbent thresholds. (A2-i) says that enter as formal is cheaper than enter as informal and immediately formalize; (A2-ii) says that entering informally is cheaper than entering formally and immediately switching to formal. (A2-ii) holds automatically whenever $c_{fi} \geq c_i^e$, and in particular whenever informal entry is (nearly) free, the empirically relevant case.

Lemma P.6 (Cost ordering). *Evaluate (16) at each origin's fee pair $(\kappa_{of}, \kappa_{oi})$ ²⁶ and define $K \equiv \beta[(1 - \lambda f)c_f - (1 - \lambda i)c_i]$. Then the costs are*

$$C_{i-inc} = (1 + \phi_f)c_{if} + K, \quad C_{f-inc} = -(1 + \phi_i)c_{fi} + K, \quad C_e = (1 + \phi_f)c_f^e - (1 + \phi_i)c_i^e + K, \quad (41)$$

and $C_e < C_{i-inc} \iff (\text{A2-i})$, while $C_{f-inc} < C_e \iff (\text{A2-ii})$. Under $c_{if}, c_{fi} > 0$ alone, $C_{f-inc} < C_{i-inc}$ always holds; (A2-i)–(A2-ii) are what places the entrant threshold strictly inside these two costs.

Remark P.7 (Cash-rich firms and the role of the audit margin). *For unconstrained firms ($R1 \times R1$) the finance advantage is irrelevant and $\text{sgn}\{\Delta_o\} = \text{sgn}\{\Psi_f(\hat{x}; R^0) - \Psi_i(\hat{x}; R^0) - \text{costs}\}$, which under $T_\lambda < 1$ is negative for large $\hat{x}$. This means that with taxes as the only non-financial difference, a sufficiently wealthy firm prefers informality. Two observations discipline this artifact. First, with $\lambda i > \lambda f$ large enough, $T_\lambda \geq 1$ and formality remains absorbing even for the cash-rich. Second, the general model restores a size-dependent cost of informality on the labor margin (the audit function $\tau_i(\ell)$ of Section 8), which caps informal scale and makes formality optimal for large firms independently of finance. The sizable mass of formal-to-informal transitions in the data, on average 286 per market and window against 310 formalizations (Section 6), suggests that for many firms the net non-financial advantage of formality is weak, that is, T_λ close to or below one; the quantitative model lets the data settle this through $(\tau_y, \tau_\ell, \lambda_s, \eta_i)$.*

P.5 The master threshold and its four instances

Combining the exit option with Proposition 7.1 yields the complete policy map of each origin state, mirroring the decision rules s_n, s_i, s_f of the quantitative model.

Every exit and entry margin reduces to one comparison, continuation in the informal sector (the relevant alternative at low $\hat{x}$, by the ordering of Proposition 7.2) against consuming the outside value. Define

$$A_s \equiv (1 - \gamma) \left(\frac{\gamma}{R_s} \right)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}, \quad A_s^0 \equiv (1 - \gamma) \left(\frac{\gamma}{R^0} \right)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}, \quad \mathcal{K}_s \equiv \beta(1 - \lambda_s)c_s, \quad (42)$$

so that $\Psi_s(\hat{x}; R_s) = A_s \hat{x}^{1/(1-\gamma)}$, and define the identities $A = A_f - A_i$ for the formality-margin slope in (15) and $A^0 = A_f^0 - A_i^0$ for the cash-rich slope of Remark P.7. Each margin is characterized by a condition of the form $G_s(\hat{x}, \tilde{n}) = m$, where $\tilde{n}$ are the funds carried into the continuation and m collects the operating and transition costs. Solving branch by branch of (38) defines the following master threshold

²⁶ $(c_{if}, 0)$ for informal incumbents, $(0, c_{fi})$ for formal incumbents, (c_f^e, c_i^e) for entrants

$$\mathcal{X}_s(\tilde{n}; m) = \begin{cases} (m/A_s^0)^{1-\gamma}, & \text{R1} \\ \frac{m + R^0 \tilde{n}}{M_s \tilde{n}^\gamma}, & \text{R2} \\ \left[\frac{m - \phi_s \tilde{n}}{A_s} \right]^{1-\gamma}, & \text{R3, if } m > \phi_s \tilde{n} \\ (1 - b_s)^\gamma \frac{m + \tilde{n} \frac{R^0 + \phi_s b_s}{1 - b_s}}{M_s \tilde{n}^\gamma}, & \text{R4,} \end{cases} \quad (43)$$

where $\mathcal{X}_s$ is continuous across the R1/R2, R2/R3 and R3/R4 boundaries, weakly decreasing in $\tilde{n}$, independent of financial conditions on R1, and satisfies $\mathcal{X}_s \rightarrow \infty$ as $\tilde{n} \rightarrow 0^+$ on R4. Notice that firms without internal funds exit or fail to enter at any productivity (financial exit). Moreover, on R2, G_s is linear in $\hat{x}$, hence the no power form. If $m \leq \phi_s \tilde{n}$ on R3 (or the analogous condition on R4), the constrained firm never exits because the quasi-rents $\phi_s \tilde{n}$ earned by internal funds keep it alive at any productivity.

Hence, the four thresholds are instances of (43):

$$\text{Exit as informal: } \hat{x}_i^X(n) = \mathcal{X}_i(n; \mathcal{K}_i), \quad n = n^{\text{exit}}, \quad (44)$$

$$\text{Exit as formal: } \hat{x}_f^X(n) = \mathcal{X}_i(n - c_{fi}; \mathcal{K}_i + c_{fi}), \quad n = n^{\text{exit}}, \quad (45)$$

$$\text{Entry as informal: } \hat{x}_i^E(k_0) = \mathcal{X}_i(k_0 - c_i^e; \mathcal{K}_i + c_i^e), \quad (46)$$

$$\text{Entry as formal: } \hat{x}_f^E(k_0) = \hat{x}_e^F(k_0) = \left[\frac{B k_0 + C_e}{A} \right]^{1-\gamma} \quad (\text{Prop. 7.1}). \quad (47)$$

A complete taxonomy of these thresholds is developed in appendix Section S.

P.6 Transition semi-elasticities

The thresholds translate into transition probabilities, the objects the Shift-Share design estimates, by integrating over the period-1 population. Consider informal incumbents with permanent component v and internal funds n (the same algebra applies to each origin state). Since $\ln \theta_1 = \ln v + \varepsilon_1$ and $\hat{x}$ is monotone in θ_1 , the probability of an informal-to-formal switch is

$$\mathbb{P}(i \rightarrow f \mid v, n) = 1 - \Phi(z_{if}), \quad z_{if} \equiv \frac{\ln \bar{\theta}_{if}(v, n) - \ln v}{\sigma_\varepsilon}, \quad (48)$$

where the productivity threshold inverts (35):

$$\ln \bar{\theta}_{if}(v, n) = \frac{1}{\rho} \left[(1 - \nu) \ln \hat{x}_i^F(n) - (1 - \rho) \ln v \right] - \frac{\sigma_\varepsilon^2}{2\rho(1 - \nu)}. \quad (49)$$

Expression (49) is common to all regions. Its derivation uses only the log-linear map (35) and the existence of a cutoff in $\hat{x}$ -units, so it converts any threshold, whichever branch

of (43) delivers it, closed-form or implicit, into a cutoff for θ_1 . The financing region enters only through the value of the threshold and through its derivatives in the chain rule below. Analogous expressions hold for $\mathbb{P}(\text{exit} \mid i) = \Phi(z_{iX})$, $\mathbb{P}(f \rightarrow i) = \Phi(z_{fi}) - \Phi(z_{fX})$, entry probabilities (with $\hat{x}_e(v)$ and thresholds in v), and all the other transitions.

Proposition P.8 (Transition semi-elasticities). *Define $p = \mathbb{P}(i \rightarrow f \mid v, n)$ from (48). Then, when both destinations borrow at an interior optimum (R3) at the margin,*

$$\frac{\partial p}{\partial \phi_f} = -\phi(z_{if}) \frac{(1-\nu)(1-\gamma)}{\rho \sigma_\varepsilon} \frac{a_f^*}{D_i} < 0, \quad \frac{\partial p}{\partial \phi_i} = \phi(z_{if}) \frac{(1-\nu)(1-\gamma)}{\rho \sigma_\varepsilon} \frac{a_i^*}{D_i} > 0, \quad (50)$$

while, when the relevant destination is collateral-constrained (R4) at the margin, the collateral responses are

$$\frac{\partial p}{\partial b_f} = \phi(z_{if}) \frac{1-\nu}{\rho \sigma_\varepsilon} \frac{\mu_f k_f^*}{N_i} \geq 0, \quad \frac{\partial p}{\partial b_i} = -\phi(z_{if}) \frac{1-\nu}{\rho \sigma_\varepsilon} \frac{\mu_i k_i^*}{N_i} \leq 0, \quad (51)$$

where N_i is the numerator of the operative branch of $\hat{x}_i^F$ at the crossing²⁷ and $\mu_s k_s^$ is the shadow value of collateral of Lemma P.5. The same structure, with the appropriate threshold and sign from Proposition 7.3, applies to every other transition probability. The factor $(1-\gamma)$ is specific to the power branch (R3) and drops on the linear branches; all responses are scaled by $\phi(z)/\sigma_\varepsilon$, which is the density of firms at the margin, so aggregate responses are largest where the firm distribution is thick near thresholds.*

Two aggregation implications worth highlighting. First, because (49) is region-free, the chain rule composes with whichever region's log-derivative is operative at the margin for the given (v, n) . Thus, the ϕ -formulas displayed are the R3 instances, the collateral formulas the R4 instances, and R1/R2 self-financed firms contribute exactly zero. Second, aggregate transition responses are therefore incidence-weighted integrals. This is $\phi(z)$ times the region-specific term, integrated over the joint distribution of (v, n) , that concentrates predicted responses among small, young, low-net-worth firms. Which is the sample counterpart of the heterogeneity documented in Section 6.

P.7 The model-implied estimand in closed form

Finally, the toy model makes precise what the Leave-One-Out Shift-Share coefficient identifies through the lens of the model. Appendix R shows as an example the detailed case of the effect on the mass of informal firms. The quantitative model computes this object numerically; the generic case presented here is schematic and serves to draw general conclusions. Let the bank-supply shock in market m shift financial conditions by

²⁷Which is equal to $D_i \hat{x}^*$ on R4×R4. See appendix Section S for more details.

$d\Xi_m = (-\chi_i\phi_i, -\chi_f\phi_f, \kappa_i b_i, \kappa_f b_f)' \zeta_m$, where ζ_m is the (exogenous) shock intensity. Aggregate borrowing is $C = \int a^* d\mu$ over active firms, and each firm-count outcome Y (stocks, entry, exit, transitions by sector) aggregates the threshold rules. To first order, the model counterpart of the IV estimand is the ratio of two equilibrium responses to the *same* financial-conditions shift:

$$\beta_Y^{\text{model}} = \frac{d \ln Y / d \zeta}{d \ln C / d \zeta} = \frac{\sum_{j \in \text{margins of } Y} \frac{\phi(z_j)}{Y} \frac{(1-\nu)(1-\gamma)}{\rho\sigma_\varepsilon} \frac{\phi a^* \chi}{D_j}}{\frac{1}{C} \left[\underbrace{\int_{\text{borrowers}} \frac{\chi_{s'} \phi_{s'} \tilde{k}_{s'}(\hat{x})}{(1-\gamma)R_{s'}} d\mu}_{\text{intensive: loan deepening}} - \underbrace{\sum_j \int w_M s_M \Delta a_j^* d\mu}_{\text{extensive: Created/swapped loans}} \right]}, \quad (52)$$

where w_M and s_M are as defined in Appendix R. Three lessons that carry to the quantitative model. First, the *numerator* is a density-weighted sum of threshold shifts: the IV coefficient is large where many firms sit near their thresholds, young, small, constrained firms, which rationalizes why the gross-flow elasticities (entry, exit, switching) exceed the net stock response, since flows add up threshold crossings while the stock nets them out, and why the responses concentrate on the formal side, the sector whose borrowing terms the shock improves. Second, the *denominator* mixes intensive responses (continuing borrowers deepen loans by $\chi_s \phi_s \tilde{k}_s / [(1-\gamma)R_s]$ each) with extensive ones (loans created and swapped at the margins), and both are built from parameters estimated off cross-sectional moments, spread levels, participation costs, collateral caps. Matching the observed first stage disciplines that financing block directly: the quantity response pins $\chi_f \phi_f$ and the collateral response pins κ_f , while the loan-rate response is kept as a measurement check rather than a target, because the observed book rate mixes repricing with near-complete turnover of the loan stock. Third, because ζ enters numerator and denominator symmetrically, β_Y^{model} is invariant to the scale of the shock for small ζ : the model counterpart can be computed from any small perturbation, and only the *incidence* $(\chi_i, \chi_f, \kappa_f)$, not the size, of the credit supply shock needs to be estimated ($\kappa_i = 0$ in the baseline: bank credit supply shocks do not shift informal collateral technology); Section 9 pins it from the stacked quantity, collateral, and switching responses and reserves the rest for validation.

Q Proofs

Proof of Lemma P.1. The objective $(1 - \tau_y^s)\theta k^\alpha \ell^\nu - (1 + \tau_\ell^s)w\ell$ is strictly concave in ℓ ($\nu < 1$); the first-order condition $\nu(1 - \tau_y^s)\theta k^\alpha \ell^{\nu-1} = (1 + \tau_\ell^s)w$ gives ℓ_s^* in (32). Substituting, using $(1 + \tau_\ell^s)w \ell^* = \nu(1 - \tau_y^s)\theta k^\alpha \ell^{*\nu}$, yields $\pi_s = (1 - \nu)(1 - \tau_y^s)\theta k^\alpha \ell^{*\nu}$ and, after substituting $\ell^{*\nu}$, expression (33). Monotonicity claims follow by inspection of the closed forms:

$\partial \ln \ell^* / \partial \ln \theta = 1/(1-\nu) > 0$, $\partial \ln \ell^* / \partial \ln k = \alpha/(1-\nu) > 0$, $\partial \ln \ell^* / \partial \ln w = -1/(1-\nu) < 0$, and both tax terms enter with negative sign. Strict concavity of π_s in k follows from $\gamma = \alpha/(1-\nu) < 1$, itself implied by $\alpha + \nu < 1$. $T < 1$ is immediate from $\tau_y, \tau_\ell > 0$. $\square$

Proof of Lemma P.2. The objective in (11) is continuous and, on each branch of $a = \max\{0, k - n\}$, strictly concave in k (the return term has curvature $\gamma < 1$; the cost terms are linear). Marginal return to capital is $g'(k) = \gamma M_s \hat{x} k^{\gamma-1} + \beta(1-\delta)$, strictly decreasing, with range $(\beta(1-\delta), \infty)$. Marginal cost is 1 for $k < n$ and $1 + \phi_s$ for $k > n$. If $g'(n) \leq 1$, i.e. $n \geq \bar{k}_s$, the self-financed optimum $\bar{k}_s$ (where $g' = 1$) is attainable without borrowing: region R1. If $1 < g'(n) \leq 1 + \phi_s$, i.e. $\tilde{k}_s \leq n < \bar{k}_s$, the kink at $k = n$ is optimal: region R2. If $g'(n) > 1 + \phi_s$, i.e. $n < \tilde{k}_s$, the firm borrows to the interior optimum $\tilde{k}_s$ (where $g' = 1 + \phi_s$) provided the collateral constraint $k - n \leq b_s k \iff k \leq n/(1 - b_s)$ permits, i.e. $\tilde{k}_s \leq n/(1 - b_s)$: region R3. Otherwise the constraint binds and $k = n/(1 - b_s)$: region R4. The boundaries in (36) restate these conditions. The derivative signs follow from $\tilde{k}_s = (\gamma M_s \hat{x} / R_s)^{1/(1-\gamma)}$ and $\bar{k}_s = (\gamma M_s \hat{x} / R^0)^{1/(1-\gamma)}$, so that $\partial \tilde{k}_s / \partial \phi_s = -\tilde{k}_s / [(1-\gamma)R_s]$, $\partial \tilde{k}_s / \partial \hat{x} = \tilde{k}_s / [(1-\gamma)\hat{x}]$ and $\partial \bar{k}_s / \partial \hat{x} = \bar{k}_s / [(1-\gamma)\hat{x}]$; from $k = n/(1 - b_s)$ in R4, whose b_s -derivative is $n/(1 - b_s)^2$; and by inspection elsewhere. Continuity at boundaries is direct substitution. $\square$

Proof of Lemma P.3. Substitute the policy (36) into (11). R1: $G = M_s \hat{x} \bar{k}^\gamma - R^0 \bar{k}$; using $\gamma M_s \hat{x} \bar{k}^{\gamma-1} = R^0$, $G = R^0 \bar{k}(1-\gamma)/\gamma = \Psi_s(\hat{x}; R^0)$ after substituting $\bar{k}$. R2: $k = n$, $a = 0$: $G = M_s \hat{x} n^\gamma - R^0 n$. R3: $G = -\tilde{k} - \phi_s(\tilde{k} - n) + M_s \hat{x} \tilde{k}^\gamma + \beta(1-\delta)\tilde{k} = \phi_s n + M_s \hat{x} \tilde{k}^\gamma - R_s \tilde{k}$, and the same substitution with R_s gives $\phi_s n + \Psi_s(\hat{x}; R_s)$. R4: substitute $k = n/(1 - b_s)$ directly. Monotonicity and convexity of Ψ_s in $\hat{x}$ follow from the exponent $1/(1-\gamma) > 1$; G inherits them. The formulas for q_s are the n -derivatives of (38); the R2 bounds follow because $g'(n) \in (1, 1 + \phi_s]$ there, and the R4 inequality because the bracketed marginal-return gap is strictly positive when the collateral constraint binds. Concavity of G in n holds piecewise and at kinks because q_s is (weakly) decreasing in n region by region as n rises through R4→R3→R2→R1. $\square$

Proof of Lemma P.5. On R1–R2, $a^* = 0$ and (ϕ_s, b_s) do not enter the realized objective: derivatives are zero, consistent with (40). On R3–R4 apply the envelope theorem to (11): the derivative of the value with respect to ϕ_s at the optimum is the partial derivative of the objective, $-a_s^*$; for b_s , the constraint $a \leq b_s k$ has multiplier $\mu_s \geq 0$ and the envelope derivative is $\mu_s k_s^*$, zero when slack (R3) and equal to the stated expression in R4, where the Kuhn–Tucker condition gives $\mu_s = [\gamma M_s \hat{x} k_s^{*\gamma-1} - R_s] / (1 - b_s) \cdot (\text{normalization})$; differentiating (38) in R4 with respect to b_s directly yields $[\gamma M_s \hat{x} k_s^{*\gamma-1} - R_s] n / (1 - b_s)^2 > 0$. Cross-sector derivatives are zero because s' -parameters do not enter sector- s payoffs. $\square$

Proof of Proposition 7.1. On R3×R3, (14) follows by subtracting the R3 branch of (38) for each destination and collecting the $\hat{x}^{1/(1-\gamma)}$, n_o , and constant terms, using (10). $A > 0$ iff

$M_f^{1/(1-\gamma)} R_f^{-\gamma/(1-\gamma)} > M_i^{1/(1-\gamma)} R_i^{-\gamma/(1-\gamma)}$, i.e. iff $M_f/M_i > (R_f/R_i)^\gamma$; since $M_f/M_i = T_\lambda$ by definition, this is Assumption 1. Strict monotonicity and convexity in $\hat{x}$ follow from $A > 0$ and the exponent $1/(1-\gamma) > 1$; since $\Delta_o \rightarrow -(Bn_o + C_o) < 0$ as $\hat{x} \rightarrow 0$ (by Assumption 2) and $\Delta_o \rightarrow \infty$ as $\hat{x} \rightarrow \infty$, a unique zero exists and solves (18). Differentiating (18) in n_o gives (19). For completeness, off R3×R3 the difference Δ_o remains continuous and strictly increasing in $\hat{x}$ wherever the formal destination is in a (weakly) more constrained region than the informal one, which holds at the crossing point under Assumption 1 because $\tilde{k}_f > \tilde{k}_i$ (shown in the proof of Corollary 7.4); hence the crossing is unique globally on the borrowing range. $\square$

Proof of Lemma P.6. Substituting each origin's fee pair into (16) gives (41) directly; the common constant K cancels from every pairwise difference. Upper comparison:

$$C_{i\text{-inc}} - C_e = (1 + \phi_f)c_{if} - [(1 + \phi_f)c_f^e - (1 + \phi_i)c_i^e] > 0 \iff (\text{A2-i}).$$

Lower comparison:

$$C_e - C_{f\text{-inc}} = (1 + \phi_f)c_f^e - (1 + \phi_i)c_i^e + (1 + \phi_i)c_{fi} > 0 \iff (\text{A2-ii}).$$

Note that $c_i^e \leq c_f^e$ alone does *not* imply (A2-ii), because $\phi_i > \phi_f$ can make $(1 + \phi_i)c_i^e$ exceed $(1 + \phi_f)c_f^e$; if $c_{fi} \geq c_i^e$, however, $(1 + \phi_i)(c_{fi} - c_i^e) \geq 0$ and $(1 + \phi_f)c_f^e > 0$ deliver (A2-ii). Finally, $C_{i\text{-inc}} - C_{f\text{-inc}} = (1 + \phi_f)c_{if} + (1 + \phi_i)c_{fi} > 0$ under positive switching fees alone. $\square$

Proof of Proposition 7.2. The three-interval structure for each origin follows from: (i) values of continuation are continuous and strictly increasing in $\hat{x}$ while exit/out values are constant in $\hat{x}$, giving unique exit/entry cutoffs; (ii) Proposition 7.1 gives the unique formality cutoff above them whenever $\hat{x}^X < \hat{x}^F$, which holds under Assumption 2 (the formality margin requires strictly positive net cost $D_o > 0$ while the exit margin turns on c_i); the formal incumbent's middle interval is switching down rather than staying informal by definition of its choice set. The ordering $\hat{x}_f^F < \hat{x}_e^F < \hat{x}_i^F$ follows from (18): thresholds are increasing in C_o , and $C_{f\text{-inc}} < C_{\text{entrant}} < C_{i\text{-inc}}$ by Assumption 2. $\square$

Proof of Proposition 7.3. Each threshold is defined by an indifference condition $\Delta(\hat{x}^*, \cdot) = 0$ between two options. By the implicit function theorem, $\partial\hat{x}^*/\partial\xi = -(\partial\Delta/\partial\xi)/(\partial\Delta/\partial\hat{x})$, and $\partial\Delta/\partial\hat{x} > 0$ at an upward crossing. The numerators are given by Lemma P.5: for the formality margins, $\partial\Delta/\partial\phi_f = -a_f^*$, $\partial\Delta/\partial\phi_i = +a_i^*$, $\partial\Delta/\partial b_f = \mu_f k_f^* \geq 0$, $\partial\Delta/\partial b_i = -\mu_i k_i^* \leq 0$. Using $\partial\Delta/\partial\hat{x} = A\hat{x}^{\gamma/(1-\gamma)}/(1-\gamma) = D_o/[(1-\gamma)\hat{x}^*]$ at the threshold (substituting $A\hat{x}^{*1/(1-\gamma)} = D_o$) delivers the log-derivatives in the first row. For the informal entry/exit margins the indifference is $G_i = \text{const}$, so $\partial\Delta/\partial\phi_i = -a_i^*$ and $\partial\Delta/\partial\hat{x} = \Psi_i' > 0$, giving the stated positive derivative (thresholds rise as informal finance becomes dearer); formal-side parameters do not enter, hence the zeros. For formal exit, the relevant comparison at low $\hat{x}$ under the ordering of Proposition 7.2 is exit versus switching down, whose value is

$U_i(\hat{x}, n - c_{fi})$; its derivatives in informal financial parameters are those of U_i (Lemma P.5), delivering the third row's informal-side entries, and formal-side parameters do not enter the comparison at all, hence its zeros. Off that ordering — if staying formal were the binding alternative at the exit margin — the cutoff would be $\mathcal{X}_f(n; \mathcal{K}_f)$ and easier formal terms would weakly lower it (≥ 0 in ϕ_f , ≤ 0 in b_f), which is the caveat recorded in the table note. $\square$

Proof of Proposition P.8. $p = 1 - \Phi(z)$ with $z = (\ln \bar{\theta} - \ln v)/\sigma_\varepsilon$ and, from (49), $\partial \ln \bar{\theta}/\partial \xi = \frac{1-\nu}{\rho} \partial \ln \hat{x}_i^F/\partial \xi$ for any financial parameter ξ . Chain-differentiate and substitute the log-derivatives from Proposition 7.3; $\phi(\cdot) > 0$ delivers the signs. For (51): on the linear branches the threshold is the ratio of a linear numerator to a slope (Section S); the quotient rule with the root identity $N_i = D_4 \hat{x}^*$ gives $\partial \ln \hat{x}_i^F/\partial b_f = -\mu_f k_f^*/N_i$ and $\partial \ln \hat{x}_i^F/\partial b_i = +\mu_i k_i^*/N_i$, and the same chain rule applies; the $(1 - \gamma)$ factor is absent because the operative branch is linear in $\hat{x}$. $\square$

Proof of Corollary 7.4. (i) Under Proposition 7.2 the binding entry margin and both exit margins compare an *informal* continuation with the outside value, so by Lemma P.5 they respond only to informal-side terms (second and third rows of Proposition 7.3): all three thresholds fall when $\varepsilon_i > 0$ or $db_i > 0$, and total entry, the mass of entrants above $\hat{x}_i^E$, rises.

(ii)–(iii) By Lemma P.5, total differentiation of the formality margin of origin o gives

$$d\Delta_o = -a_f^* d\phi_f + a_i^* d\phi_i = \varepsilon_f a_f^* - \varepsilon_i a_i^*,$$

so $\hat{x}_o^F$ falls, formality gains ground, if and only if $\varepsilon_f a_f^* > \varepsilon_i a_i^*$. Since $p_{i \rightarrow f}$ falls in $\hat{x}_i^F$ while $p_{f \rightarrow i}$ rises in $\hat{x}_f^F$, this is (20) at the $i \rightarrow f$ margin and, reversed, (21) at the $f \rightarrow i$ margin. For the borrowing gaps: at the $i \rightarrow f$ margin the formal option carries funds $n - c_{if}$ and the informal option n , so

$$a_f^* - a_i^* = [\tilde{k}_f - (n - c_{if})] - [\tilde{k}_i - n] = \tilde{k}_f - \tilde{k}_i + c_{if}.$$

For the entrant margin the fee pair (c_f^e, c_i^e) gives $a_f^* - a_i^* = \tilde{k}_f - \tilde{k}_i + c_f^e - c_i^e$, so the same argument applies to $\hat{x}_e^F$.

Under Assumption 1, $T_\lambda > (R_f/R_i)^\gamma$ implies $T_\lambda R_i/R_f > (R_i/R_f)^{1-\gamma} > 1$, hence $\tilde{k}_f/\tilde{k}_i = (T_\lambda R_i/R_f)^{1/(1-\gamma)} > 1$: the gap is strictly positive, and $\varepsilon_f = \varepsilon_i$ delivers (20). At the $f \rightarrow i$ margin the fee switches sides, $a_i^* - a_f^* = \tilde{k}_i - \tilde{k}_f + c_{fi}$, giving the statement in (iii). $\square$

Proof of Proposition 7.5. (i) The entry threshold $\bar{v}$ falls by Proposition 7.3 (second row), so the entry probability $1 - F_v(\bar{v})$ rises for every k_0 with a constrained margin. (ii) For a log-normal v the truncated mean $m(\bar{v}) \equiv \mathbb{E}[v \mid v \geq \bar{v}]$ satisfies $m'(\bar{v}) = h(\bar{v}) [m(\bar{v}) - \bar{v}] > 0$, where $h = f_v/(1 - F_v)$ is the hazard and $m(\bar{v}) > \bar{v}$; hence a lower threshold strictly lowers the truncated mean, and first-order stochastic dominance of the truncated distributions

delivers the same for every quantile and for average value added (an increasing transform of v). (iii) Write the cohort shutdown rate as $S(\bar{v}) = \mathbb{E}[\pi^X(v) \mid v \geq \bar{v}]$ with π^X strictly decreasing (by (22) and $\partial k_2/\partial v \geq 0$). Then $S'(\bar{v}) = h(\bar{v})[S(\bar{v}) - \pi^X(\bar{v})] < 0$ because $\pi^X(\bar{v}) > \pi^X(v)$ for all $v > \bar{v}$ implies $\pi^X(\bar{v}) > S(\bar{v})$. A lower $\bar{v}$ therefore raises the cohort shutdown rate. $\square$

R Explicit instance of the estimand: informality mass

Fix the outcome $Y = N'_i$, the mass choosing the informal sector at the transition stage (survival $1 - \lambda_i$ multiplies the measured stock but cancels from $d \ln Y$). Six thresholds delimit it: informal incumbents contribute the band $[\hat{x}_i^X, \hat{x}_i^F)$, formal incumbents the switch-down band $[\hat{x}_f^X, \hat{x}_f^F)$, and entrants the interval $[\bar{v}^E, \bar{v}^F)$ in type units. Assume the empirically relevant financing regions (R3 at every one-sided margin, R3×R3 at the formality margins); on R4-financed margins the formulas below pick up the collateral terms $(\mu k^*, \kappa_s b_s)$ per the taxonomy appendix. For each margin M , contracting Proposition 7.3 with the shock $(d\phi_s, db_s) = (-\chi_s \phi_s, \kappa_s b_s) d\zeta$ gives the log-shift per unit of intensity,

$$s_M = -(1 - \gamma) \chi_i \phi_i \frac{a_{i,M}^*}{N_M}, \quad M \in \{X_i, X_f, E\} \text{ (one-sided, i-continuation),}$$

$$s_{F,o} = -\frac{1 - \gamma}{D_o} \left[\chi_f \phi_f a_{of}^* - \chi_i \phi_i a_{oi}^* \right], \quad o \in \{i, f, e\} \text{ (formality margins),}$$

with $(\tilde{n}_M, m_M)$ each margin's funds–cost pair, $N_M = m_M - \phi_i \tilde{n}_M$ the operative R3 numerator, and all a^* 's evaluated at the crossing. The density weights at the margins are $w_M = \frac{1-\nu}{\rho\sigma_\varepsilon} \phi(z_M)$ for incumbent margins and $w_M = (1 - \nu) f_v(\bar{v}_M) \bar{v}_M$ for entrant margins (entry decisions are deterministic in v , so the relevant density is that of the type distribution). Writing $\iota_M = +1$ if an upward crossing of M moves a firm *into* Y and -1 if out of it ($\iota = +1$ at X_i, X_f, E ; $\iota = -1$ at F_i, F_f, F_e), the reduced form is

$$\frac{d \ln N'_i}{d\zeta} = -\frac{1}{N'_i} \sum_M \iota_M \int w_M s_M d\mu_{o(M)}, \quad (53)$$

each integral taken over the origin population carrying margin M . The first stage decomposes into the loan-size response of continuing borrowers and the loans created or swapped at the same six margins:

$$\frac{d \ln C}{d\zeta} = \frac{1}{C} \left[\sum_{s' \in \{i, f\}} \int_{\text{borrowers to } s'} \frac{\chi_{s'} \phi_{s'} \tilde{k}_{s'}(\hat{x})}{(1 - \gamma) R_{s'}} d\mu - \sum_M \int w_M s_M \Delta a_M^* d\mu_{o(M)} \right], \quad (54)$$

where Δa_M^* is the borrowing gained on an upward crossing: the new loan $a_{i,M}^*$ at X_i, X_f, E (exit and non-entry carry no loan), and the destination swap $\Delta a_{F,o}^* = \tilde{k}_f - \tilde{k}_i + \kappa_{of} - \kappa_{oi}$ at the formality margins — precisely the borrowing gaps of Corollary 7.4. The model coefficient is the ratio $\beta_{N'_i}^{\text{model}} = (d \ln N'_i / d\zeta) / (d \ln C / d\zeta)$; every term in (53)–(54) is linear in the incidence $(\chi_i, \chi_f, \kappa_f)$, so the scale of ζ cancels.

S Complete threshold taxonomy

This appendix reports the fully substituted closed form of every exit and entry threshold on every financing region. Throughout, $\mathcal{K}_s \equiv \beta(1 - \lambda_s)c_s$ (so that $K = \mathcal{K}_f - \mathcal{K}_i$), $A_s = (1 - \gamma)(\gamma/R_s)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}$ and $A_s^0 = (1 - \gamma)(\gamma/R^0)^{\frac{\gamma}{1-\gamma}} M_s^{\frac{1}{1-\gamma}}$, with $A = A_f - A_i$ and $A^0 = A_f^0 - A_i^0$. Each one-sided margin (exit or entry against the outside option, informal continuation being the relevant alternative by Proposition 7.2) is the solution of $G_i(\hat{x}, \tilde{n}) = m$ with the margin-specific pair $(\tilde{n}, m)$:

margin	$\tilde{n}$	m	comparison
exit as informal	n	$\mathcal{K}_i$	$U_i(\hat{x}, n) \geq n$
exit as formal	$n - c_{fi}$	$\mathcal{K}_i + c_{fi}$	$U_i(\hat{x}, n - c_{fi}) \geq n$
entry as informal	$k_0 - c_i^e$	$\mathcal{K}_i + c_i^e$	$U_i(\hat{x}, k_0 - c_i^e) \geq k_0$

with $n = n^{\text{exit}}$ of (10). Substituting the four branches of (38):

Exit as informal.

$$\hat{x}_i^X(n) = \begin{cases} (\mathcal{K}_i/A_i^0)^{1-\gamma}, & \text{R1} \\ \frac{\mathcal{K}_i + R^0 n}{M_i n^\gamma}, & \text{R2} \\ \left[\frac{\mathcal{K}_i - \phi_i n}{A_i} \right]^{1-\gamma}, & \text{R3, if } \mathcal{K}_i > \phi_i n \\ (1 - b_i)^{\gamma-1} \frac{\mathcal{K}_i(1 - b_i) + n(R^0 + \phi_i b_i)}{M_i n^\gamma}, & \text{R4.} \end{cases} \quad (55)$$

Exit as formal (against switching down).

$$\hat{x}_f^X(n) = \begin{cases} [(\mathcal{K}_i + c_{fi})/A_i^0]^{1-\gamma}, & \text{R1} \\ \frac{\mathcal{K}_i + c_{fi} + R^0(n - c_{fi})}{M_i (n - c_{fi})^\gamma}, & \text{R2} \\ \left[\frac{\mathcal{K}_i + (1 + \phi_i)c_{fi} - \phi_i n}{A_i} \right]^{1-\gamma}, & \text{R3, if numerator} > 0 \\ (1 - b_i)^{\gamma-1} \frac{(\mathcal{K}_i + c_{fi})(1 - b_i) + (n - c_{fi})(R^0 + \phi_i b_i)}{M_i (n - c_{fi})^\gamma}, & \text{R4.} \end{cases} \quad (56)$$

Entry as informal. Replace (n, c_{fi}) by (k_0, c_i^e) in (56); in particular on R3,

$$\hat{x}_i^E(k_0) = \left[\frac{\mathcal{K}_i + (1 + \phi_i)c_i^e - \phi_i k_0}{A_i} \right]^{1-\gamma}, \quad (57)$$

and in type units $\bar{v}(k_0) = \hat{x}_i^E(k_0)^{1-\nu} \exp\{-\sigma_\varepsilon^2/(2(1 - \nu))\}$.

Entry as formal (the entrant formality margin). This margin compares two continuation values, each on its own region, with funds $\tilde{n}_f = k_0 - c_f^e$ and $\tilde{n}_i = k_0 - c_i^e$. Because $G_s(\cdot, \tilde{n})$ is a power function of $\hat{x}$ on R1/R3 but *linear* in $\hat{x}$ on R2/R4, closed forms obtain on the diagonal region pairs:

$$\text{R3} \times \text{R3: } \hat{x}_e^F(k_0) = \left[\frac{B k_0 + C_e}{A} \right]^{1-\gamma}, \quad C_e = (1 + \phi_f)c_f^e - (1 + \phi_i)c_i^e + K; \quad (58)$$

$$\text{R1} \times \text{R1: } \hat{x}_e^F = (C_e^0/A^0)^{1-\gamma}, \quad C_e^0 = c_f^e - c_i^e + K, \quad (59)$$

(direction of the inequality per the trichotomy of Remark P.7);

$$\text{R2} \times \text{R2: } \hat{x}_e^F = \frac{K + \beta(1 - \delta)(c_f^e - c_i^e)}{M_f \tilde{n}_f^\gamma - M_i \tilde{n}_i^\gamma}, \quad \text{formal above iff the denominator} > 0 \quad (60)$$

(at kink, only cost is forgone resale value, hence $\beta(1 - \delta)$, not R^0);

$$\text{R4} \times \text{R4: } \hat{x}_e^F = \frac{K + R_f h_f - R_i h_i + (1 + \phi_i)\tilde{n}_i - (1 + \phi_f)\tilde{n}_f}{M_f h_f^\gamma - M_i h_i^\gamma}, \quad (61)$$

with $h_s \equiv \tilde{n}_s/(1 - b_s)$ the capital installed at the binding cap.

Off-diagonal pairs (one destination borrowing while the other self-finances, e.g. R3×R2) mix $\hat{x}^{1/(1-\gamma)}$ and $\hat{x}$ terms: the indifference condition remains strictly monotone in $\hat{x}$ at the crossing and has a unique root, but no closed form; it is solved implicitly. The same diagonal/off-diagonal taxonomy applies verbatim to the incumbent formality margins $\hat{x}_i^F, \hat{x}_f^F$ with the fee pairs of Lemma P.6.

Branch selection. Which branch applies is a self-consistency requirement: compute the candidate threshold from each branch and retain the one whose funds fall in that branch's bracket *evaluated at the candidate*,

$$\text{R1: } \tilde{n} \geq \bar{k}_s(\hat{x}^*); \quad \text{R2: } \tilde{k}_s(\hat{x}^*) \leq \tilde{n} < \bar{k}_s(\hat{x}^*); \quad \text{R3: } (1 - b_s)\tilde{k}_s(\hat{x}^*) \leq \tilde{n} < \tilde{k}_s(\hat{x}^*); \quad \text{R4: below.}$$

Exactly one branch is self-consistent: G_s is continuous and strictly increasing in $\hat{x}$ ($\partial G_s / \partial \hat{x} = M_s k^{*\gamma} > 0$ by the envelope theorem), and at fixed $\tilde{n}$ the firm moves monotonically through R1→R2→R3→R4 as $\hat{x}$ rises, so $G_s(\cdot, \tilde{n})$ crosses any constant m exactly once.

T Discussion of modeling assumptions

Holding the wage fixed is the exact analogue of the empirical identification, which is within market and year and purged of aggregate shocks, and it is additionally supported by local wage stickiness at the five year horizon. Debt is within period. Firms borrow to install capital and repay out of the following period's resources, so net worth stays out of the state vector, at the cost of abstracting from leverage dynamics. The measurement counterpart is that model borrowing maps to gross new credit within a year and the cap b_s

to the ratio of credit lines to capital among borrowers. Loans are safe because the cap guarantees repayment, so ϕ_s is an intermediation cost rather than a default premium, and reinterpreting part of it as a default premium would change the welfare accounting but not the firm’s problem. The financial primitives themselves are measured rather than derived from deeper frictions.²⁸ Allowing the productivity process to differ by sector is essential, since the census autocovariances imply higher innovation variance for informal firms, and imposing a common process would force the switching costs to absorb the difference.

U Presets and additional model exhibits

This appendix collects the preset table, the measurement choices behind it, and two estimation exhibits referenced from Section 9.

Table 29: Preset parameters

Parameter	Value	Source
β discount factor	0.95	real rate near five percent
δ depreciation	0.10	standard
w wage	1	normalization
τ_y effective output tax	0.048	corporate income tax collections over formal value added
τ_ℓ effective payroll wedge	0.16	social security and payroll taxes over the formal wage bill
α, ν technology	0.17, 0.59	census cost shares
ρ_i, ρ_f persistence	0.95, 0.95	census autocovariances
$\sigma_{\varepsilon,i}, \sigma_{\varepsilon,f}$ innovations	0.14, 0.21	census autocovariances
$\sigma_{u,i}, \sigma_{u,f}$ transitory real	0.83, 0.53	half of the excess variance at horizon zero
σ_v permanent dispersion	0.56	between firm construction
λ_i, λ_f death rates	0.067, 0.056	flat tail of exit by size
ϕ_i, ϕ_f spreads	0.07, 0.05	Median R04C registry loan spreads over CETES 28 by borrower IMSS registration, 2023 (classification validated in Figure 5)
b_i, b_f collateral capacities	0.85, 0.90	ninetieth percentile of leverage among borrowers; the formal read of 1.25 is capped at 0.90 for physical feasibility.
η_i audit scale	2.71	informal labor share inversion

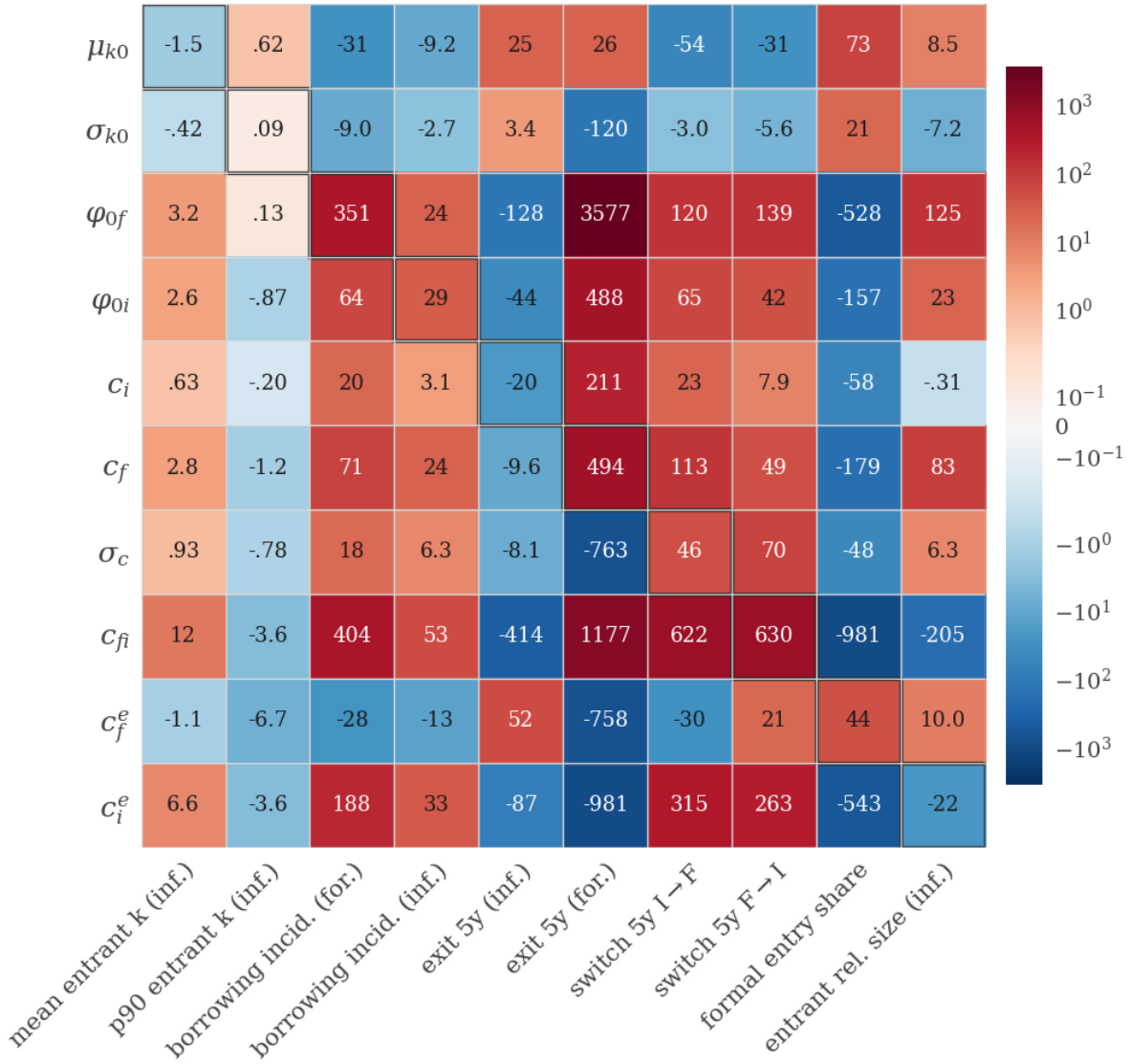
²⁸Limited enforcement generates endogenous borrowing limits (Albuquerque and Hopenhayn, 2004; Clementi and Hopenhayn, 2006), costly intermediation and default generate endogenous spreads (Cooley and Quadrini, 2001; Arellano et al., 2012), and D’Erasmus and Boedo (2012) embed endogenous credit in an economy with informality. Each such theory delivers exactly the two objects assumed here, a quantity limit and a price wedge, and predicts the estimated sector ranking. Treating them as measured primitives, as in Midrigan and Xu (2014) and Moll (2014), keeps the mapping from the quasi experimental evidence transparent. The cost is a caveat on the counterfactuals. Under a large formalization the informal primitives would themselves improve, so experiments holding them fixed likely understate the financial channel, in the direction Brooks and DAVIS (2020) document.

Four presets deserve a word. The technology parameters come from cost shares because a regression of output on inputs is contaminated by the response of inputs to productivity, and the regression labor coefficient of 1.26 is reported only as a diagnostic of that bias. The productivity process is identified from the autocovariances of measured productivity at horizons of five and ten years, which are free of measurement error, with the horizon zero equation left to determine the sum of transitory and error variance.²⁹ Death rates come from the flat top of the exit by size profile, since endogenous exit concentrates among small firms while exogenous death is size blind, and the gradient below the tail is left as validation. The spreads are the parameters themselves, because financing costs are linear at the margin, so the observed sector level loan rate net of the risk free rate requires no estimation, and the caps exploit the atom that the model places at maximal leverage among constrained borrowers, read at the ninetieth percentile to absorb measurement error in imputed capital.³⁰

²⁹The naive inversion that uses the horizon zero autocovariance biases persistence down and innovation variance up, and it is re run as robustness. Autocovariances cannot split the transitory sum into a real component and error, since both load only on horizon zero, but behavior can, because only the real component moves exit, switching, and borrowing. The baseline assigns half of the excess to the real component and re runs the chain at one quarter and three quarters. The logic of a split disciplined by behavior is familiar from [Blundell et al. \(2008\)](#), with firm turnover in the role of consumption.

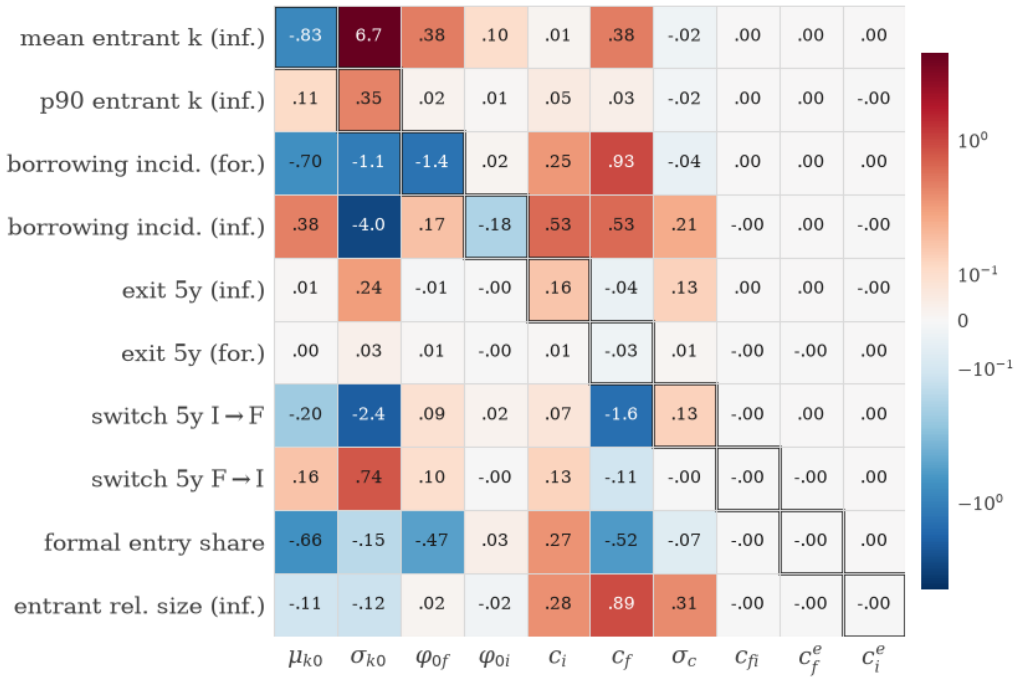
³⁰The informal cap uses the most informal cell the lender survey contains and is therefore an upper bound, a caveat the robustness battery takes up. Effective rather than statutory tax rates are used because the model abstracts from worker level registration, and the statutory rates serve as the robustness bracket.

Figure 12: Sensitivity of estimates to moments



The measure of [Andrews et al. \(2017\)](#). The formal exit rate, whose Jacobian row is nearly flat, is the single most influential target, and the small fees are sensitive to many moments at once; both patterns are reflected in the standard errors of Table 11.

Figure 13: Jacobian of moments with respect to parameters



Elasticity units. Rows are moments and columns are parameters. Own responses are strong for the cost and wealth parameters, the entrant wealth dispersion is the one global column, and the small fees' columns are near zero, matching their standard errors.

Figure 14: Fit of the targeted cross sectional moments

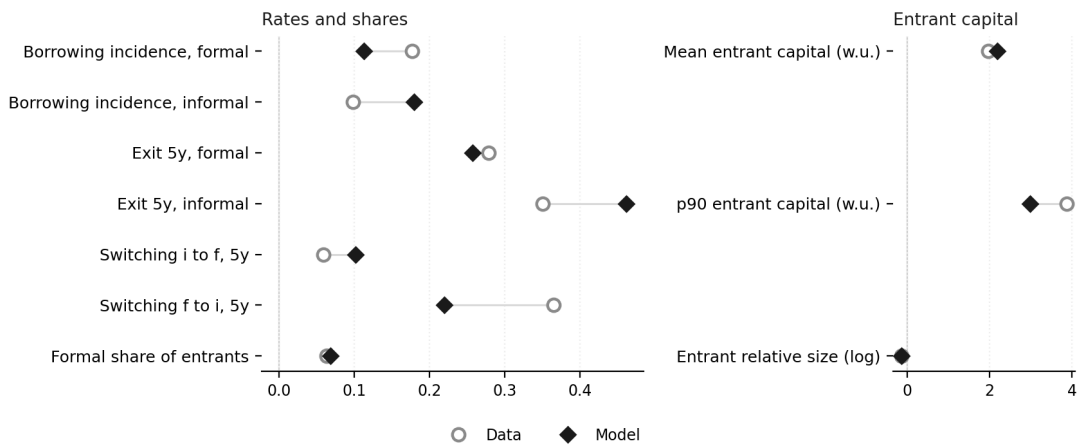
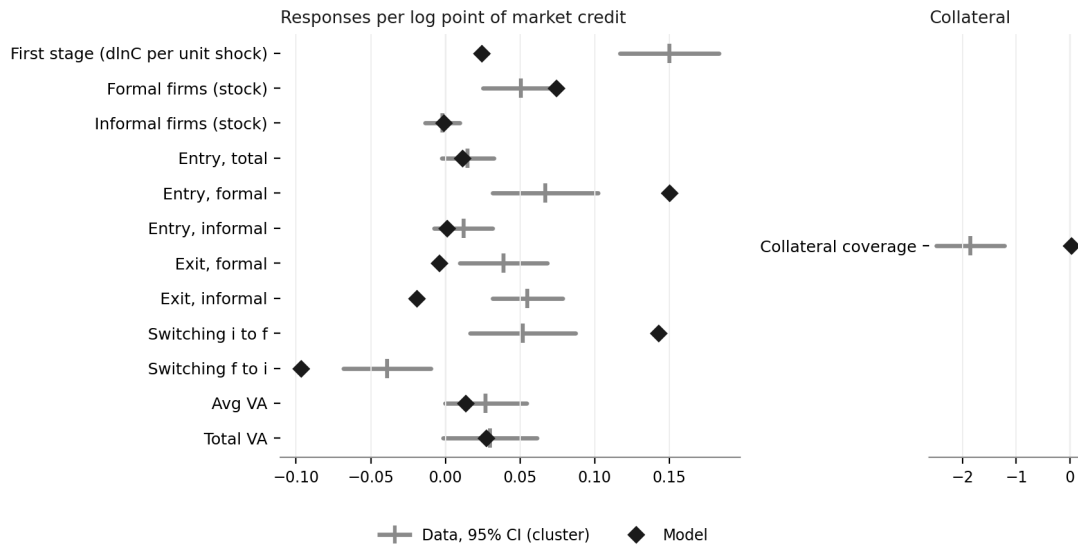
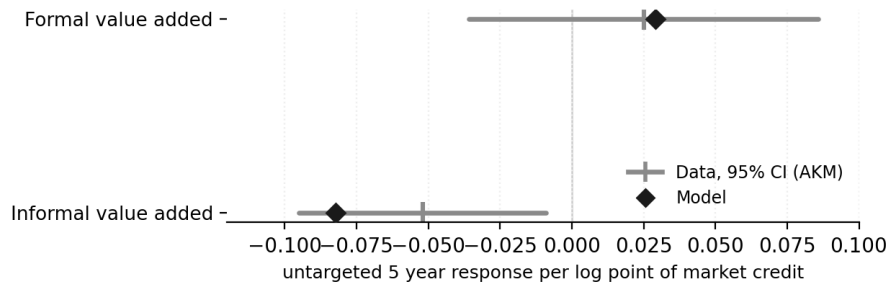


Figure 15: Fit of the targeted quasi experimental coefficients



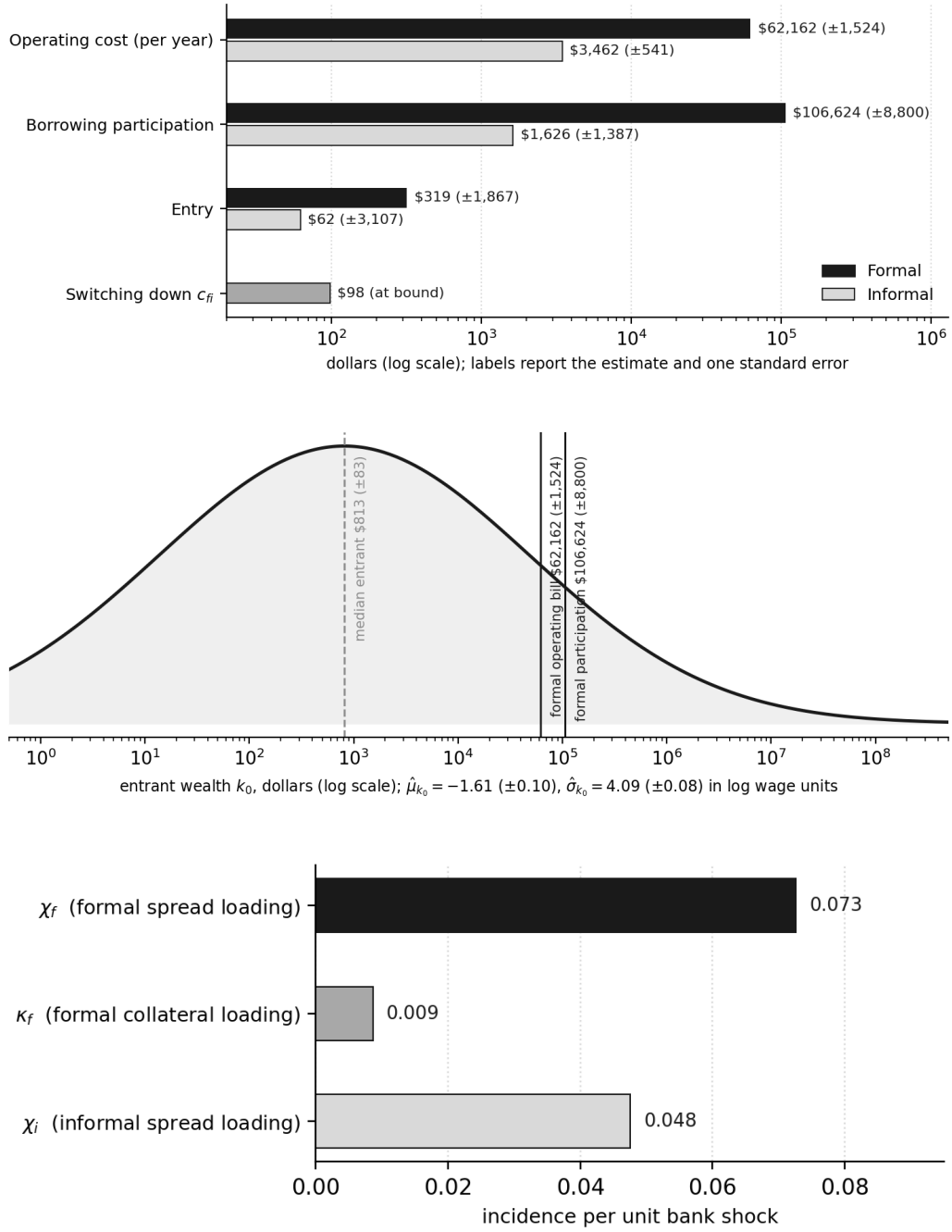
Model coefficients against the empirical estimates with their cluster-robust confidence bands.

Figure 16: Untargeted validation of the sectoral value added responses



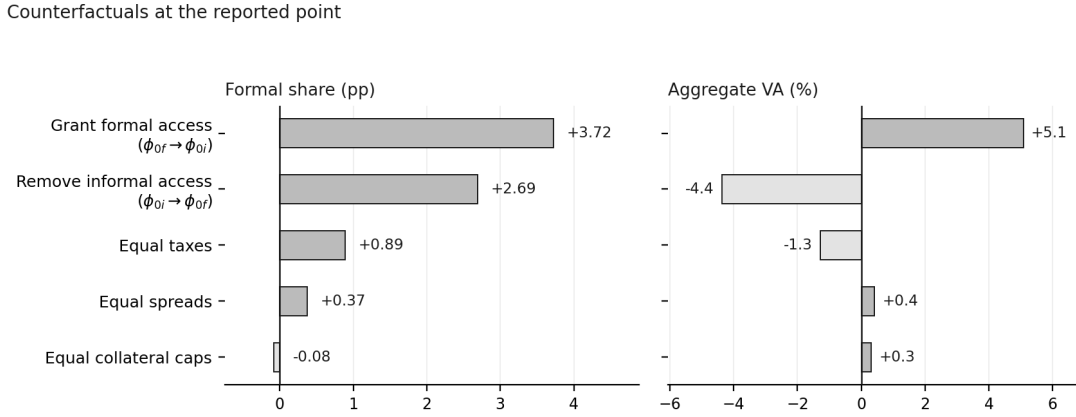
The model, estimated without these two responses, predicts value added moving toward formal firms.
The empirical counterparts are 0.025 and -0.052 .

Figure 17: Estimated parameters in money units



The top panel shows the operating and financial participation costs by sector, the middle panel the entrant wealth distribution against the formal bill, and the bottom panel the incidence of the credit supply shock. Conversions use the median annual wage of 77.5 thousand pesos and 19 pesos per dollar.

Figure 18: The counterfactual policy menu



V Measurement

The three data sources live on separated secure servers, so every moment is defined once, in the microdata language of its source, and the model reproduces the measurement process rather than an idealized object. Table 30 collects the conventions.

Four conventions deserve emphasis. First, the census is quinquennial, so exit and switching are observed as five year outcomes in which death and exit are indistinguishable and round trips are invisible. Rather than annualizing with a constant hazard formula, which is reported only as a diagnostic, the estimation computes every dynamic moment from the model's five year transition kernel with the same definitions the extraction scripts apply to the census. Second, census employment includes owners and unpaid family workers. The model's owner unit and its self employment corner exist precisely so that the model size distribution is comparable to the census one. Third, financing is measured as a flow. The survey reports credit raised during the reference year and the approved line of the main credit, and the model's within period borrowing and cap are their exact analogues, so no debt stock imputation is needed, with the stock based variant serving as a bracket. Fourth, all monetary moments are converted to multiples of the census median annual wage, the model's numeraire, and all pricing moments are spreads over the 28 day CETES rate, matching the convention that Φ_s prices external funds over the risk free rate.

Table 30: Alignment of model objects with the data

Object	Data measurement	Model counterpart
Formality status	EC, positive social security contributions (J300A > 0), with zero income firms holding zero contributions coded formal	sector $s = f$
Employment	EC, total persons engaged including owners and unpaid workers (H001A)	$1 + \ell^h$, the owner unit plus hires
Profit	EC, value added minus remunerations (A131A - J000A), with contributions and penalties not netted out	$y - w \ell^h$
Capital	EC, end of year fixed assets at book value (Q000A)	installed k
External funds	ENAFIN, credit raised during the year (P35*), and EC credit incidence (0511A)	transition stage borrowing a
Formal leverage cap	ENAFIN, approved credit line over imputed capital (P38_4B/ $\hat{K}$), ninetieth percentile among borrowers, with the debt stock variant (P24/ $\hat{K}$) as a bracket	atom of a/k' at b_f
Informal leverage cap	ENAFIN proxy informal cell, micro size crossed with persona fisica, ninetieth percentile of leverage among borrowers	atom of a/k' at b_i
Loan pricing	R04C registry median spreads over CETES 28 by borrower IMSS registration, 2023, with the ENAFIN median rate as a cross check	marginal cost ϕ_s over the risk free rate
Firm dynamics	EC waves five years apart, with exit defined as absence at the next wave, switching among survivors, and entrants defined as age two or less	the five year transition kernel under identical definitions
Units	thousands of 2023 pesos	multiples of the median annual wage of 77.5 thousand pesos, with $w = 1$